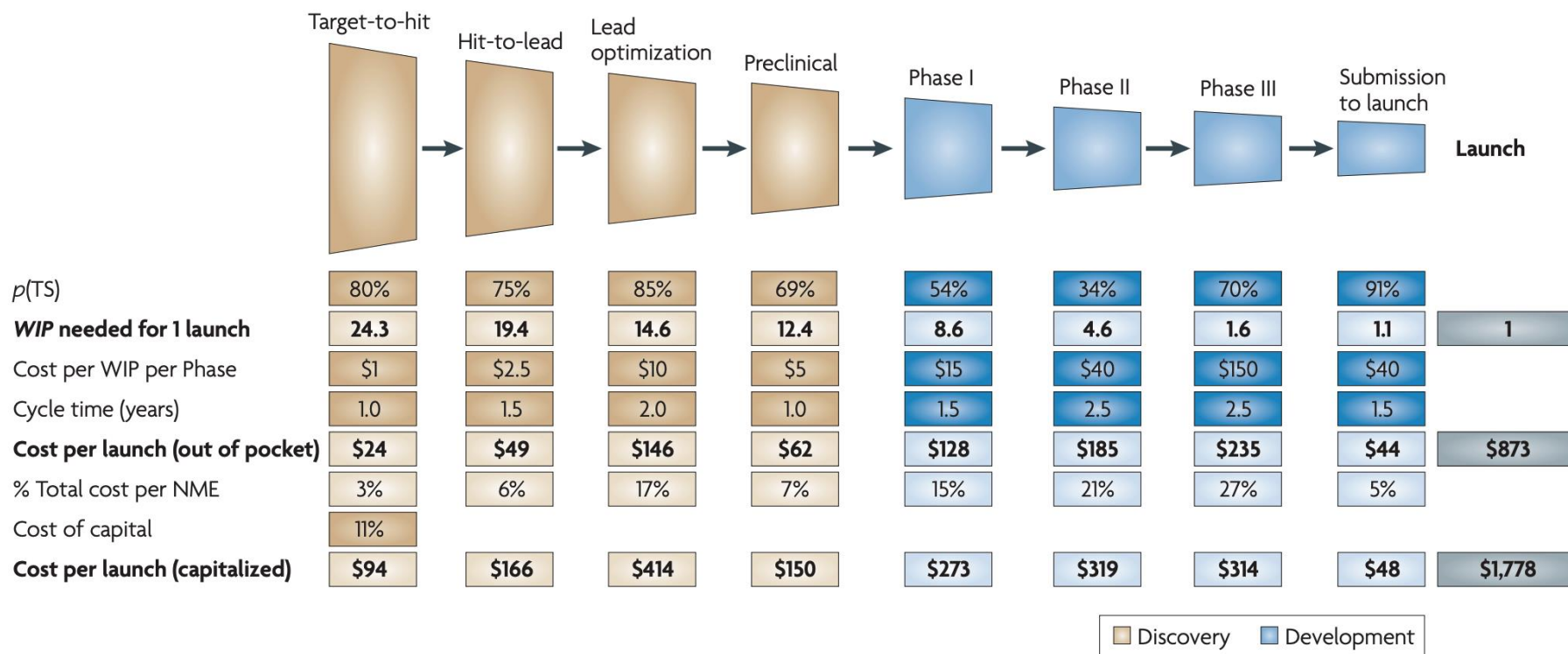


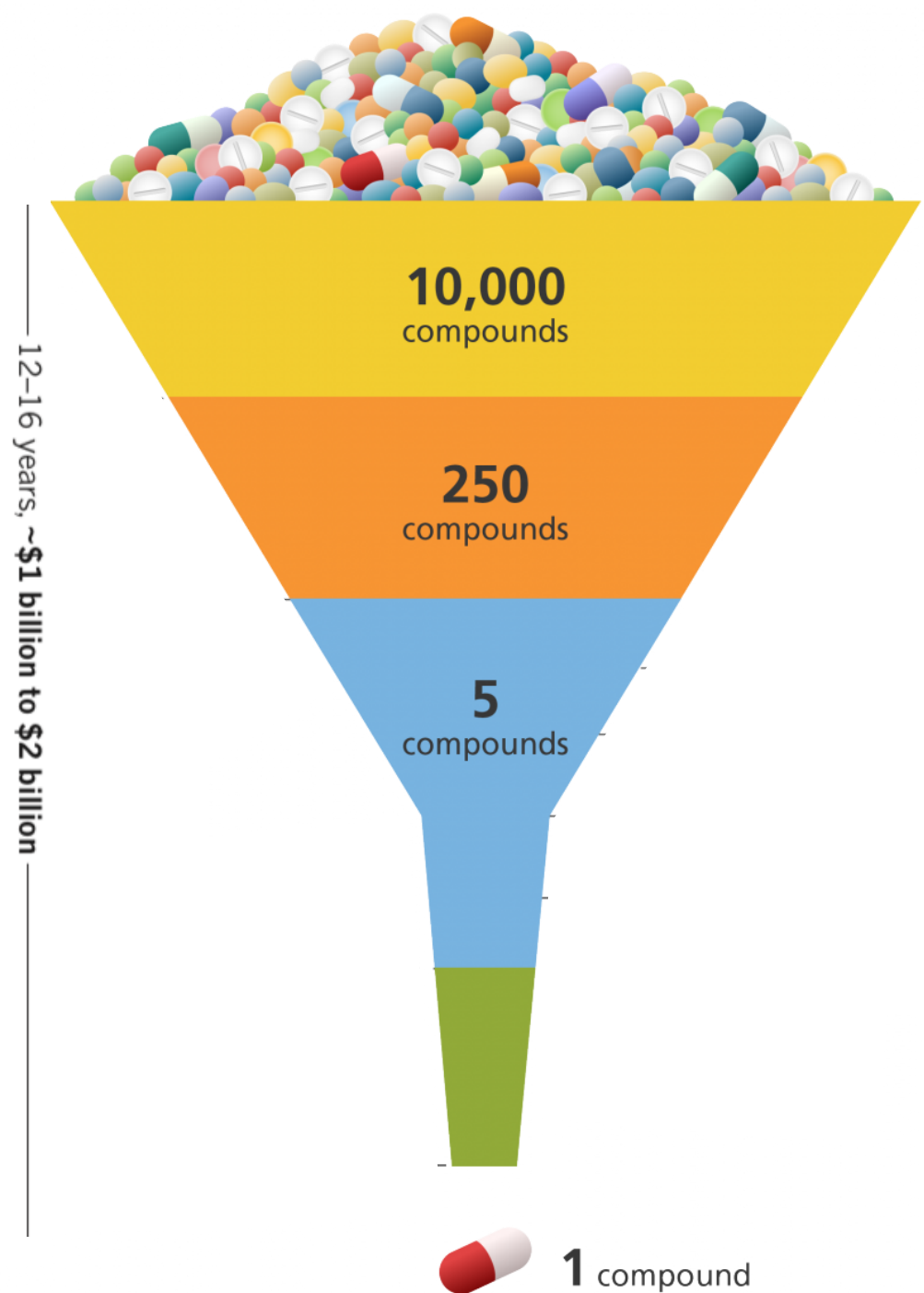
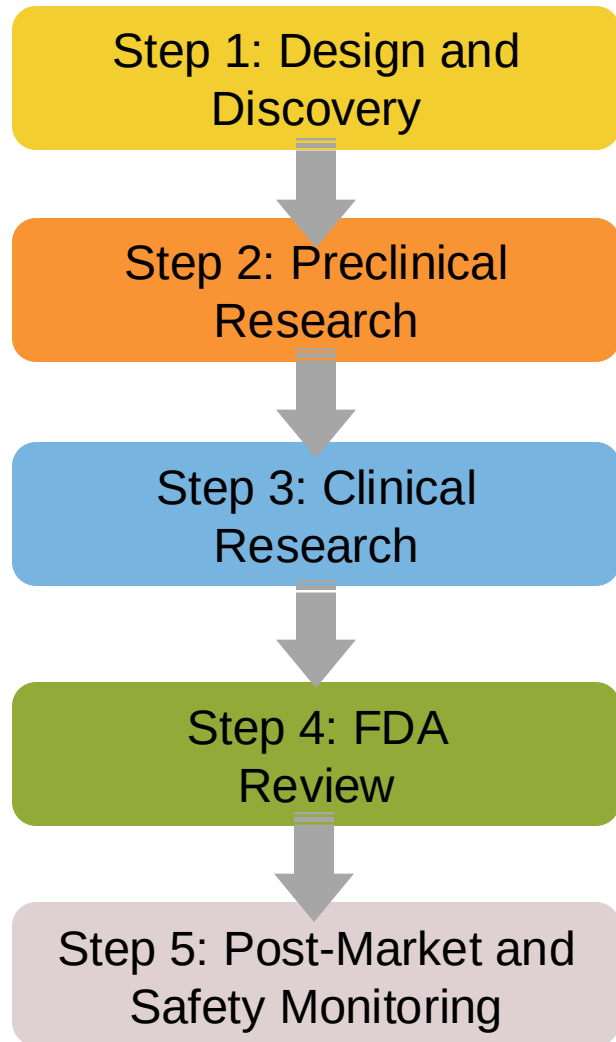
Phases of drug discovery from initial stage (target-to-hit) to final stage (launch)

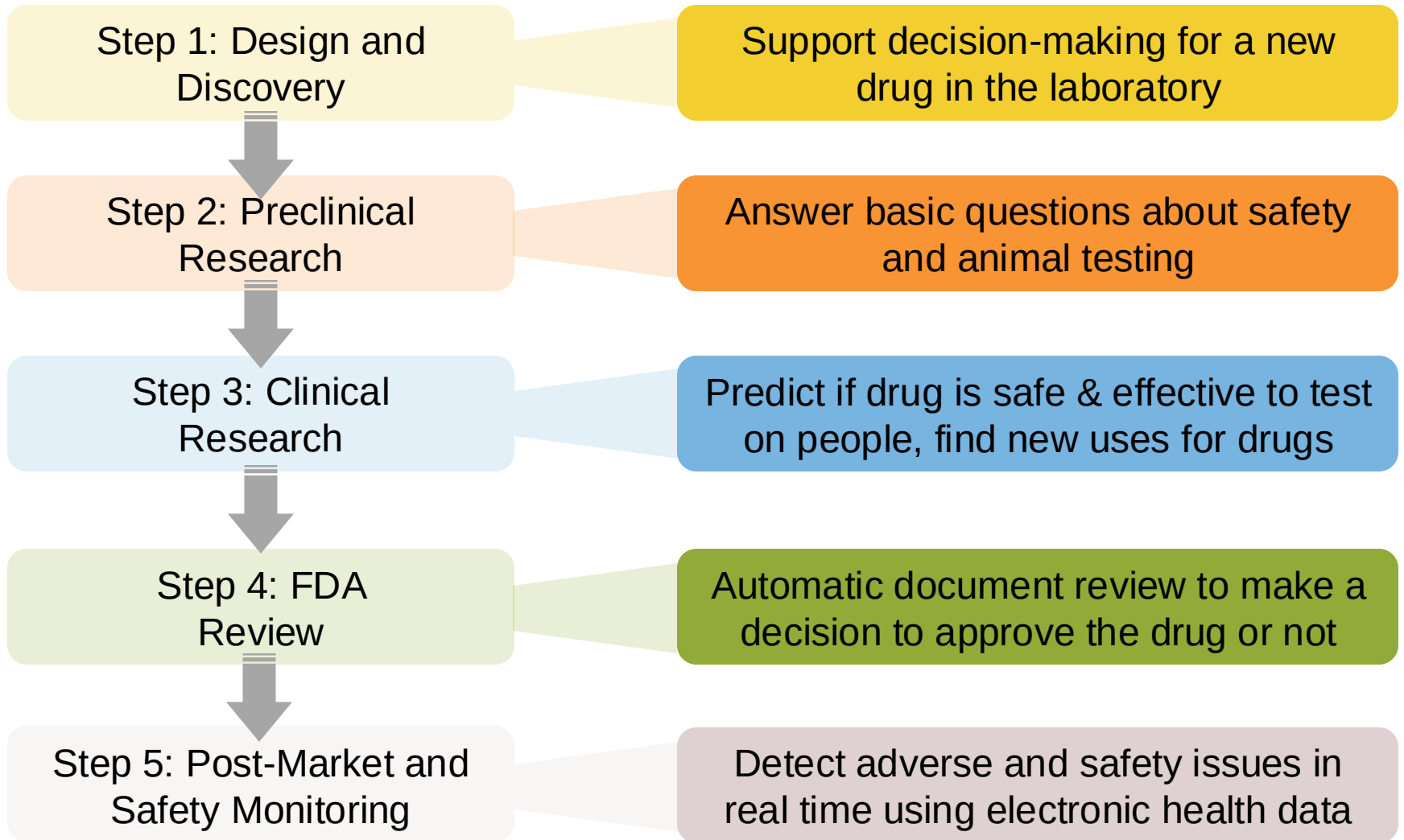


$p(TS)$ – probability of successful transition from one stage to the next; NME – new molecular entity; WIP – work in process

Drug-like chemical space
 10^{60} chemical compounds

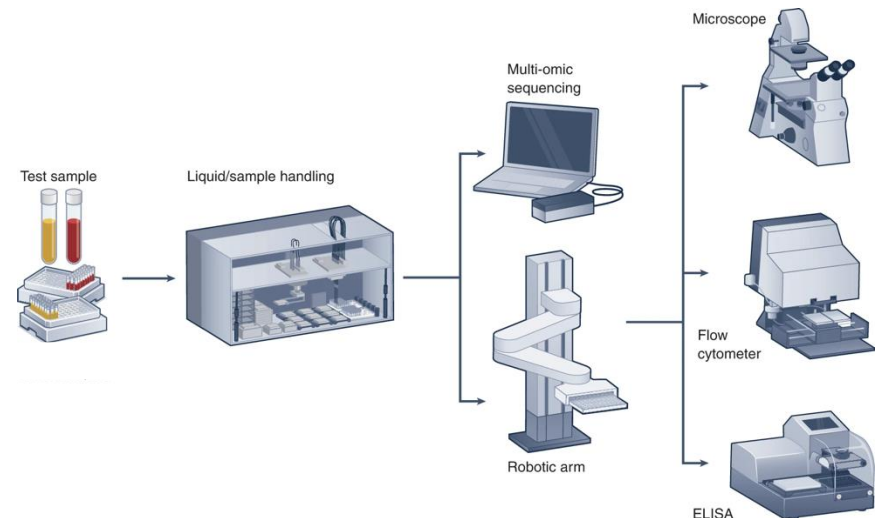
**Drugs available
to humans
 $\sim 10^4$ •**





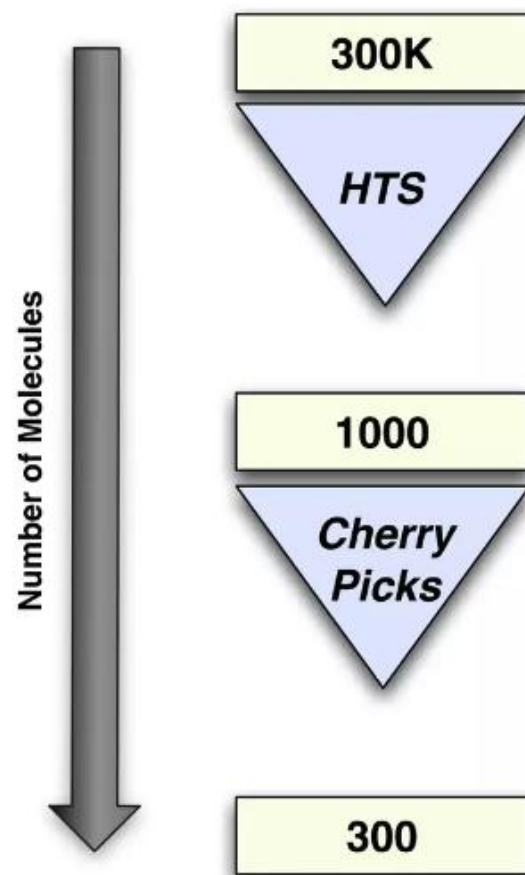
High throughput screening (HTS)

- Test thousands to hundreds of thousands of compounds in one or more assays
 - Biochemical, genetic, and pharmacological assays
- Integrate with robotics for self-driving lab
- **Goal:** Rapidly identify novel modulators of biological systems
 - Cellular basis of diseases
 - Therapeutic agents



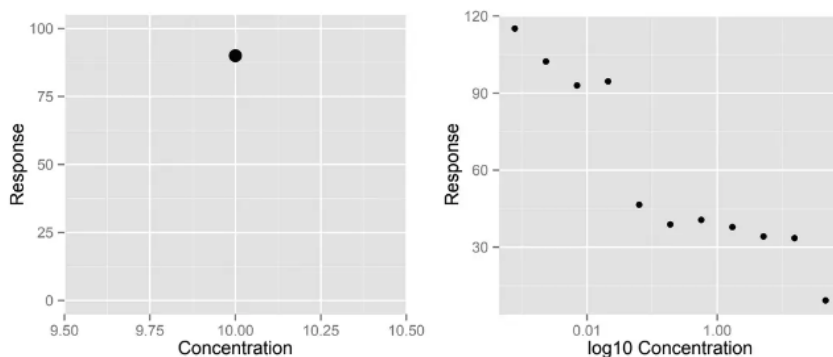
Goals of high throughput screening

- Rapidly screen large collections of compounds (chemical libraries)
- Efficiently identify active compounds
 - Test them in slower, accurate, expensive screens
- Use the data to learn what types of compounds tend to be active

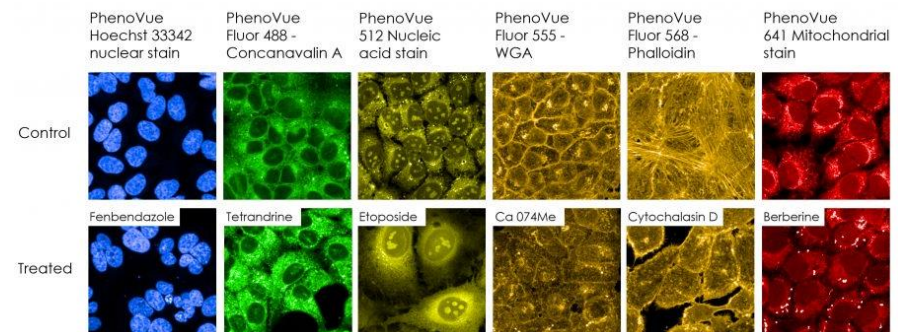


HTS data types

- Categorical: active/inactive or toxic/nontoxic
- Continuous: single-point or dose-response
- Multiple readouts:
 - Might read at different wavelengths or time points
 - More complex when dealing with images



Single-point vs. dose-response readouts



Cell painting for phenotypic drug discovery

HTS: Machine learning setup

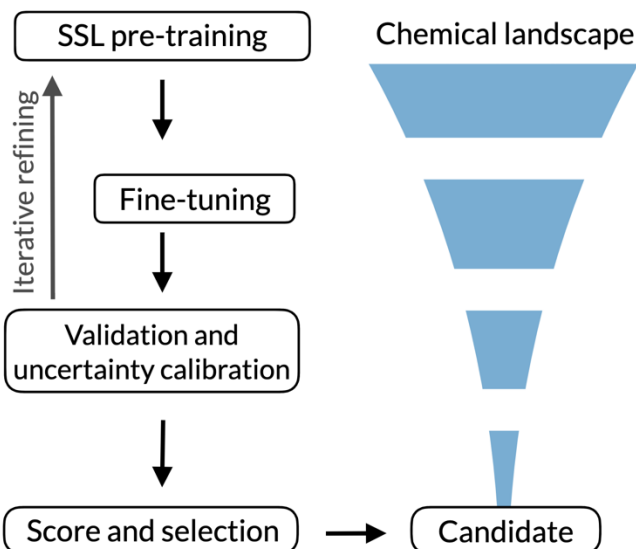
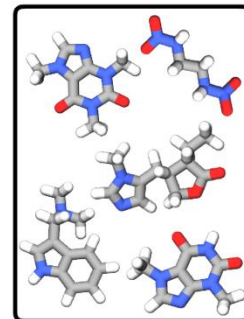
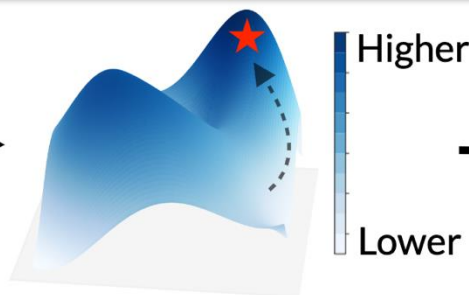
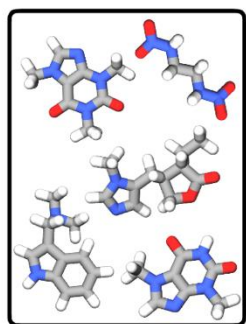
- HTS tests the activity of molecules:

$$\textit{Activity} = f(\textit{Structure})$$

- We need to describe the molecular structure
 - Various discrete or real-valued descriptors
 - Surfaces (3D)
 - Binary fingerprints
 - Learned molecular embeddings

In-silico screening and optimization of molecular structure

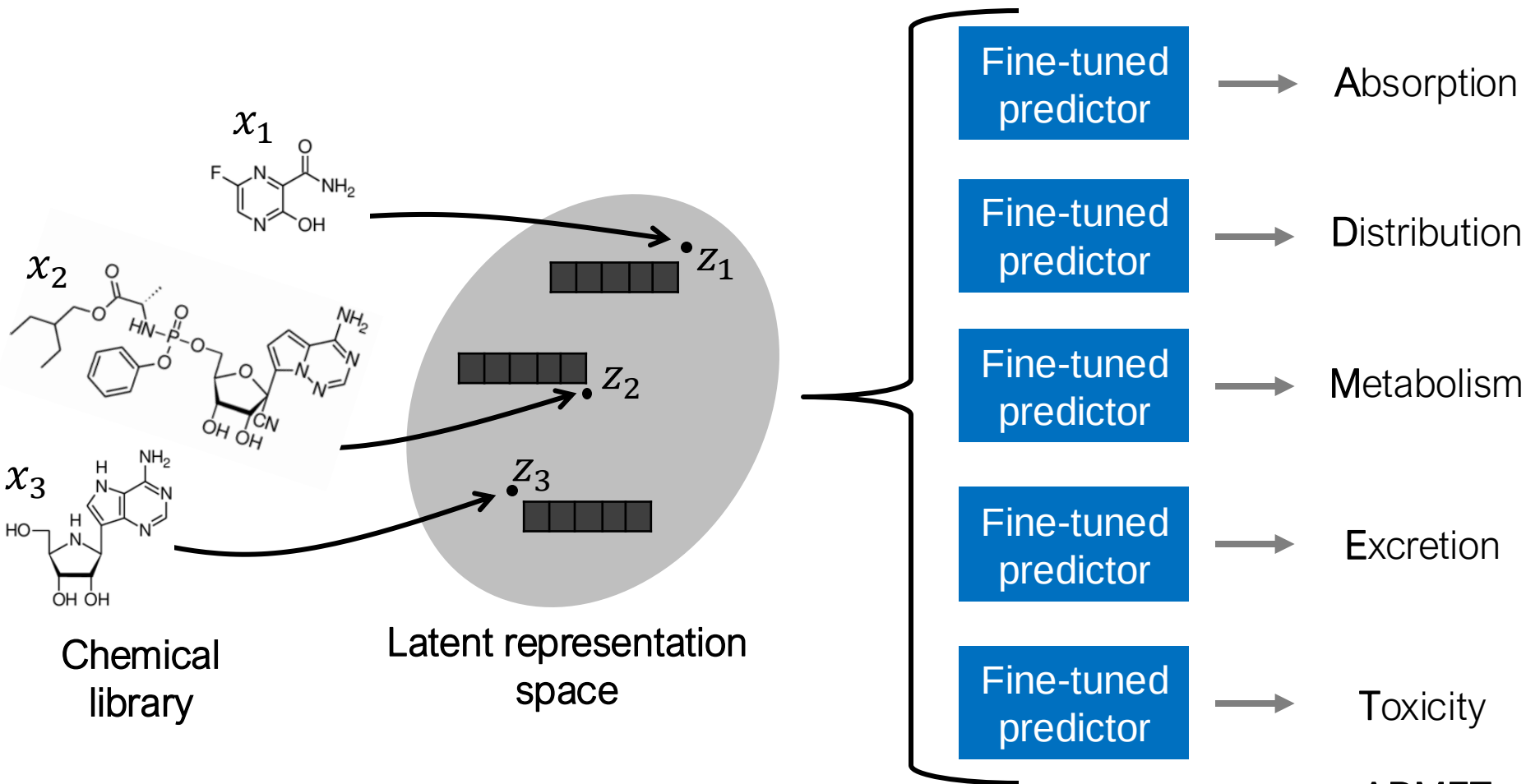
Use computational models to suggest what compounds to screen



----> Exploration route

★ Objective

Molecular property prediction



Self-supervised learning: Molecules with similar molecular structure get embedded close together. Various representations: Neural fingerprints, Attentive fingerprints, SMILES descriptors, Graphormer, Transformer-M, and others

What can we use molecular representations for?

- **Search**

- Given a potent active molecule, find similar ones (or dissimilar but also potent)

- **Prediction of various endpoints**

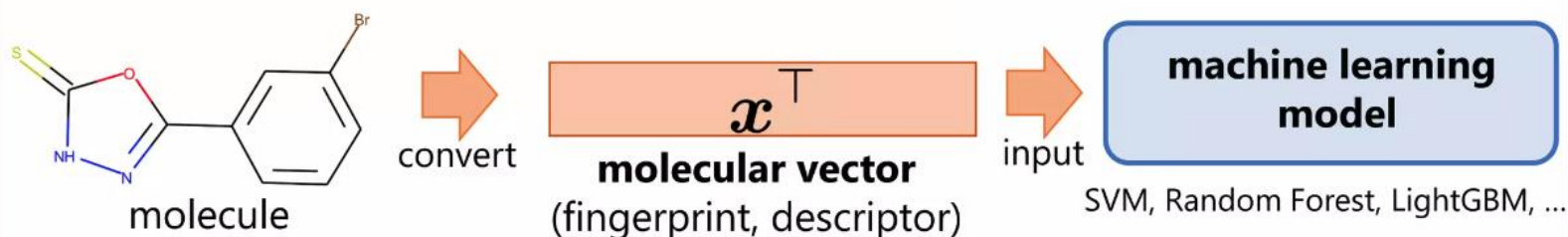
- Given a set of active and inactive molecules, build a model to predict which members from a chemical library will be active

- **Clustering**

- Given a set of molecules, do they cluster into structurally different groups?

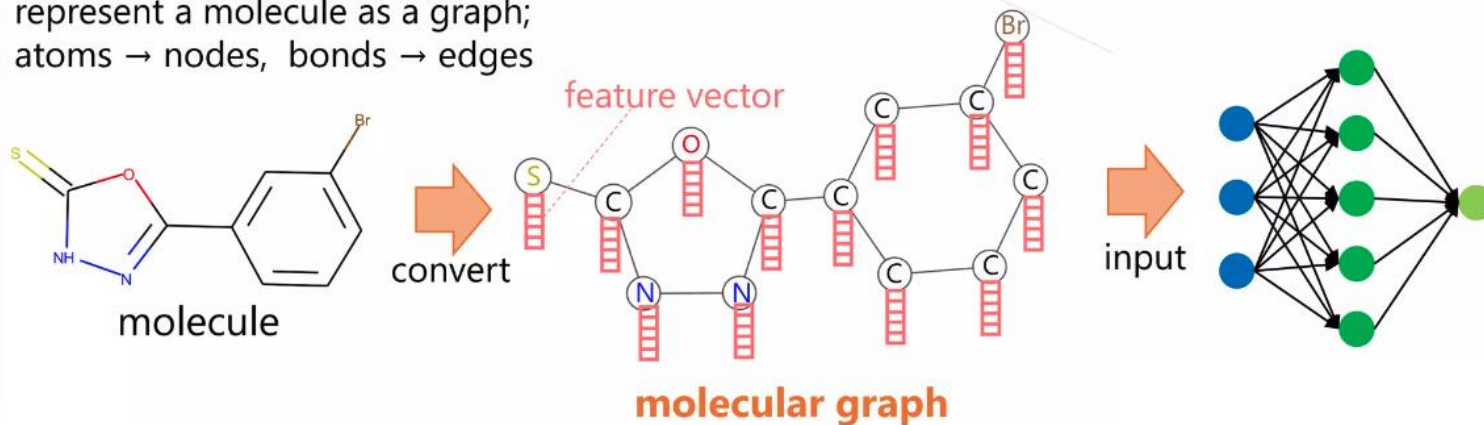
Two strategies for producing molecular representations

Traditional approach

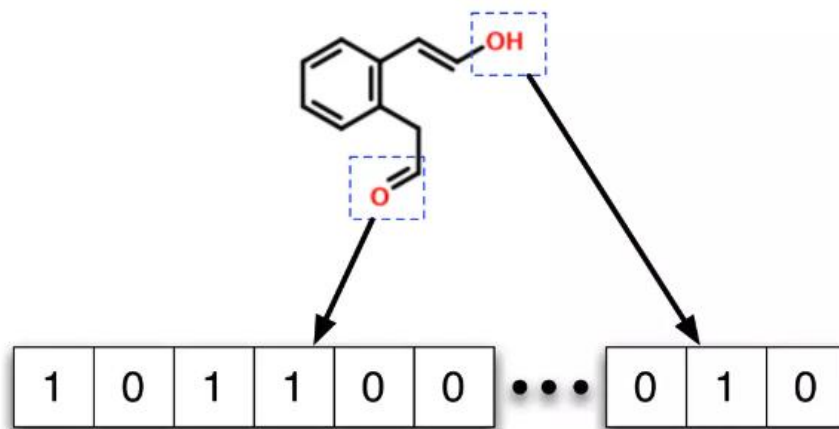


Graph convolutional network (GCN) approach

represent a molecule as a graph;
atoms $\rightarrow$ nodes, bonds $\rightarrow$ edges



Fingerprint representations



- Lots of types of fingerprints
- Keyed fingerprints indicate the presence or absence of a structural feature
- Length can vary from 166 to 4096 bits or more
- Fingerprints usually compared to each other using the Tanimoto metric

Towards neural fingerprints

Algorithm 1 Circular fingerprints

```
1: Input: molecule, radius  $R$ , fingerprint length  $S$ 
2: Initialize: fingerprint vector  $\mathbf{f} \leftarrow \mathbf{0}_S$ 
3: for each atom  $a$  in molecule
4:    $\mathbf{r}_a \leftarrow g(a)$   $\triangleright$  lookup atom features
5: for  $L = 1$  to  $R$   $\triangleright$  for each layer
6:   for each atom  $a$  in molecule
7:      $\mathbf{r}_1 \dots \mathbf{r}_N = \text{neighbors}(a)$ 
8:      $\mathbf{v} \leftarrow [\mathbf{r}_a, \mathbf{r}_1, \dots, \mathbf{r}_N]$   $\triangleright$  concatenate
9:      $\mathbf{r}_a \leftarrow \text{hash}(\mathbf{v})$   $\triangleright$  hash function
10:     $i \leftarrow \text{mod}(r_a, S)$   $\triangleright$  convert to index
11:     $\mathbf{f}_i \leftarrow 1$   $\triangleright$  Write 1 at index
12: Return: binary vector  $\mathbf{f}$ 
```

Algorithm 2 Neural graph fingerprints

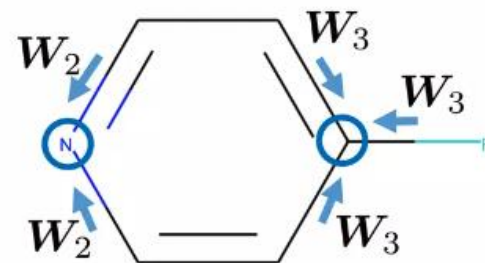
```
1: Input: molecule, radius  $R$ , hidden weights  $H_1^1 \dots H_R^5$ , output weights  $W_1 \dots W_R$ 
2: Initialize: fingerprint vector  $\mathbf{f} \leftarrow \mathbf{0}_S$ 
3: for each atom  $a$  in molecule
4:    $\mathbf{r}_a \leftarrow g(a)$   $\triangleright$  lookup atom features
5: for  $L = 1$  to  $R$   $\triangleright$  for each layer
6:   for each atom  $a$  in molecule
7:      $\mathbf{r}_1 \dots \mathbf{r}_N = \text{neighbors}(a)$ 
8:      $\mathbf{v} \leftarrow \mathbf{r}_a + \sum_{i=1}^N \mathbf{r}_i$   $\triangleright$  sum
9:      $\mathbf{r}_a \leftarrow \sigma(\mathbf{v} H_L^N)$   $\triangleright$  smooth function
10:     $\mathbf{i} \leftarrow \text{softmax}(\mathbf{r}_a W_L)$   $\triangleright$  sparsify
11:     $\mathbf{f} \leftarrow \mathbf{f} + \mathbf{i}$   $\triangleright$  add to fingerprint
12: Return: real-valued vector  $\mathbf{f}$ 
```

Figure 2: Pseudocode of circular fingerprints (*left*) and neural graph fingerprints (*right*). Differences are highlighted in blue. Every non-differentiable operation is replaced with a differentiable analog.

Neural fingerprint representations

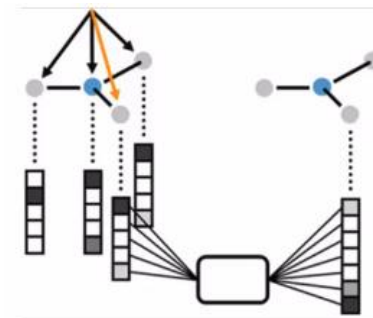
1) Neural graph fingerprints

- Generate molecular fingerprints with a neural network
- Update atom features using only adjacent atoms
- Use different weights for node degrees



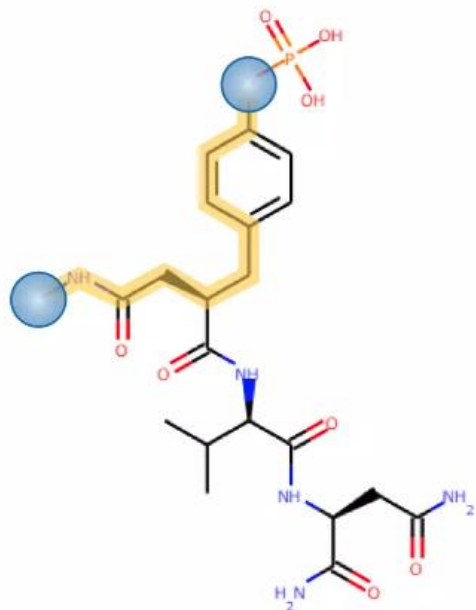
2) Molecular graphs

- Update atom features by convolutional and pooling layers using adjacent atoms

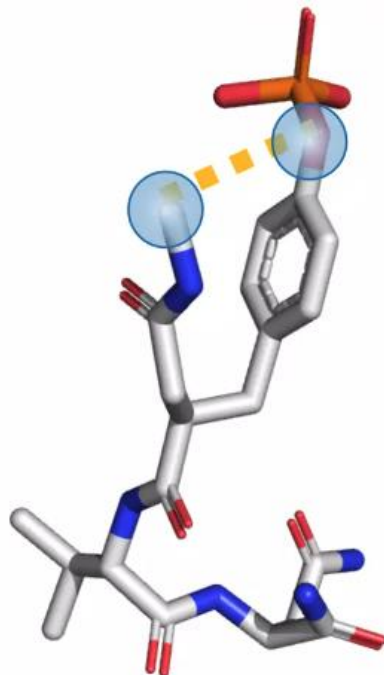


- They did not consider property of edges (bonds)
- They did not consider atoms other than 1-neighbor

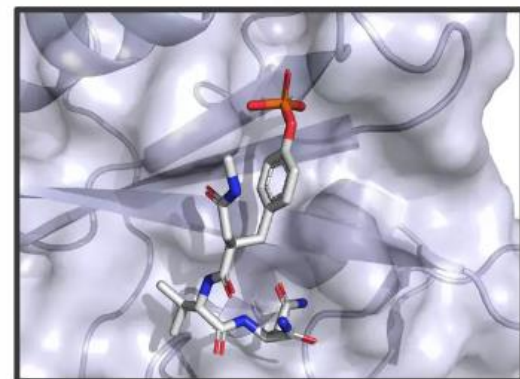
Graphs vs. 3D structures



Molecular Graph



3D Structure



The distance on the graph does not necessarily correlate with the Euclidean distance between atoms in the 3D structure

Need to consider modifying the definition of graph distance

Datasets

22 datasets with ADMET endpoints

A: Absorption

Caco2 (Cell Permeability)
HIA (Intestinal Absorption)
Pgp (P-glycoprotein)
Bioavailability
Lipophilicity
Solubility

E: Excretion

Half Life
Clearance (Hepatocyte)
Clearance (Microsome)

D: Distribution

BBB (Blood-Brain Barrier)
PPBR (Plasma Protein Binding)
VDss (Volume of Distribution)

T: Toxicity

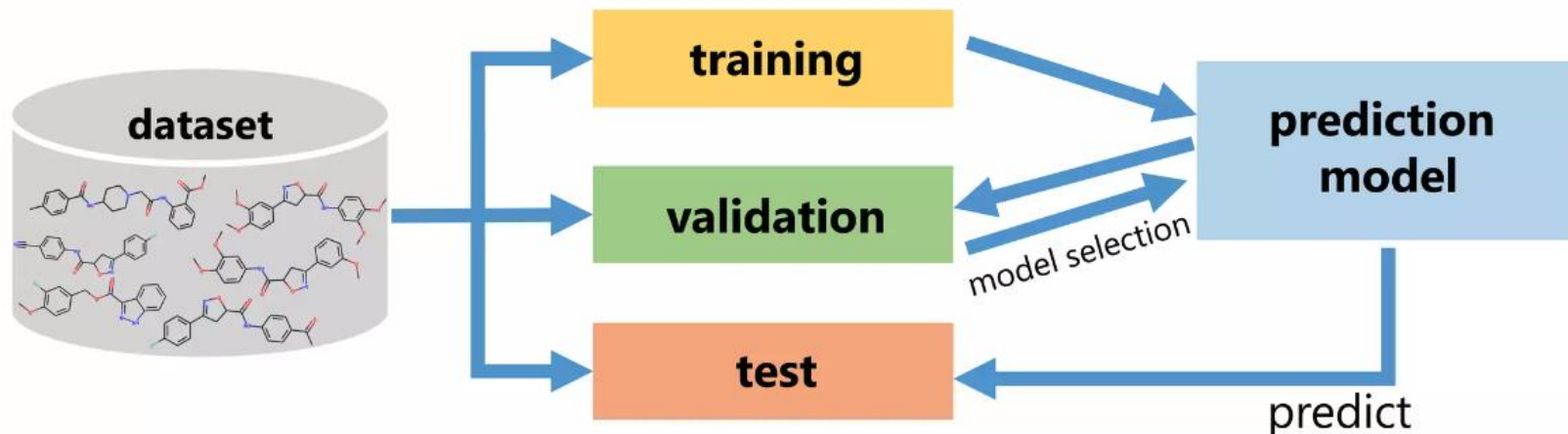
LD50 (Acute Toxicity)
hERG blocker
Ames Mutagenicity
Drug Induced Liver Injury

M: Metabolism

CYP2C9/2D6/3A4 Inhibition
CYP2C9/2D6/3A4 Substrate



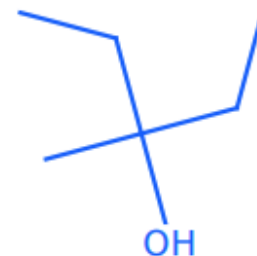
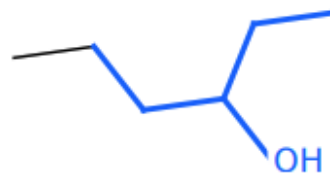
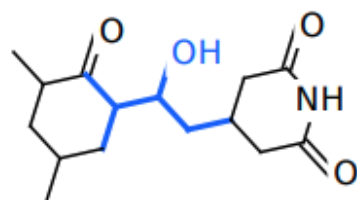
Experimental setup



- Demonstrate that fingerprints are interpretable
 - Show substructures which most activate individual features in a fingerprint vector
 - **Fingerprint features** can each only be activated by a single fragment of a single radius, except for accidental collisions
 - In contrast, **neural fingerprint features** can be activated by variations of the same structure, making them more interpretable, and allowing shorter feature vectors.

Results: Examining neural fingerprints

Fragments most
activated by
pro-solubility
feature



Fragments most
activated by
anti-solubility
feature

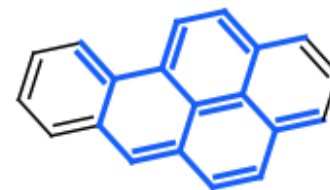
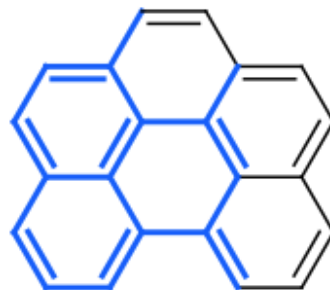
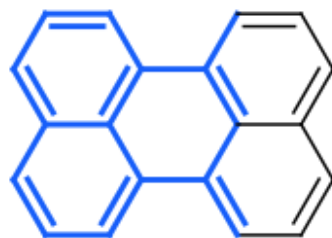
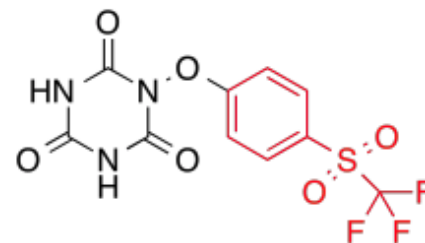
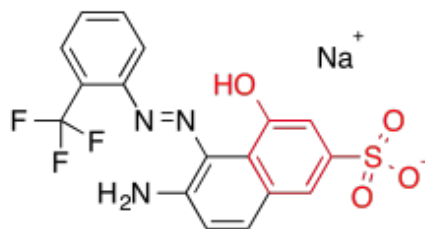
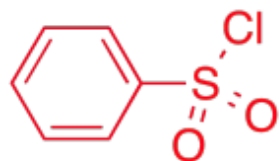


Figure 4: Examining fingerprints optimized for predicting solubility. Shown here are representative examples of molecular fragments (highlighted in blue) which most activate different features of the fingerprint. *Top row:* The feature most predictive of solubility. *Bottom row:* The feature most predictive of insolubility.

Results: Examining neural fingerprints

Fragments most
activated by
toxicity feature
on SR-MMP
dataset



Fragments most
activated by
toxicity feature
on NR-AHR
dataset

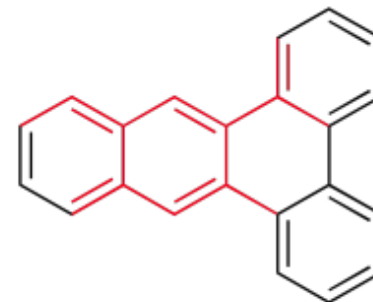
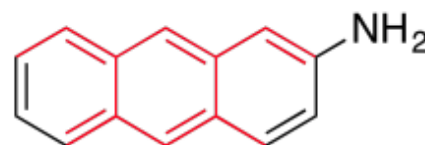
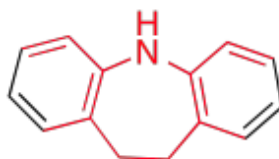


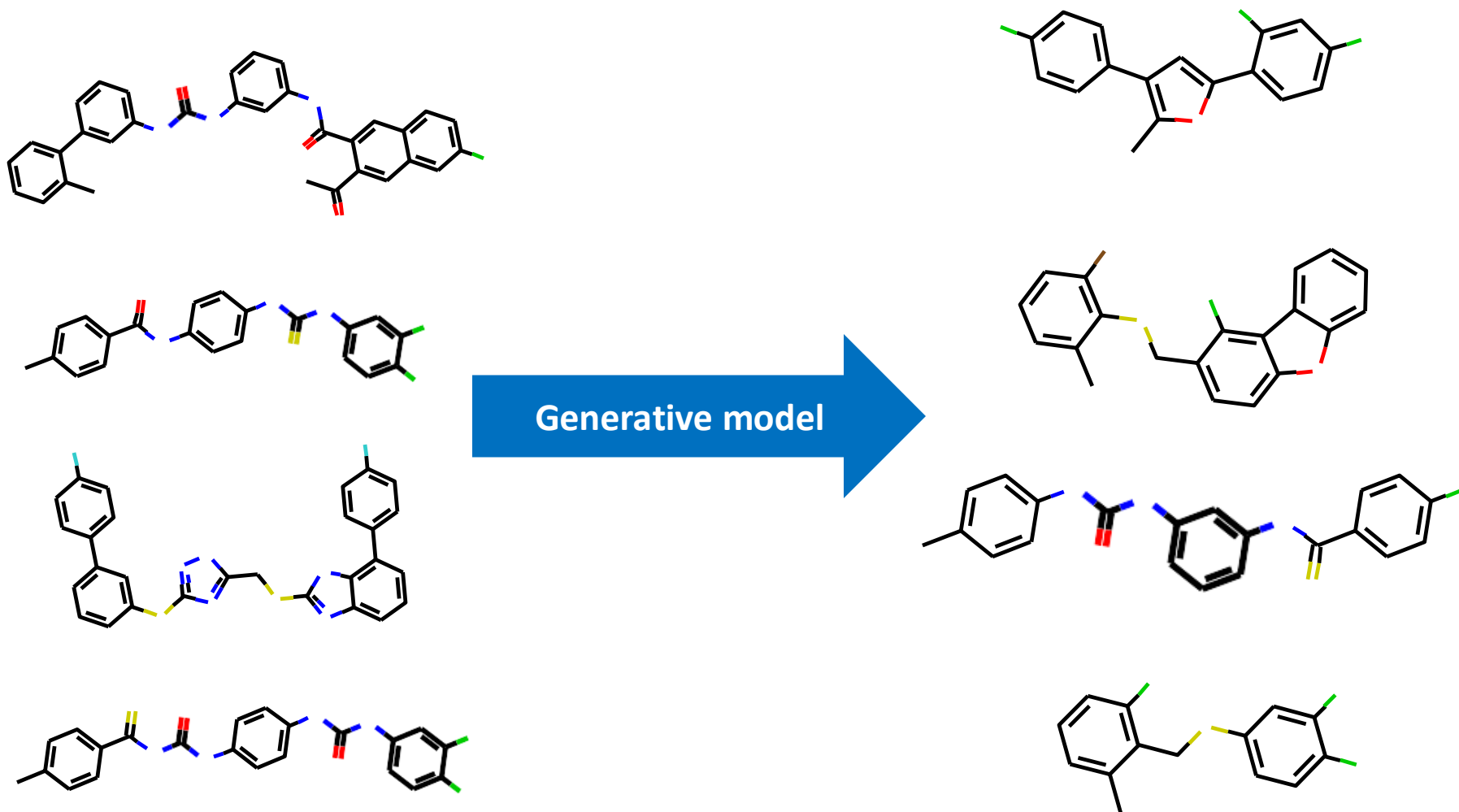
Figure 5: Visualizing fingerprints optimized for predicting toxicity. Shown here are representative samples of molecular fragments (highlighted in red) which most activate the feature most predictive of toxicity. *Top row:* the most predictive feature identifies groups containing a sulphur atom attached to an aromatic ring. *Bottom row:* the most predictive feature identifies fused aromatic rings, also known as polycyclic aromatic hydrocarbons, a well-known carcinogen.

Results: Molecular property prediction

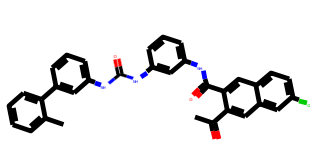
Raw Feature Type		Expert-Curated Methods		SMILES	Molecular Graph-Based Methods (state-of-the-Art in ML)				
Dataset	Metric	Morgan	RDKit2D	CNN	NeuralFP	GCN	AttentiveFP	AttrMasking	ContextPred
	# Params.	1477K	633K	227K	480K	192K	301K	2067K	2067K
TDC.Caco2 (↓)	MAE	0.908±0.060	0.393±0.024	0.446±0.036	0.530±0.102	0.599±0.104	<u>0.401±0.032</u>	0.546±0.052	0.502±0.036
TDC.HIA (↑)	AUROC	0.807±0.072	0.972±0.008	0.869±0.026	0.943±0.014	0.936±0.024	0.974±0.007	0.978±0.006	<u>0.975±0.004</u>
TDC.Pgp (↑)	AUROC	0.880±0.006	0.918±0.007	0.908±0.012	0.902±0.020	0.895±0.021	0.892±0.012	0.929±0.006	0.923±0.005
TDC.Bioav (↑)	AUROC	0.581±0.086	0.672±0.021	0.613±0.013	0.632±0.036	0.566±0.115	0.632±0.039	0.577±0.087	0.671±0.026
TDC.Lipo (↓)	MAE	0.701±0.009	0.574±0.017	0.743±0.020	0.563±0.023	<u>0.541±0.011</u>	0.572±0.007	0.547±0.024	0.535±0.012
TDC.AqSol (↓)	MAE	1.203±0.019	<u>0.827±0.047</u>	1.023±0.023	0.947±0.016	0.907±0.020	0.776±0.008	1.026±0.020	1.040±0.045
TDC.BBB (↑)	AUROC	0.823±0.015	0.889±0.016	0.781±0.030	0.836±0.009	0.842±0.016	0.855±0.011	<u>0.892±0.012</u>	0.897±0.004
TDC.PPBR (↓)	MAE	12.848±0.362	9.994±0.319	11.106±0.358	9.292±0.384	10.194±0.373	<u>9.373±0.335</u>	10.075±0.202	9.445±0.224
TDC.VD (↑)	Spearman	0.493±0.011	0.561±0.025	0.226±0.114	0.258±0.162	0.457±0.050	0.241±0.145	<u>0.559±0.019</u>	0.485±0.092
TDC.CYP2D6-I (↑)	AUPRC	0.587±0.011	0.616±0.007	0.544±0.053	0.627±0.009	0.616±0.020	0.646±0.014	<u>0.721±0.009</u>	0.739±0.005
TDC.CYP3A4-I (↑)	AUPRC	0.827±0.009	0.829±0.007	0.821±0.003	0.849±0.004	0.840±0.010	0.851±0.006	<u>0.902±0.002</u>	0.904±0.002
TDC.CYP2C9-I (↑)	AUPRC	0.715±0.004	0.742±0.006	0.713±0.006	0.739±0.010	0.735±0.004	0.749±0.004	<u>0.829±0.003</u>	0.839±0.003
TDC.CYP2D6-S (↑)	AUPRC	0.671±0.066	0.677±0.047	0.485±0.037	0.572±0.062	0.617±0.039	0.574±0.030	<u>0.704±0.028</u>	0.736±0.024
TDC.CYP3A4-S (↑)	AUROC	0.633±0.013	<u>0.639±0.012</u>	0.662±0.031	0.578±0.020	0.590±0.023	0.576±0.025	0.582±0.021	0.609±0.025
TDC.CYP2C9-S (↑)	AUPRC	0.380±0.015	0.360±0.040	0.367±0.059	0.359±0.059	0.344±0.051	0.375±0.032	<u>0.381±0.045</u>	0.392±0.026
TDC.Half-Life (↑)	Spearman	0.329±0.083	0.184±0.111	0.038±0.138	0.177±0.165	<u>0.239±0.100</u>	0.085±0.068	0.151±0.068	0.129±0.114
TDC.CL-Micro (↑)	Spearman	0.492±0.020	0.586±0.014	0.252±0.116	0.529±0.015	0.532±0.033	0.365±0.055	<u>0.585±0.034</u>	0.578±0.007
TDC.CL-Hepa (↑)	Spearman	0.272±0.068	0.382±0.007	0.235±0.021	0.401±0.037	0.366±0.063	0.289±0.022	<u>0.413±0.028</u>	0.439±0.026
TDC.hERG (↑)	AUROC	0.736±0.023	0.841±0.020	0.754±0.037	0.722±0.034	0.738±0.038	<u>0.825±0.007</u>	0.778±0.046	0.756±0.023
TDC.AMES (↑)	AUROC	0.794±0.008	0.823±0.011	0.776±0.015	0.823±0.006	0.818±0.010	0.814±0.008	0.842±0.008	<u>0.837±0.009</u>
TDC.DILI (↑)	AUROC	0.832±0.021	0.875±0.019	0.792±0.016	0.851±0.026	0.859±0.033	<u>0.886±0.015</u>	0.919±0.008	0.861±0.018
TDC.LD50 (↓)	MAE	0.649±0.019	<u>0.678±0.003</u>	0.675±0.011	0.667±0.020	0.649±0.026	0.678±0.012	0.685±0.025	0.669±0.030

- No single method performs the best across all scenarios
- Pre-training boost performance
- Pre-trained graph models yield strongest predictors overall

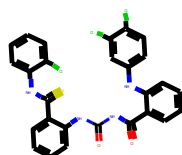
Molecular graph generation



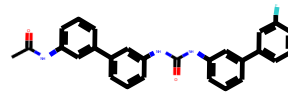
Molecular graph generation



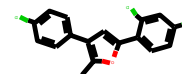
5.30



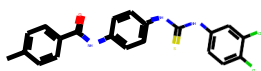
4.93



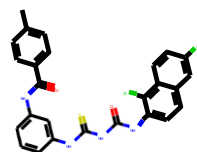
4.49



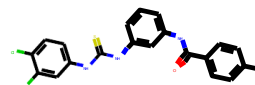
4.45



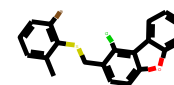
4.42



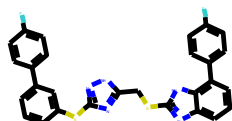
4.40



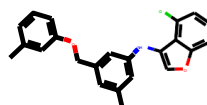
4.37



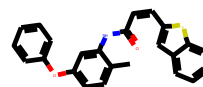
4.30



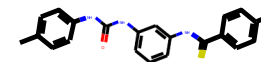
4.23



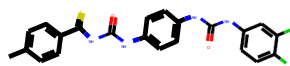
4.18



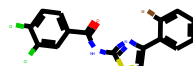
4.17



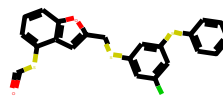
4.08



4.07



4.04



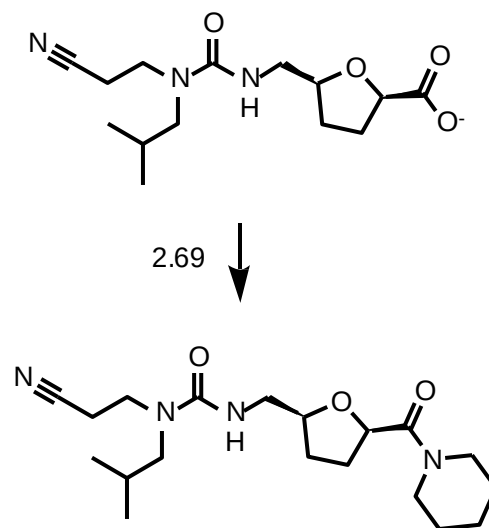
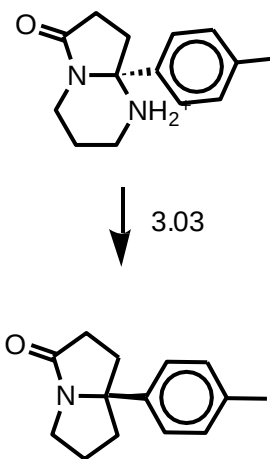
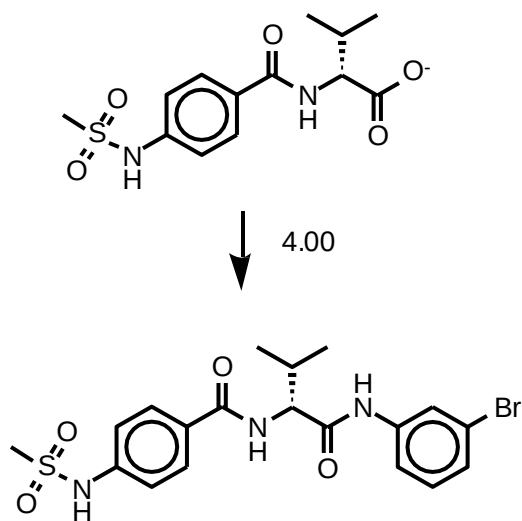
4.04



4.03

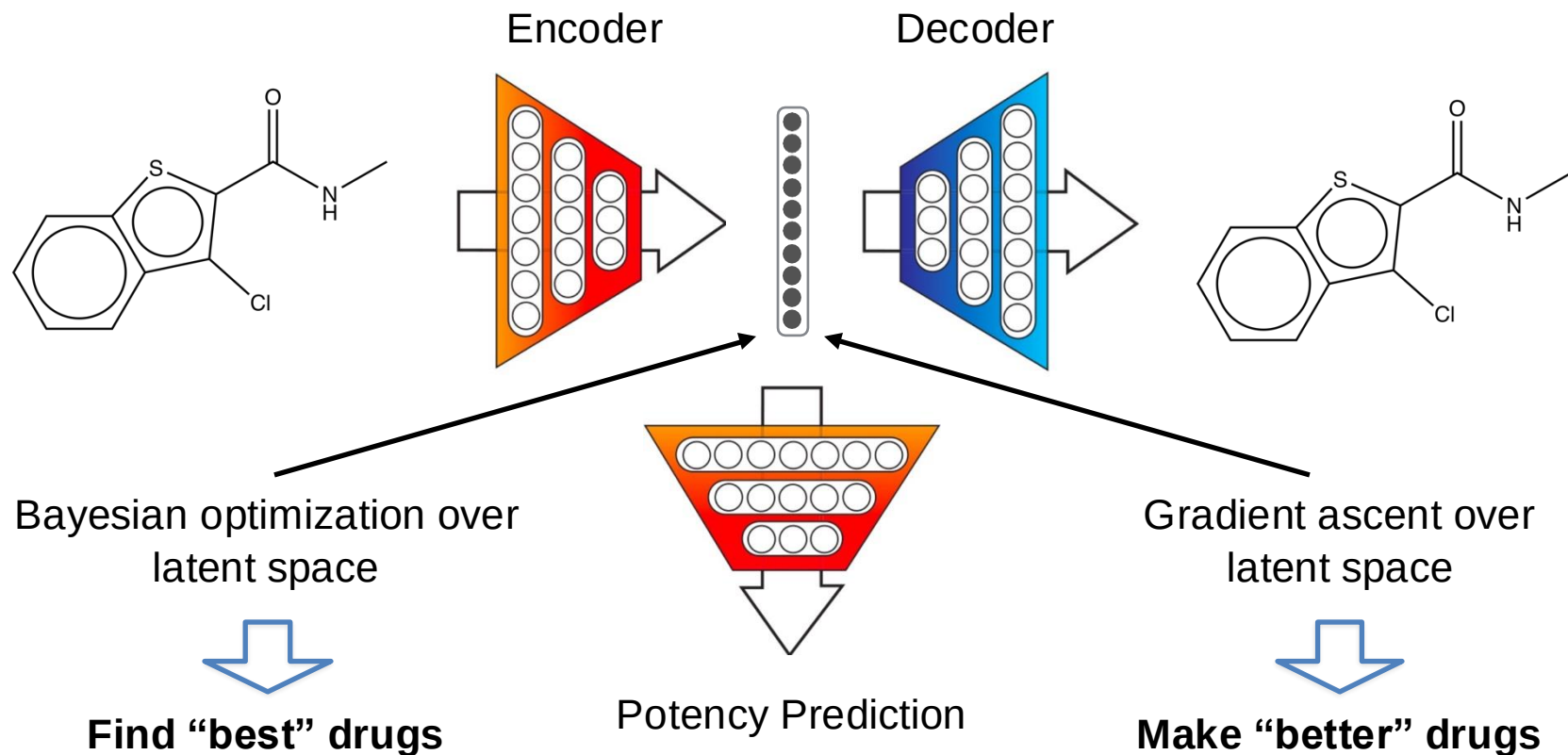
Generate molecules with high potency

Molecular graph generation

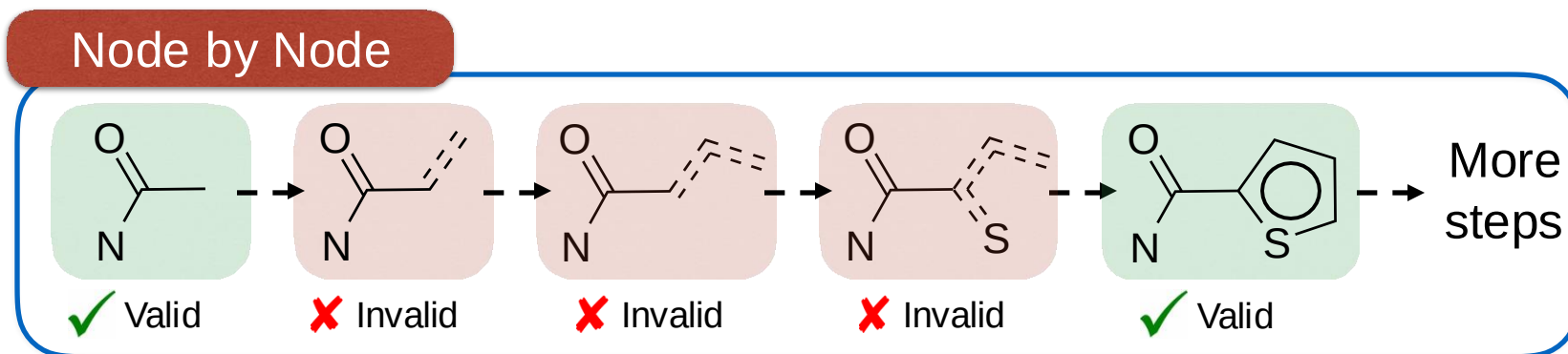


Modify molecules to increase potency

Molecular variational autoencoder

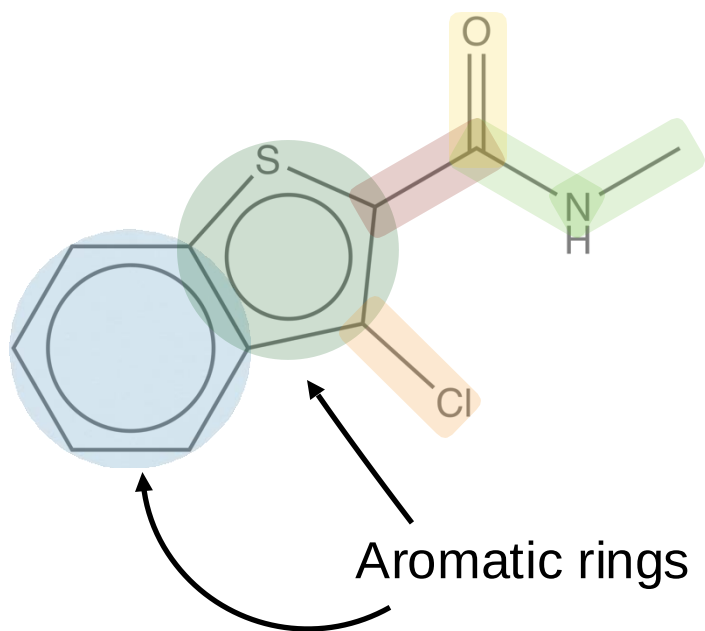


How to generate graphs?

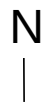
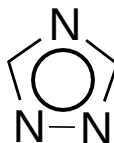
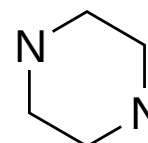
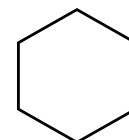
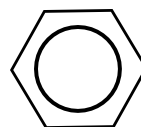
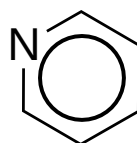


- Not every graphs is chemically valid
- Invalid intermediate states → hard to validate
- Very long intermediate steps → difficult to train (Li et al., 2018)

Functional groups

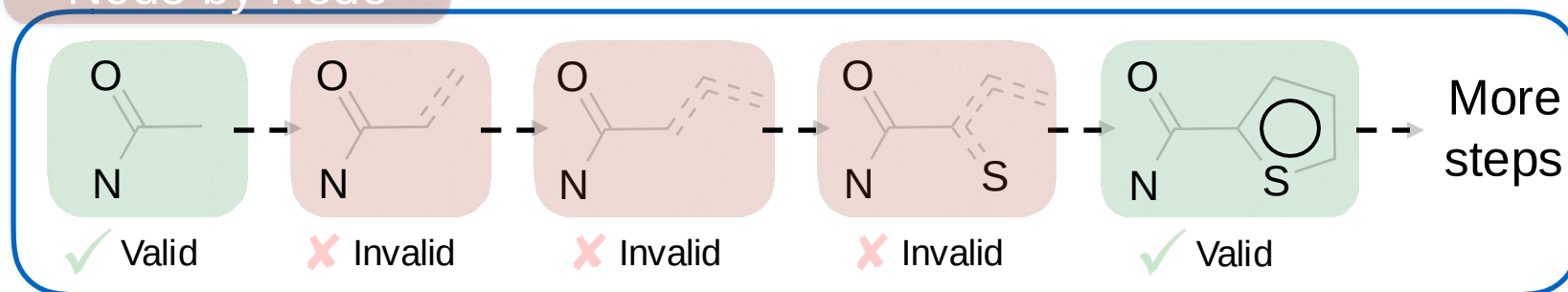


Functional Groups

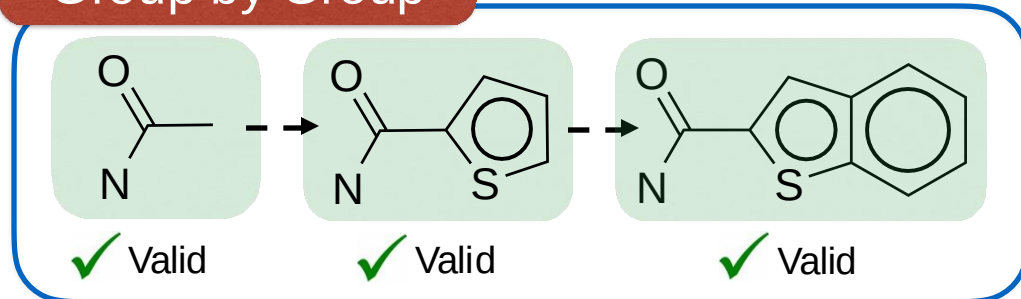


How to generate graphs?

Node by Node

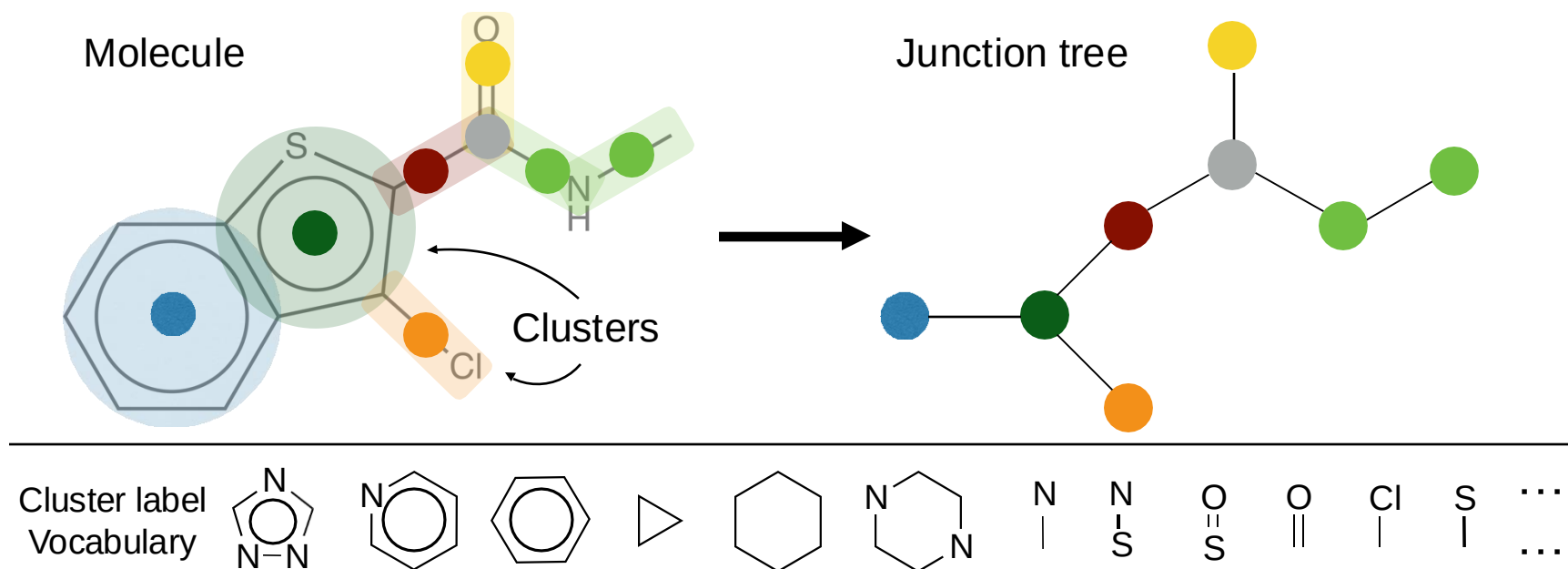


Group by Group



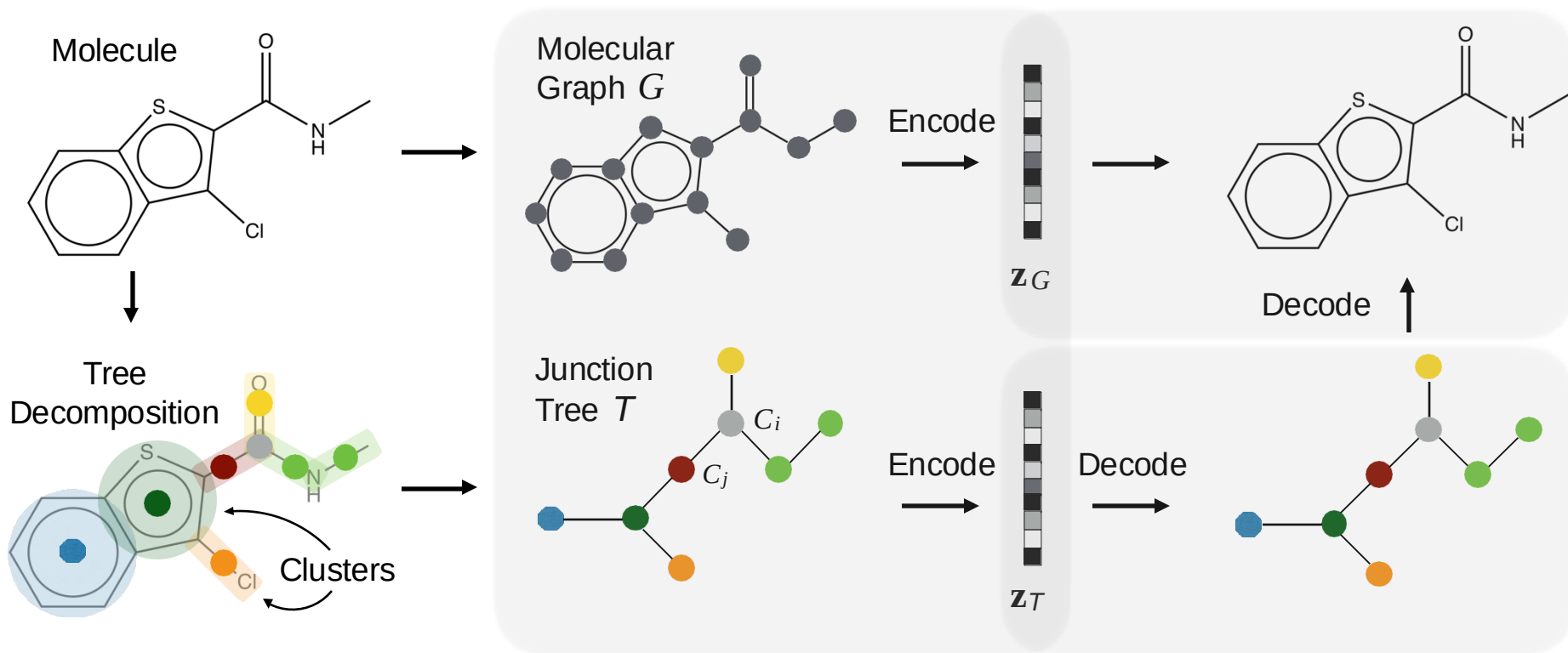
- Shorter action sequence
- Easy to check validity

Tree decomposition

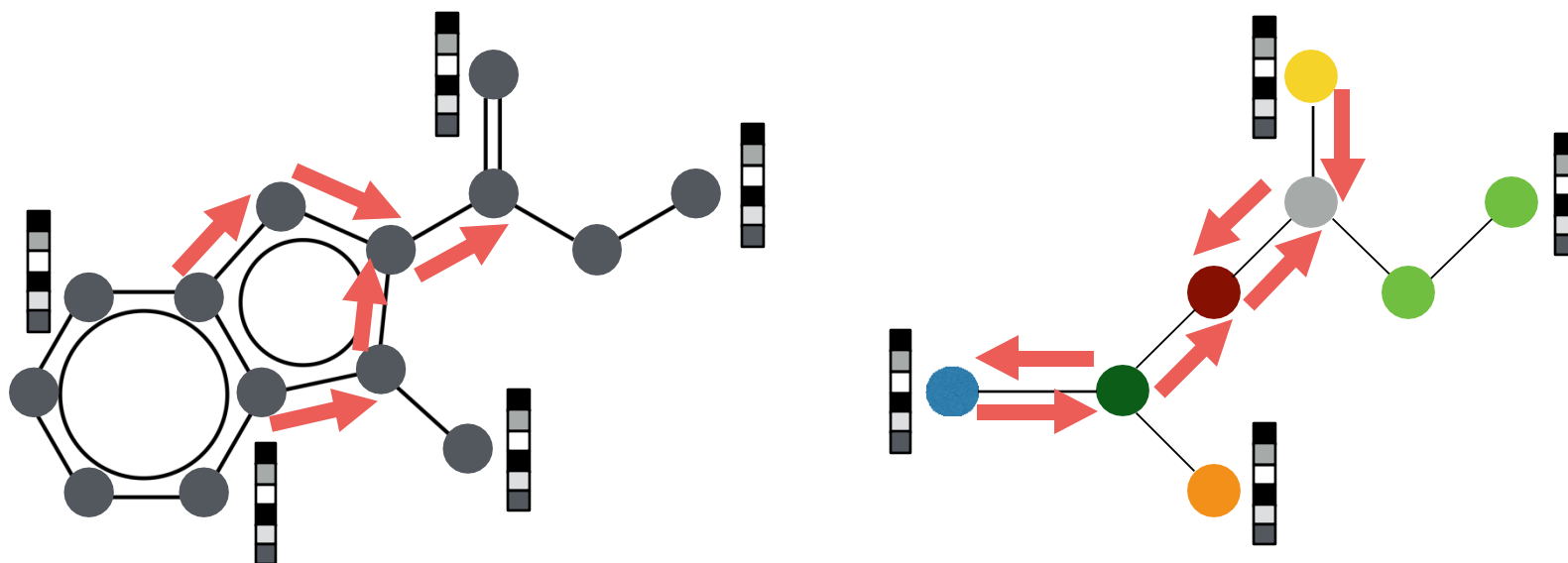


- Generate junction tree $\rightarrow$ Generate graph group by group
- Vocabulary size: less than 800 given 250K molecules

Approach: Junction-tree variational autoencoder

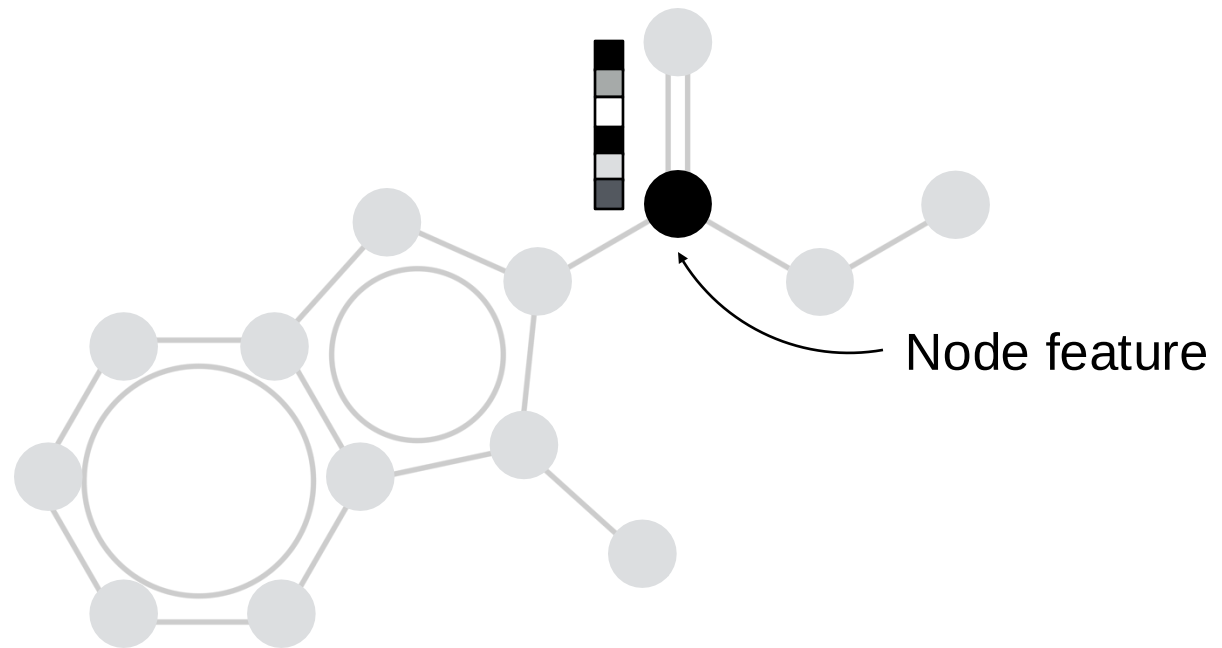


Graph and tree encoders

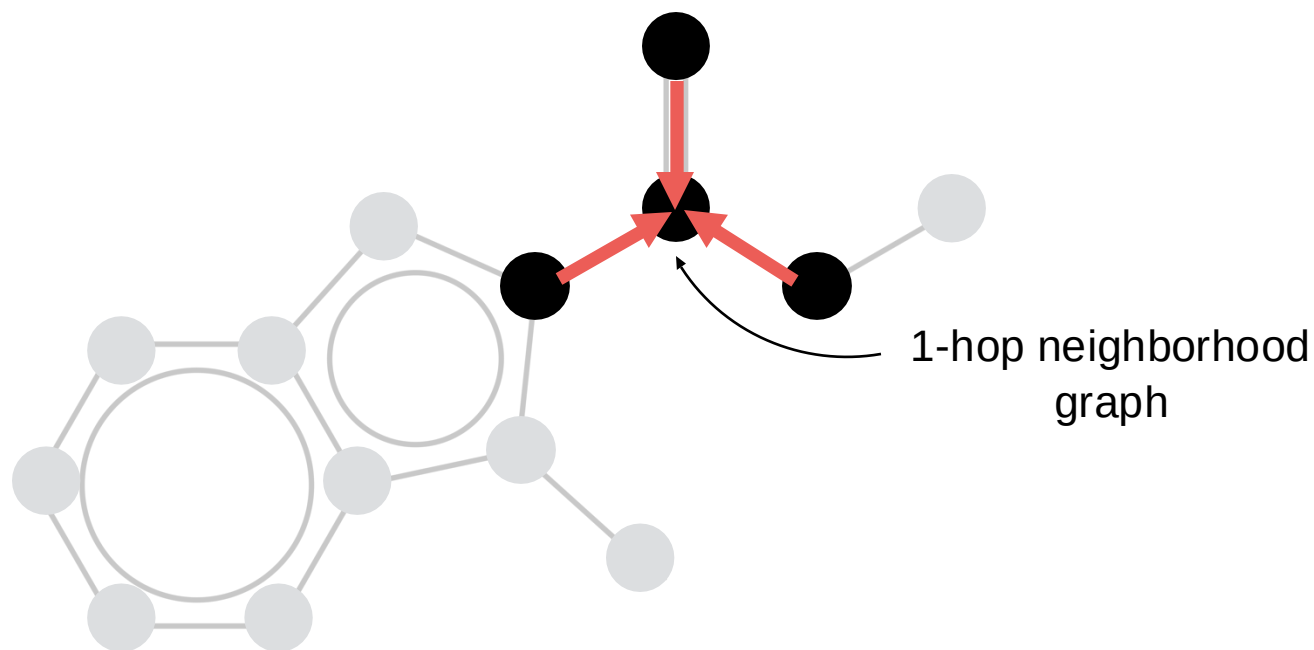


Neural Message Passing Network (MPN)

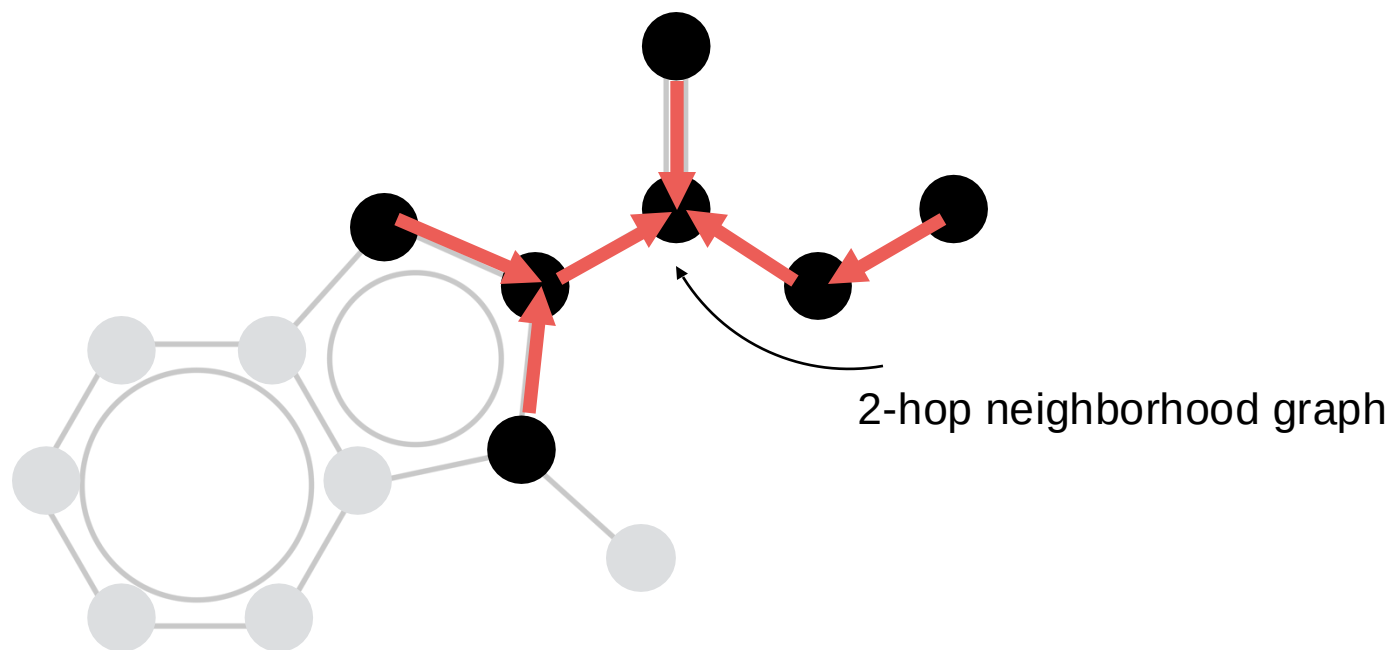
Graph encoding



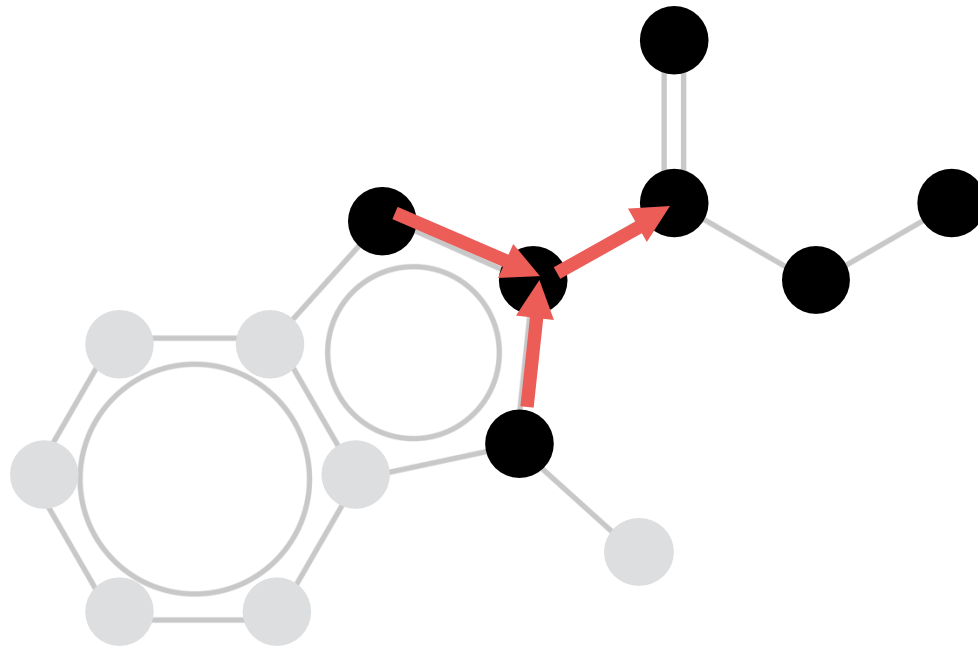
Graph encoding



Graph encoding

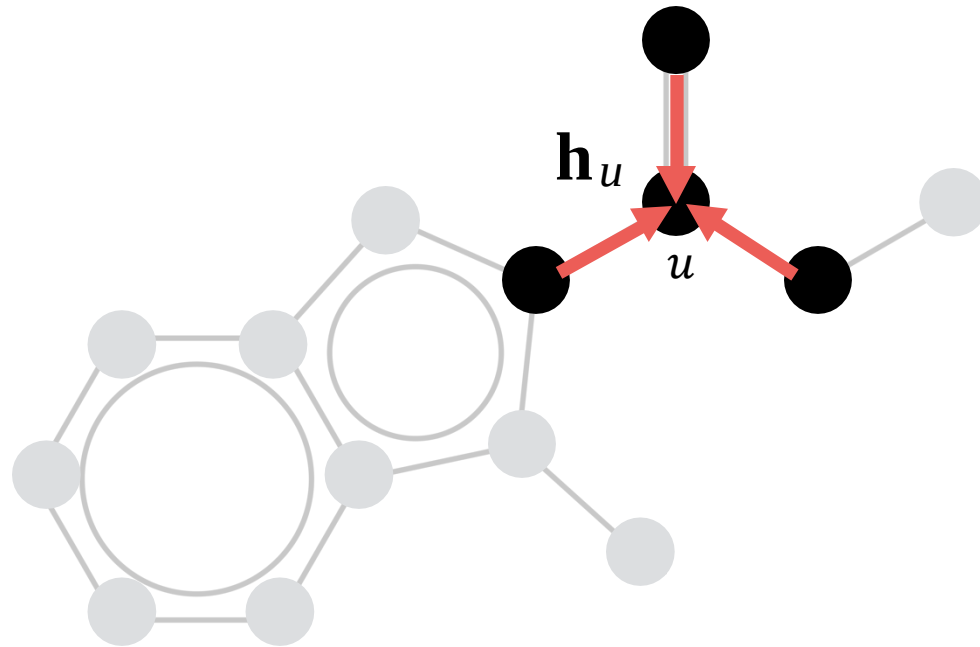


Graph encoding



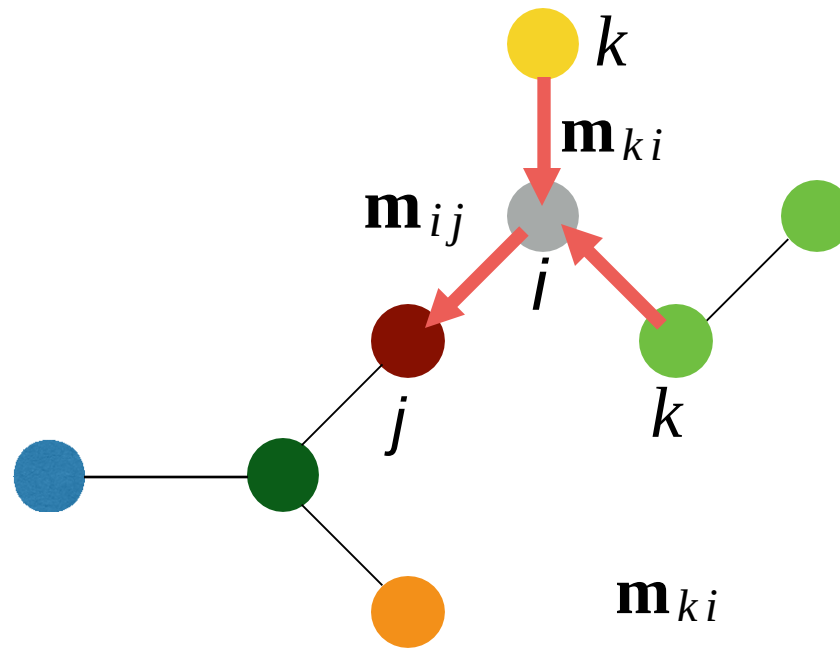
$$\nu_{uv}^{(t)} = \tau(\underbrace{\mathbf{W}_1^g \mathbf{x}_u}_{\text{Messages}} + \underbrace{\mathbf{W}_2^g \mathbf{x}_{uv}}_{\text{Node feature}} + \underbrace{\mathbf{W}_3^g}_{\text{Edge feature}} \sum_{w \in N(u) \setminus v} \nu_{wu}^{(t-1)})$$

Graph encoding



$$\mathbf{h}_u = \tau(\mathbf{U}_1^g \mathbf{x}_u + \sum_{v \in N(u)} \mathbf{U}_2^g \boldsymbol{\nu}_{vu}^{(T)})$$

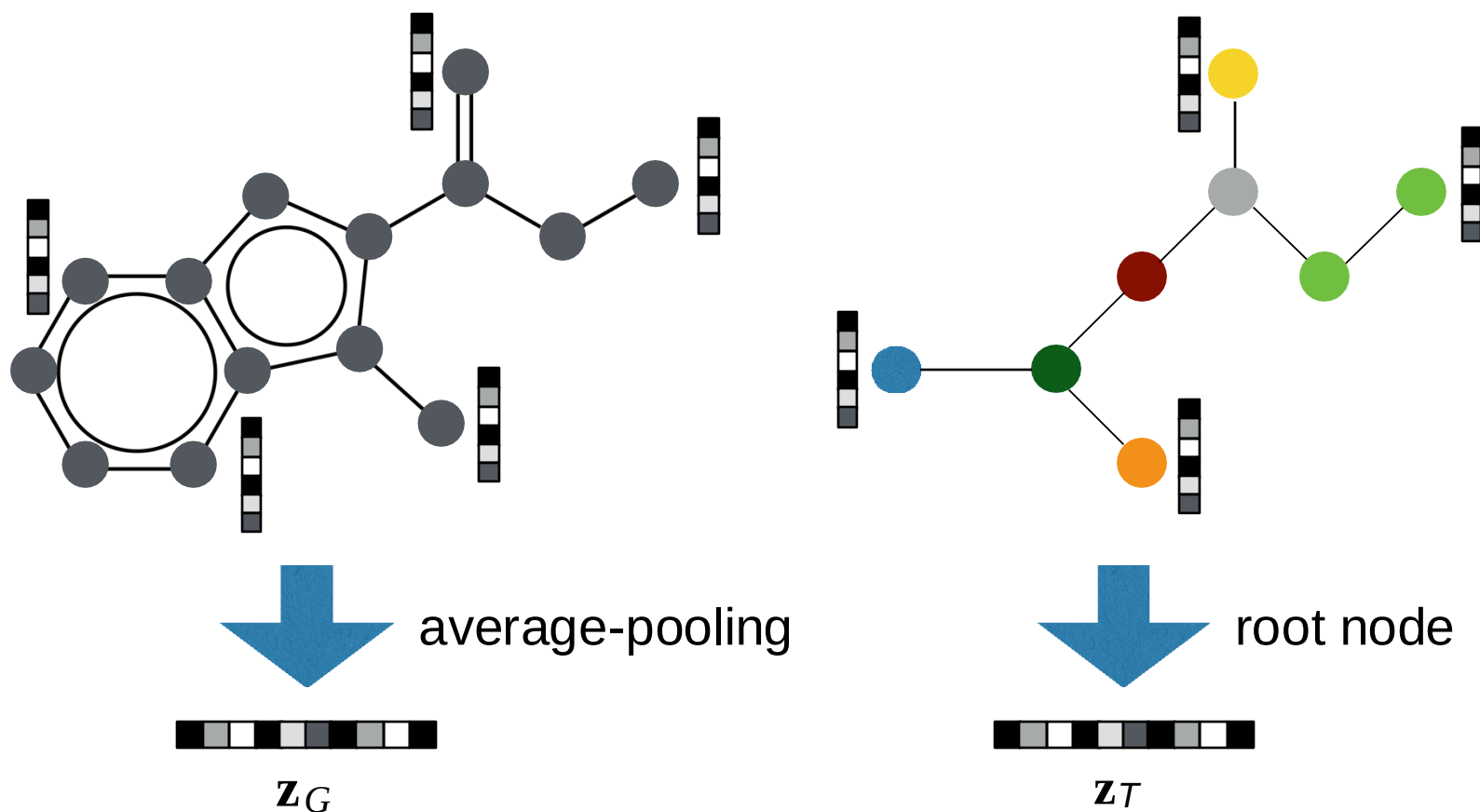
Tree encoding



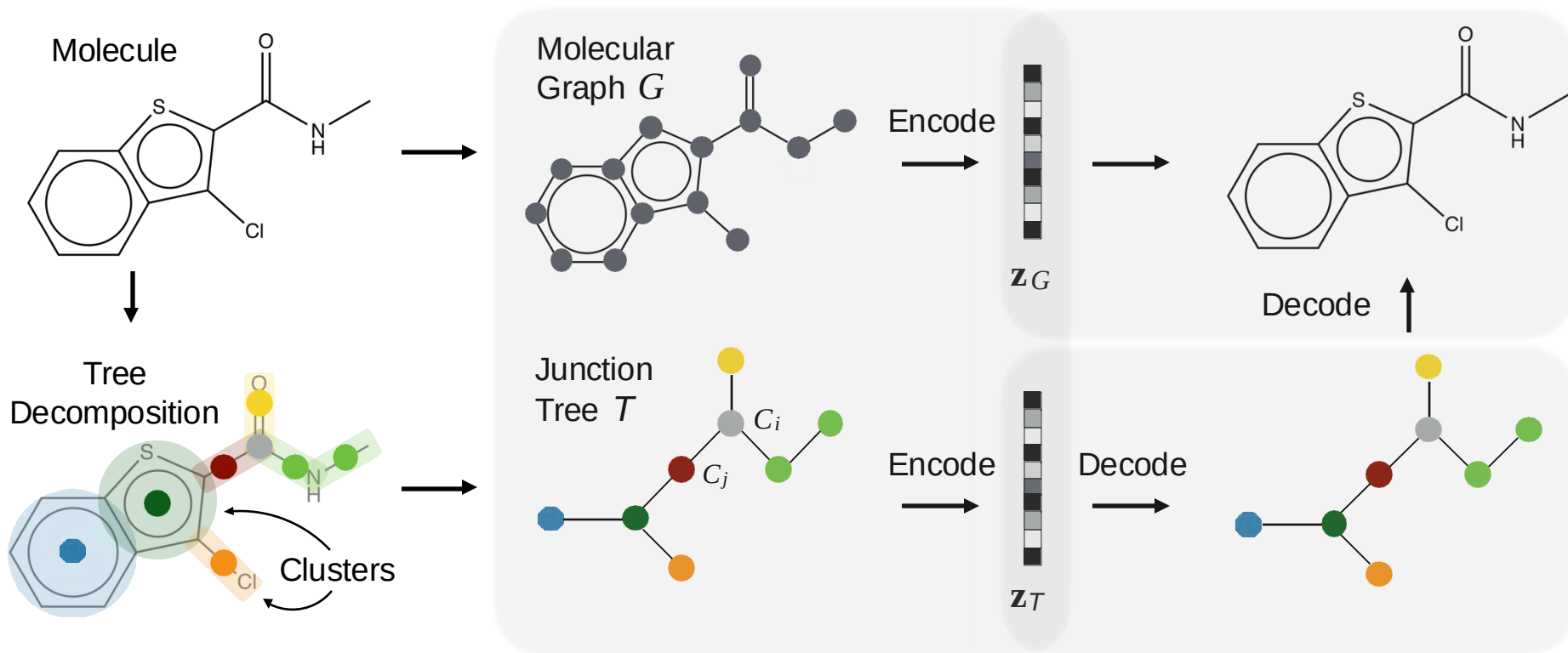
$$\mathbf{m}_{ij} = \text{GRU}(\mathbf{x}_i, \{\mathbf{m}_{ki}\}_{k \in N(i) \setminus j})$$

To capture long range interactions

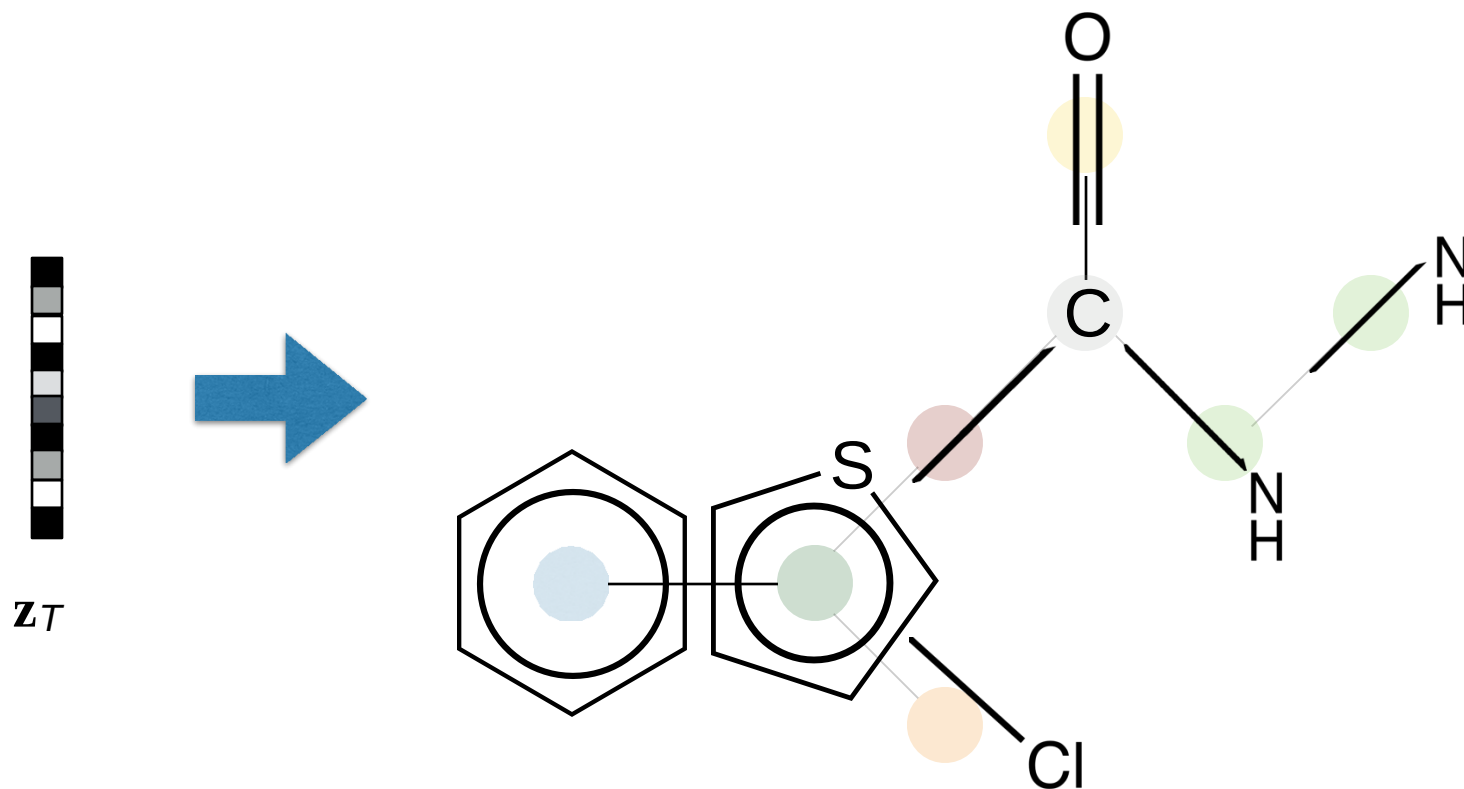
Graph and tree encoders



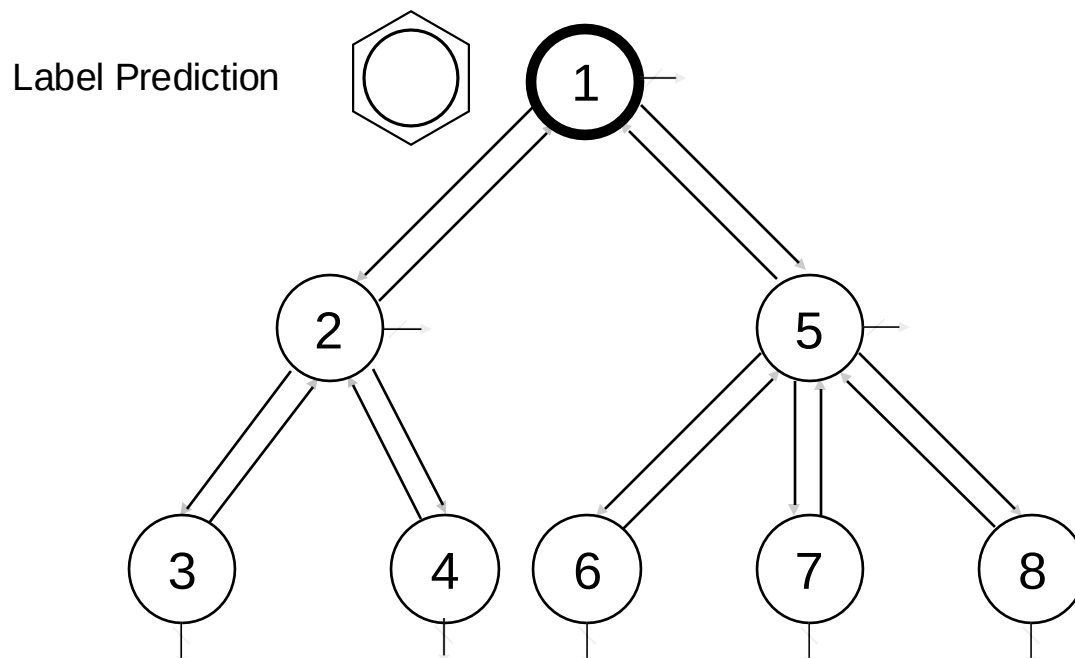
Approach: Junction-tree variational autoencoder



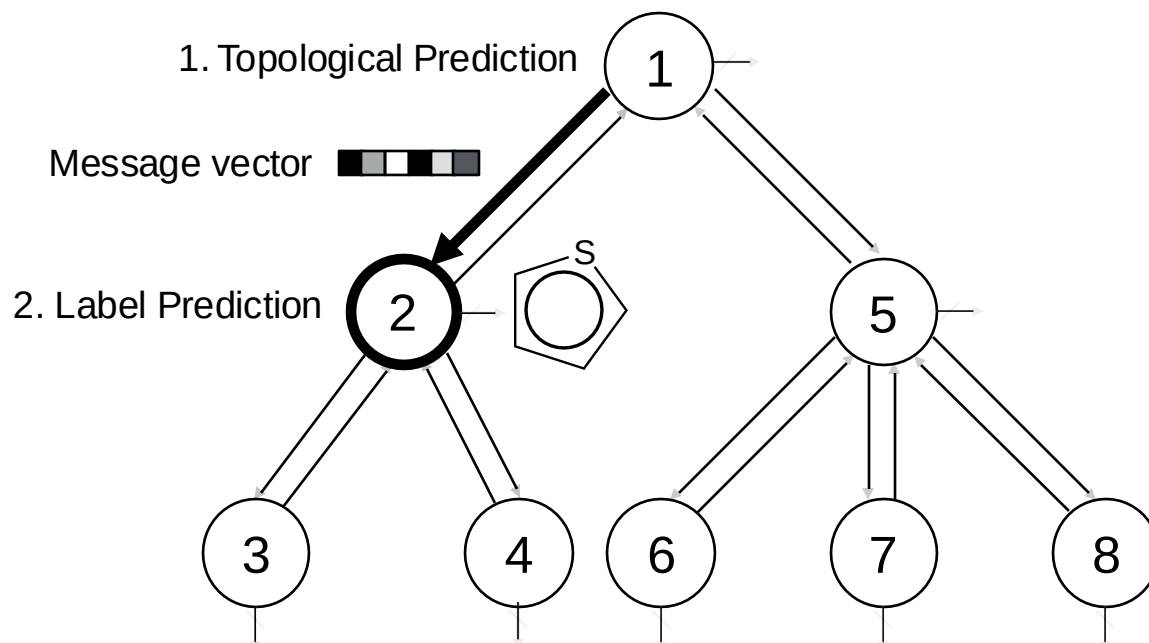
Tree decoder



Tree decoder



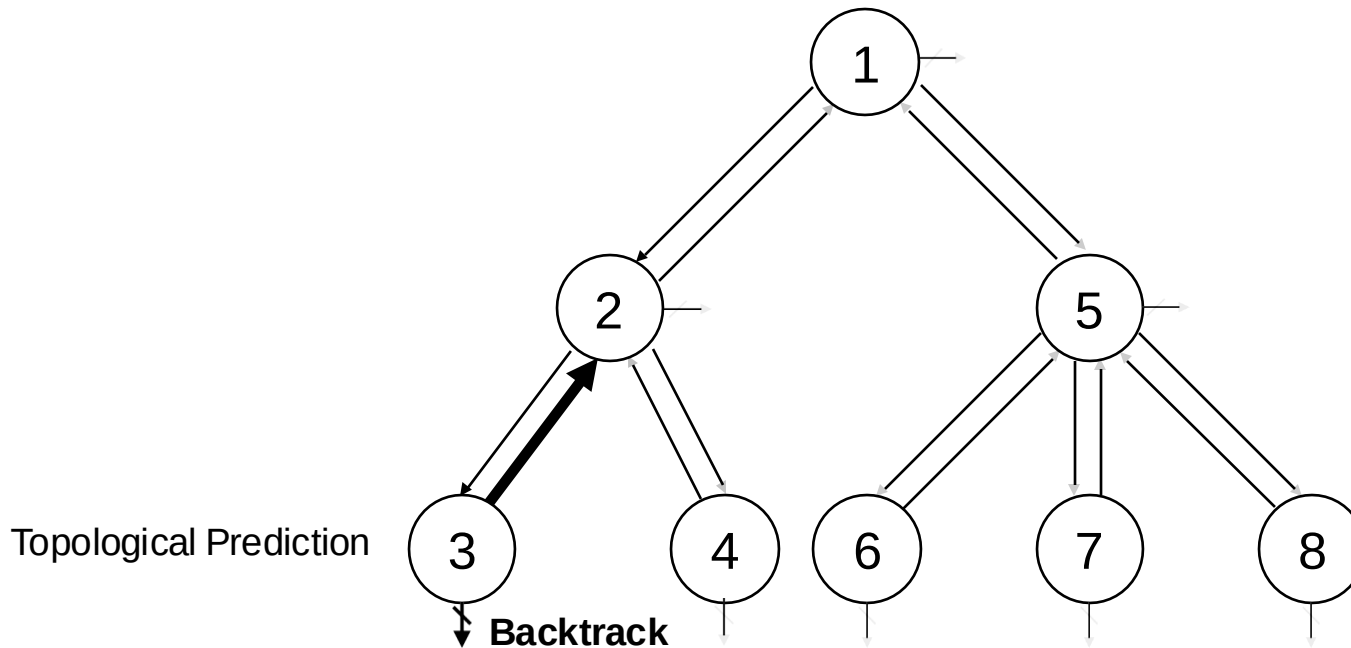
Tree decoder



Topological Prediction: Whether to expand a child or backtrack?

Label Prediction: What is the label of a node?

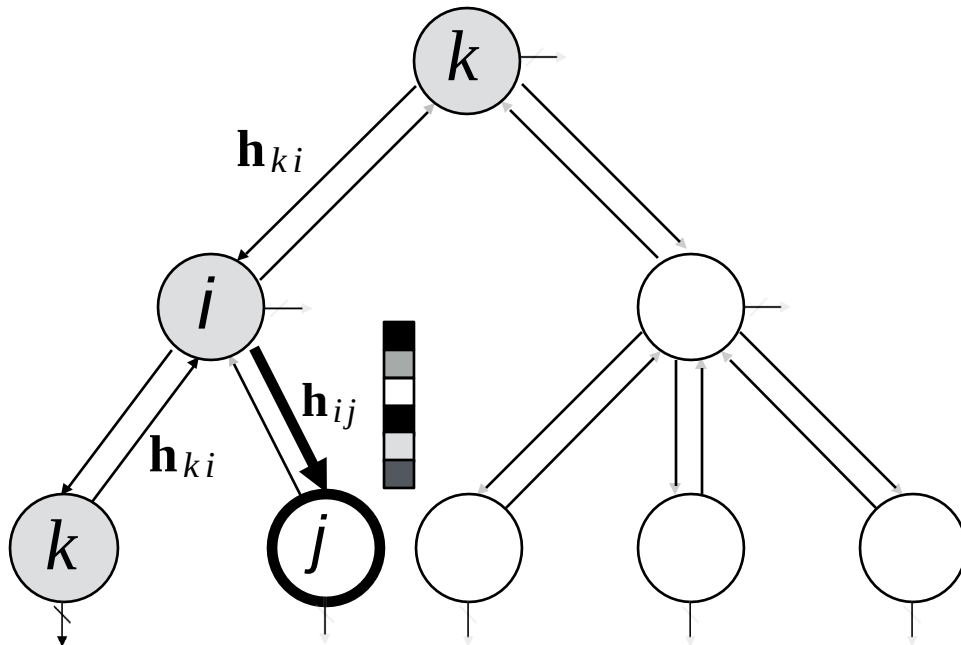
Tree decoder



Topological Prediction: Whether to expand a node or backtrack?

Label Prediction: What is the label of a node?

Tree decoder



$$\mathbf{h}_{ij} = \text{GRU}(\mathbf{x}_i, \{\mathbf{h}_{ki}\}_{k \in N_t(i) \setminus j})$$

Encodes the entire subtree of current state

Label Prediction



Feedforward
NN

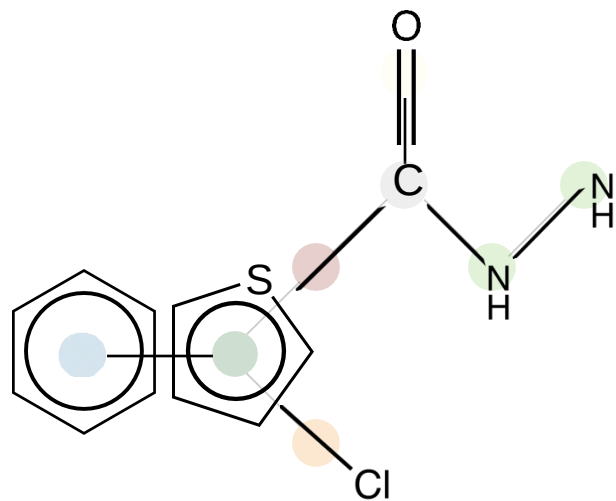


$\mathbf{h}_{ij}$

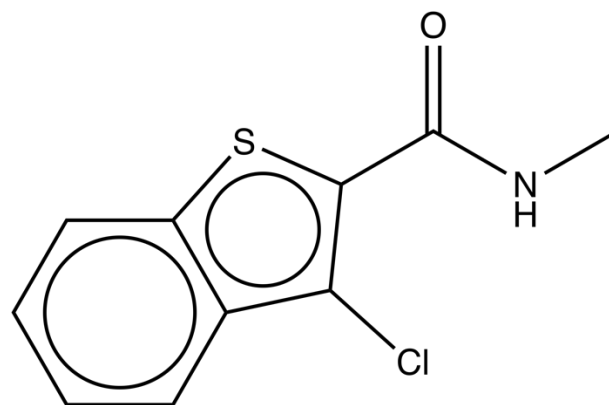
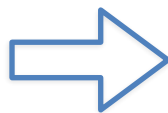


$\mathbf{z}_T$

Graph decoder

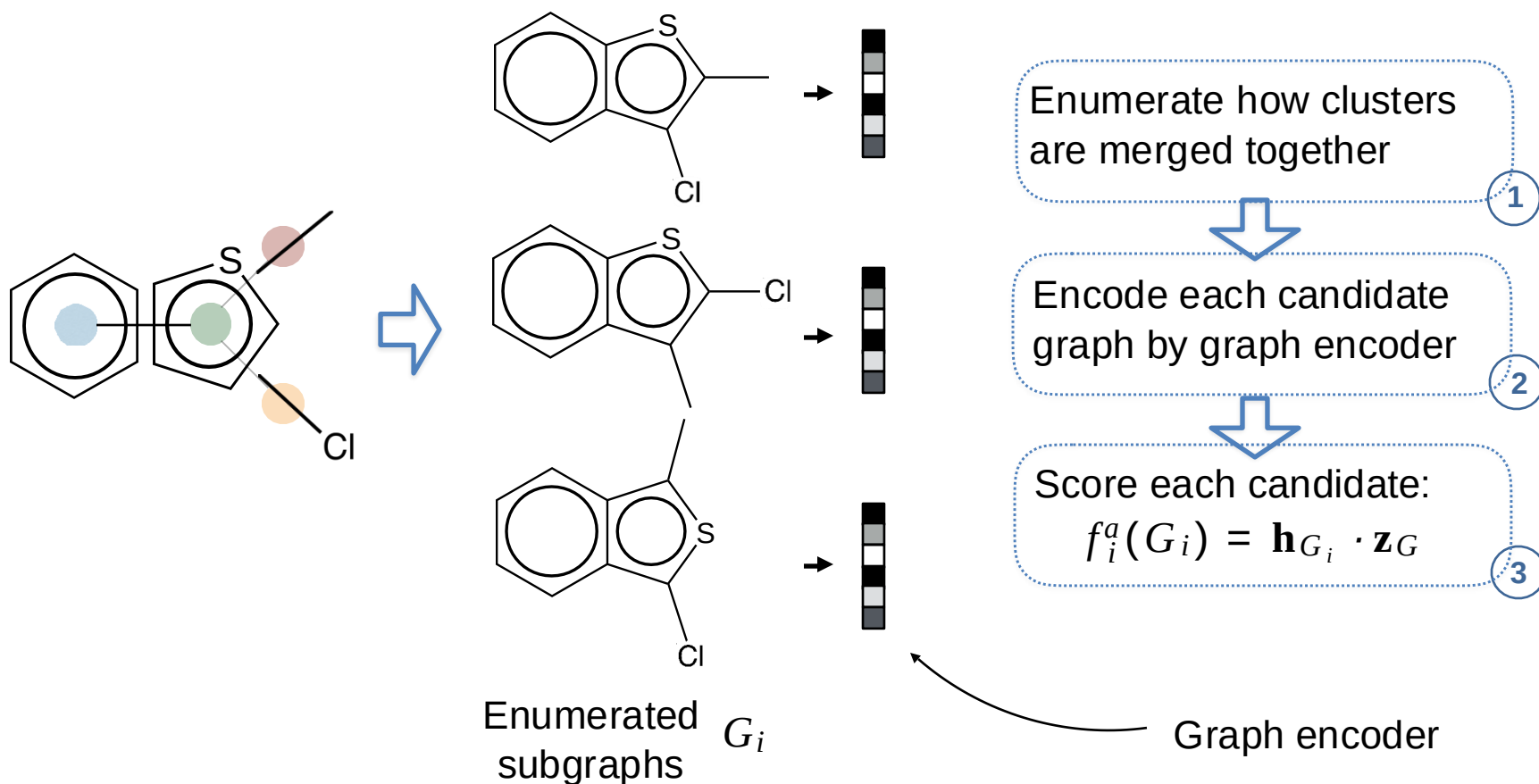


Predicted Junction Tree



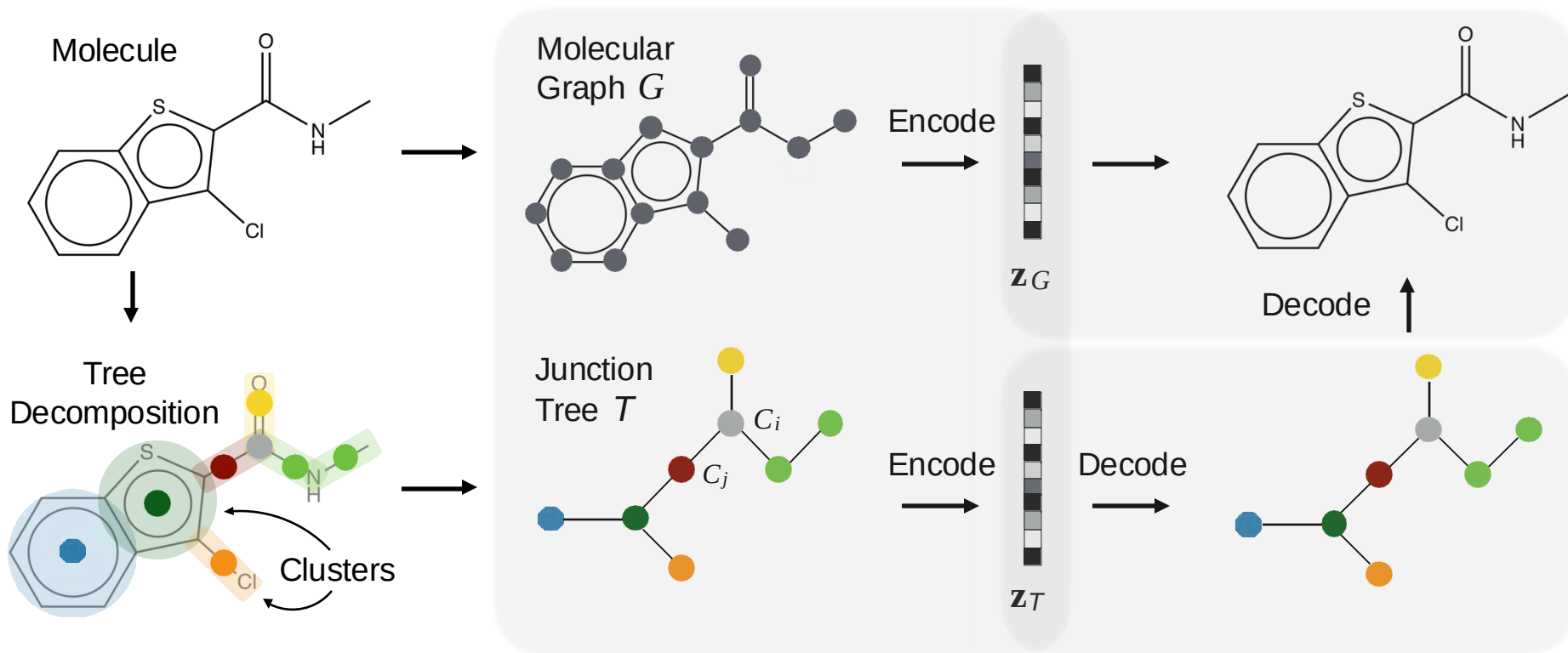
Molecular Graph

Graph decoder



$$\mathcal{L}_g(G) = \sum_i \left[f^a(G_i) - \log \sum_{G'_i \in \mathcal{G}_i} \exp(f^a(G'_i)) \right] \quad (16)$$

Recap: Junction-tree variational autoencoder



Experiments

- **Data:** 250K compounds from ZINC dataset
- **Molecule Generation:** How many molecules are valid when sampled from Gaussian prior?
- **Molecule Optimization**
 - **Global:** Find the best molecule in the entire latent space.
 - **Local:** Modify a molecule to increase its potency

Baselines

SMILES string based:

1. Grammar VAE (GVAE) (Kusner et al., 2017);
2. Syntax-directed VAE (SD-VAE) (Dai et al., 2018)

Graph based:

1. Graph VAE (Simonovsky & Komodakis, 2018)
2. DeepGMG (Li et al., 2018)

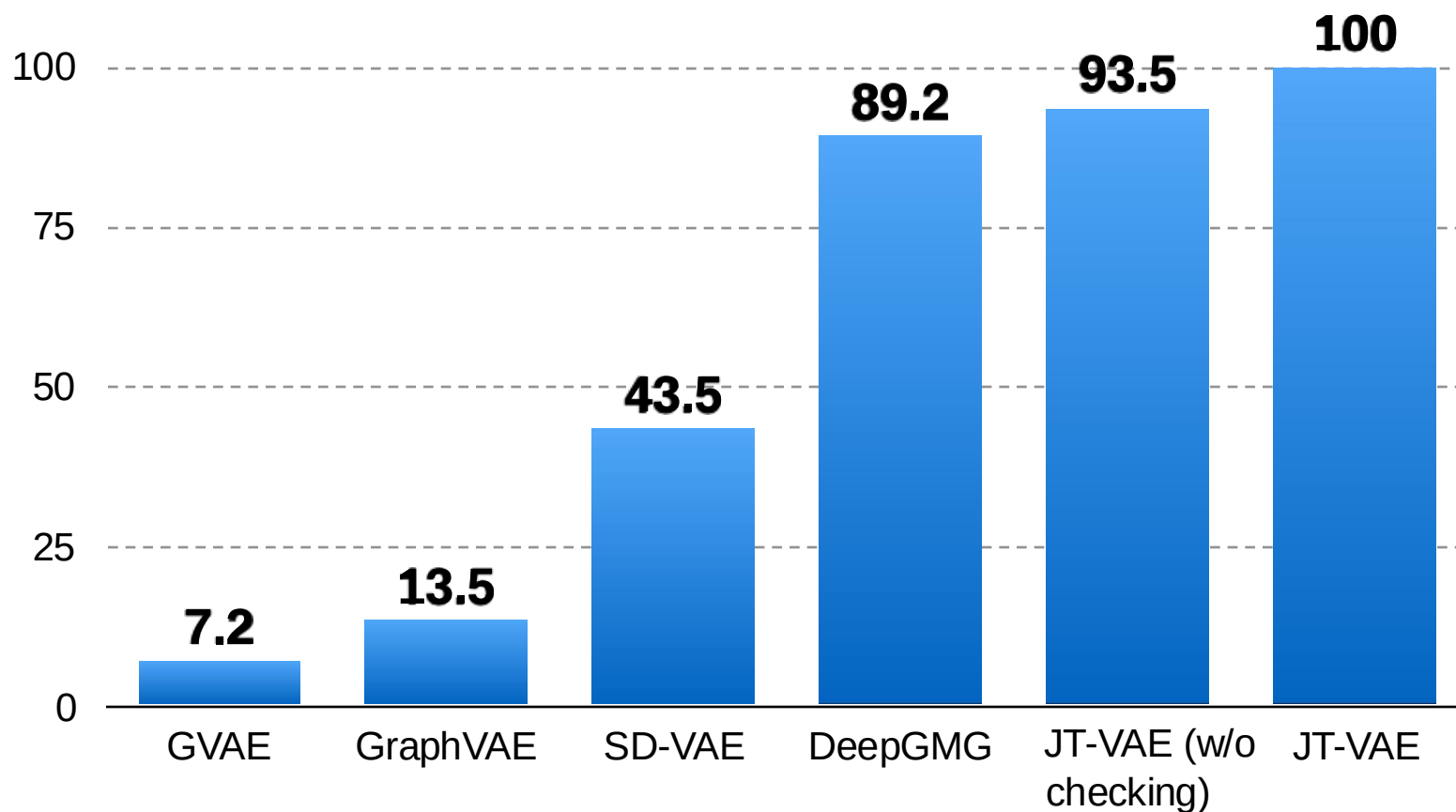
[2] Li et al., Learning Deep Generative Models of Graphs, 2018

5 Kusner et al., Grammar Variational Autoencoder, 2017

6 Dai et al., Syntax-directed Variational Autoencoder for structured data, 2018

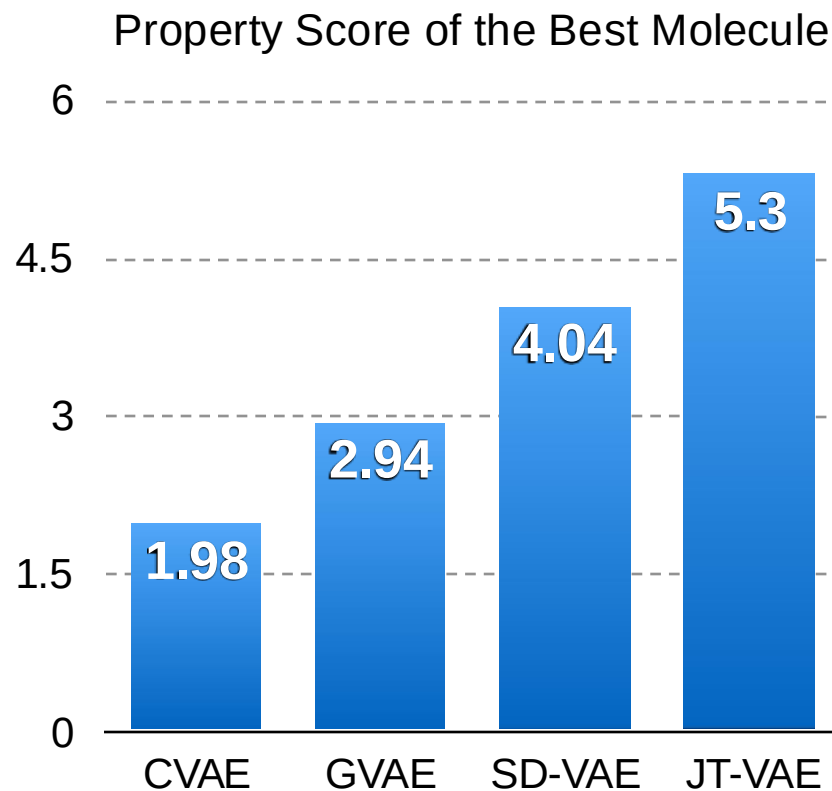
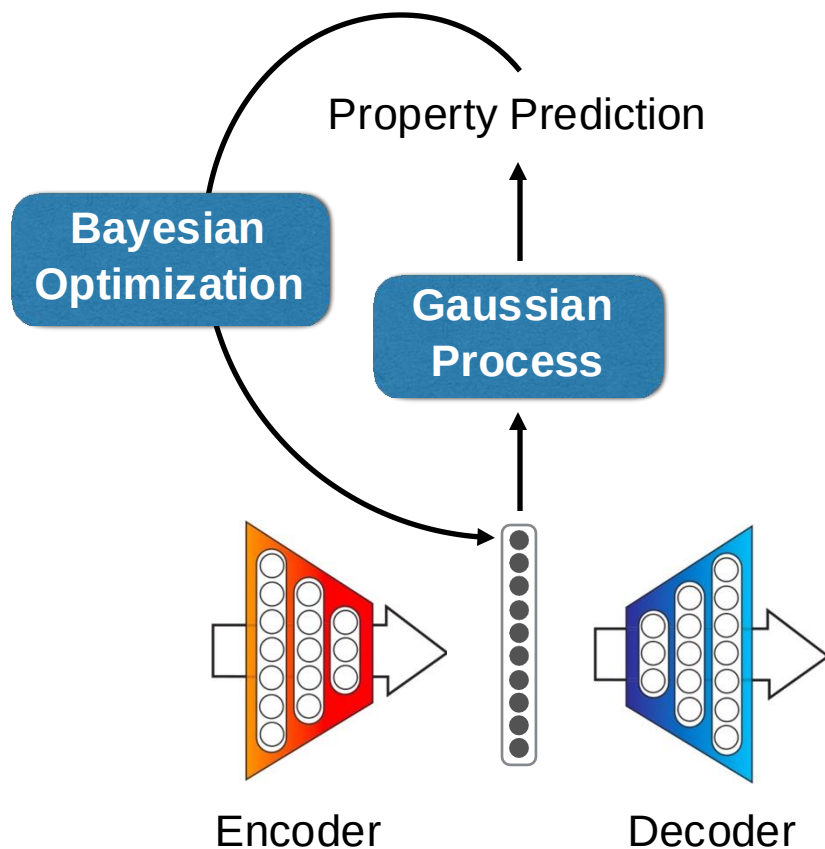
7 Simonovsky & Komodakis, GraphVAE: Towards generation of small graphs using variational autoencoders

Molecule generation (Validity)



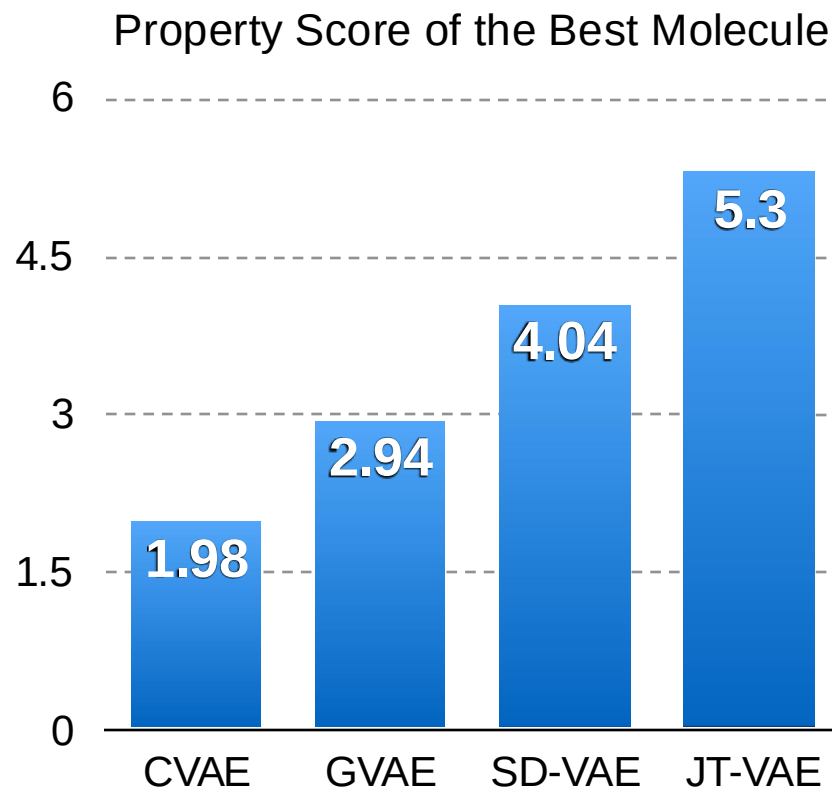
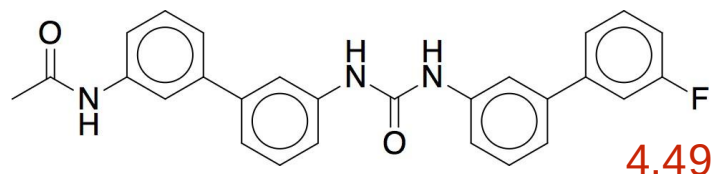
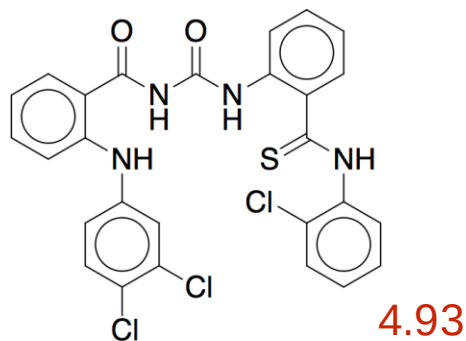
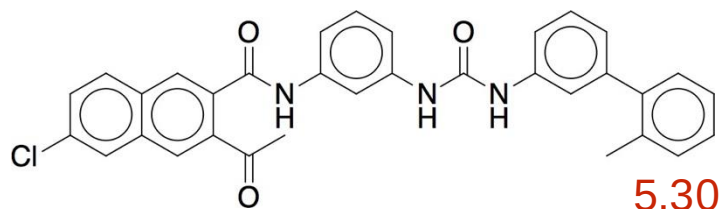
Sampled molecules

Molecule optimization (Global)



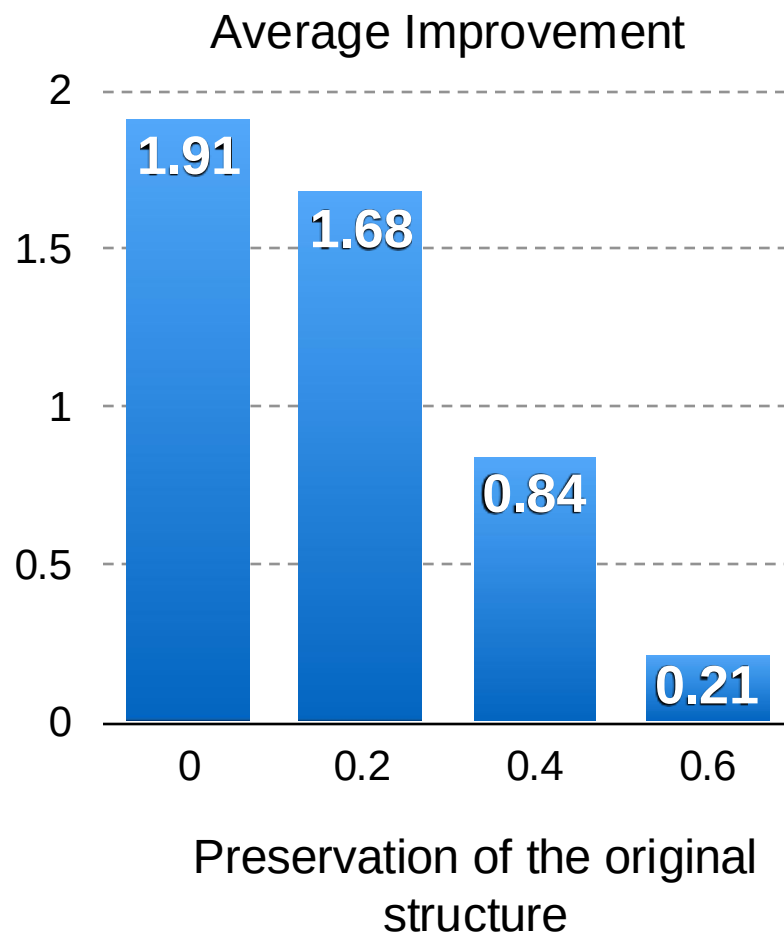
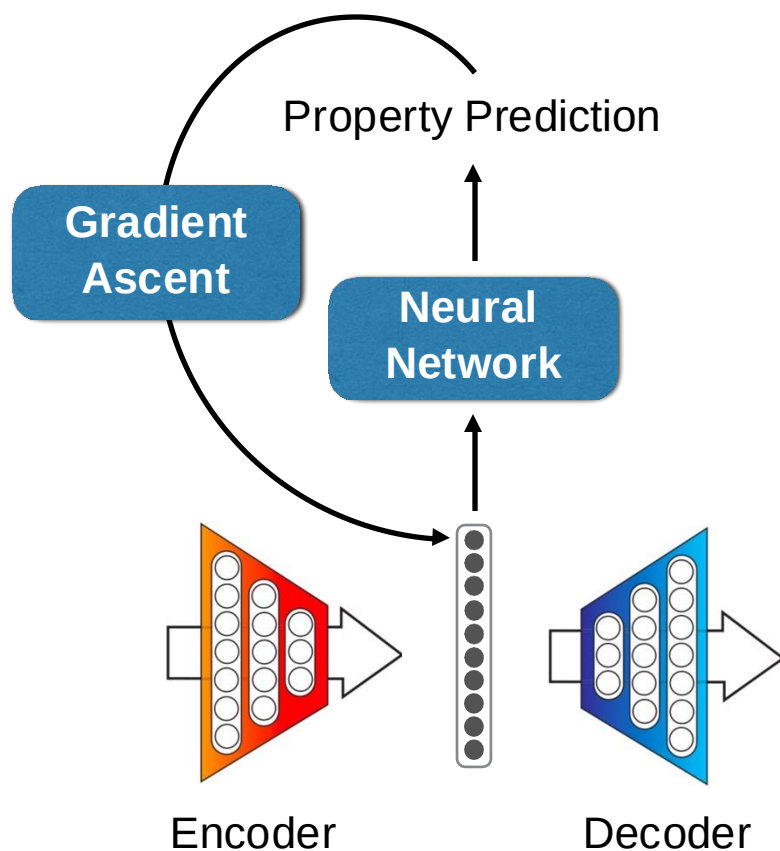
Property: Solubility + Ease of Synthesis

Molecule optimization (Global)

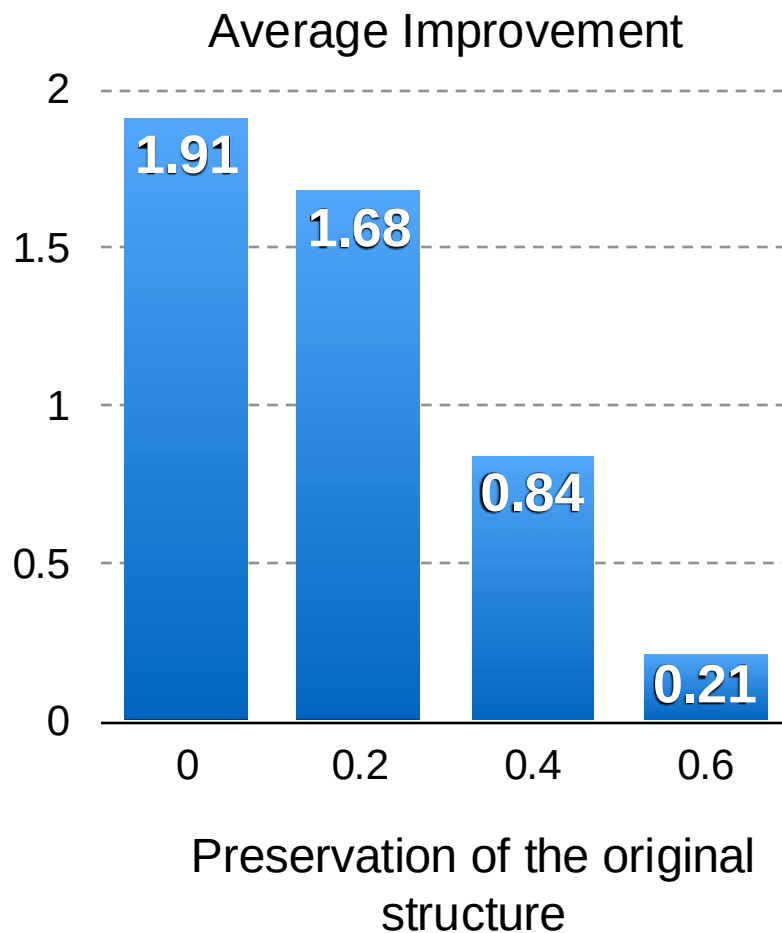
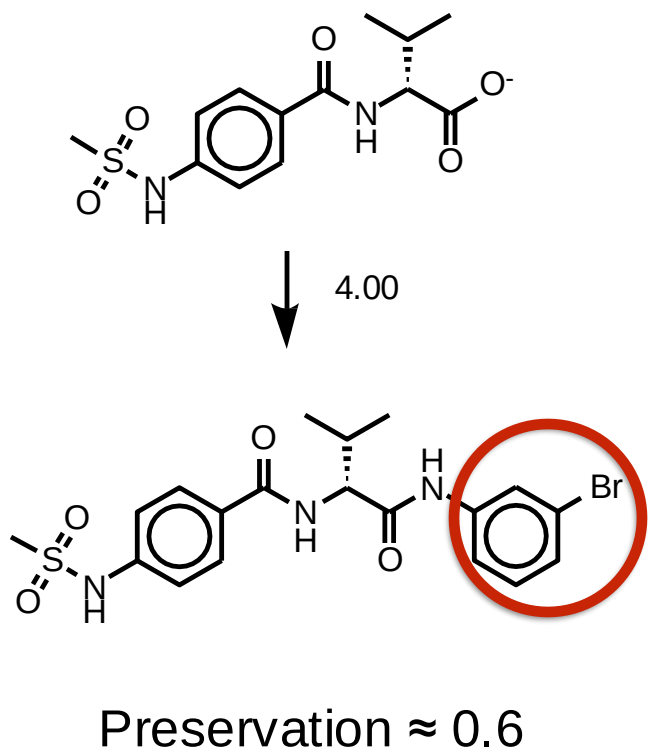


Property: Solubility + Ease of Synthesis

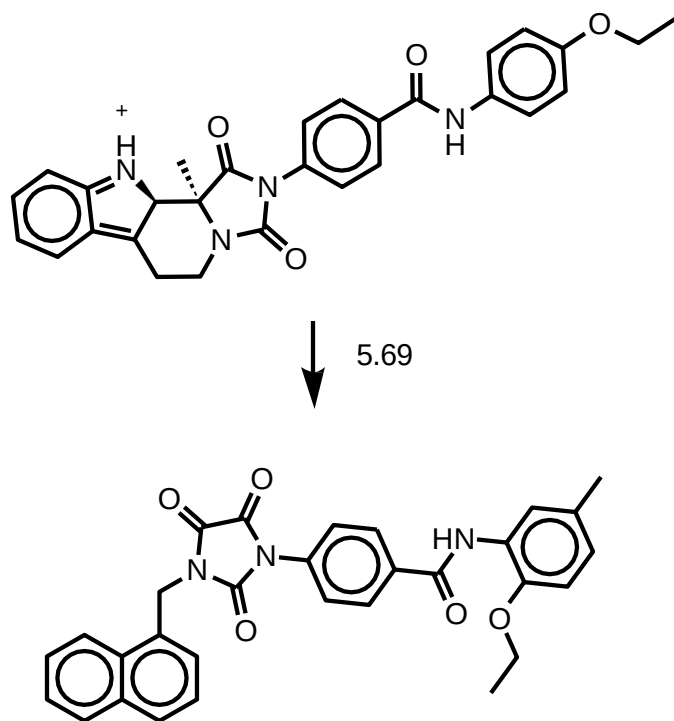
Molecule optimization (Local)



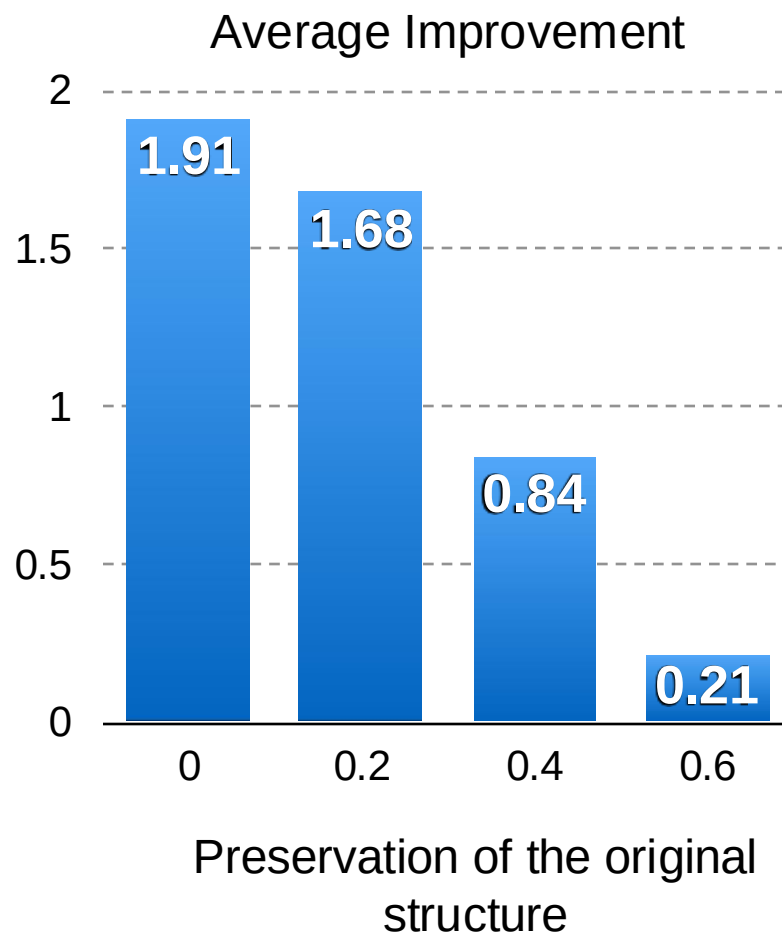
Molecule optimization (Local)



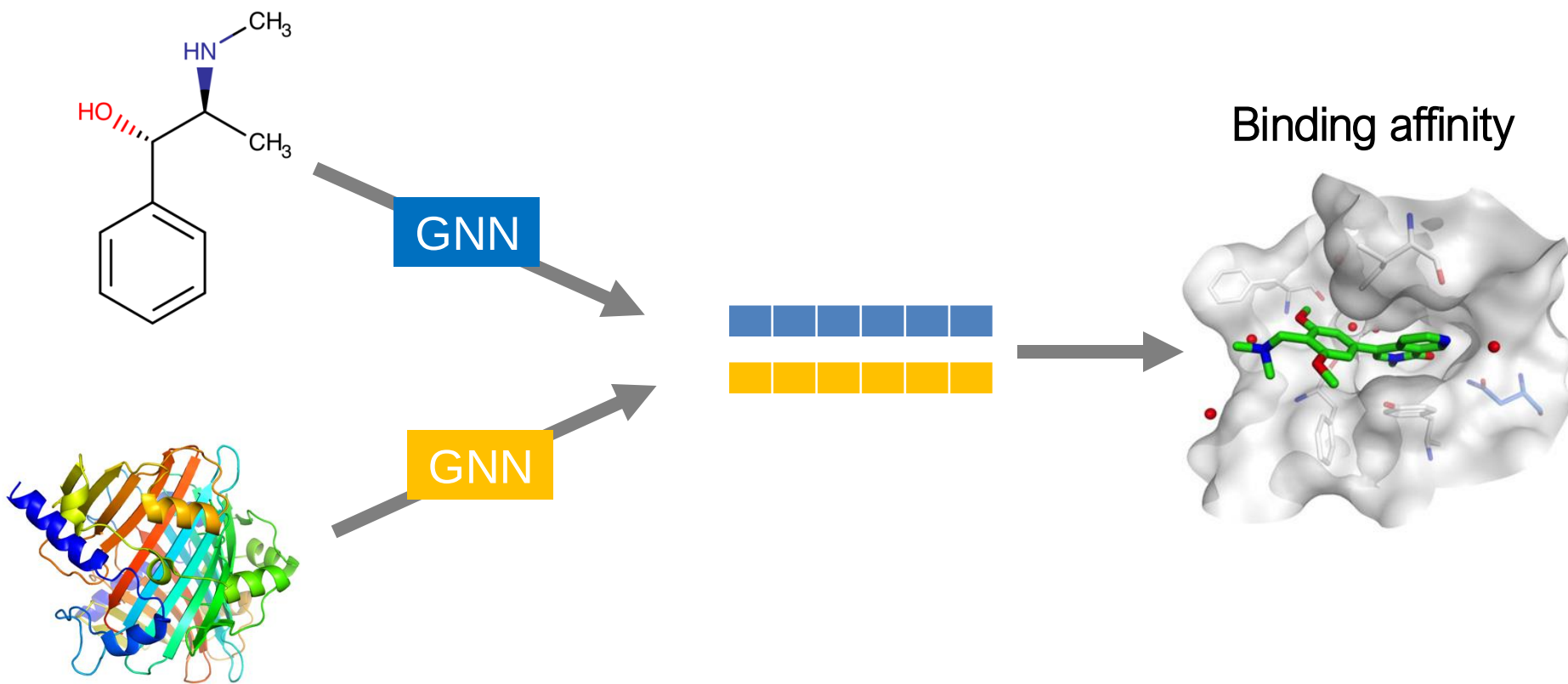
Molecule optimization (Local)



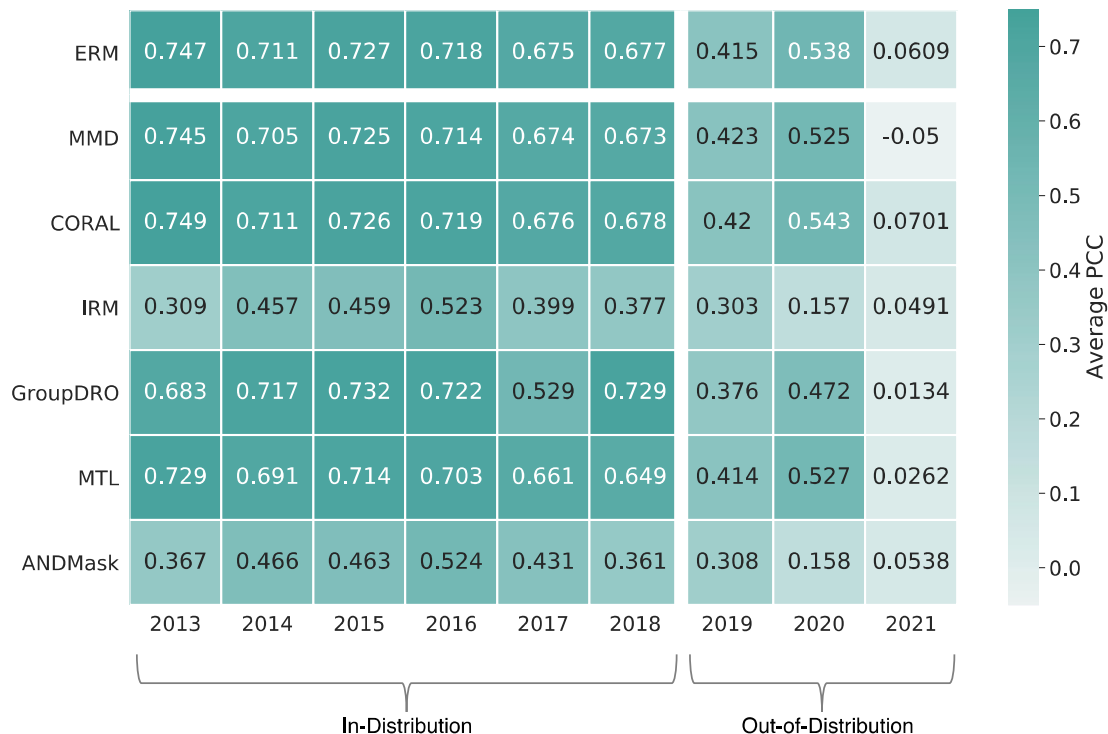
Preservation ≈ 0.4



Geometric modeling of binding



Results: Binding affinity prediction



AMINO ACID SEQUENCE
MEVRPKESWNHADPVHCEPTESVPGKPSVNADEEVGGPQICRVCGDKATGYHFNVTCEGCKGFFRRAMKRNARLRCPPFRKGACEITRKTRR
QCQACRLRKCLCESGMKKEMIMSDAEVERRALIKRKSERTGTQPLGVQGLTEEQRMMIRELMDAQMTFTDTSFHKFRLPGVLSSGCEL
PESLQAPSREEAAKWSQVRKDLCSLKVSLQLRGEDGSVMYKFPADSGGKEIFSLPHMADMSTYMFKGIISFAKVISYFRDLPEDQISLL

MOLECULE

AFFINITY PREDICTION MODEL TYPE
Daylight-AAC

ADMET PREDICTION MODEL TYPE
MPNN

CLEAR SUBMIT

CANONICAL SMILES
CC(C)clnc(cs1)CN(C)C(=O)N[C@@H](C(C)C)C(=O)N[C@@H](Cc2ccccc2)C[C@H](O)[C@H](Cc4ccccc4)NC(=O)OCc3scnc3

BINDING AFFINITY (IC50)
624.84 nM

BINDING AFFINITY (PIC50)
6.20

PREDICTED ADMET PROPERTY

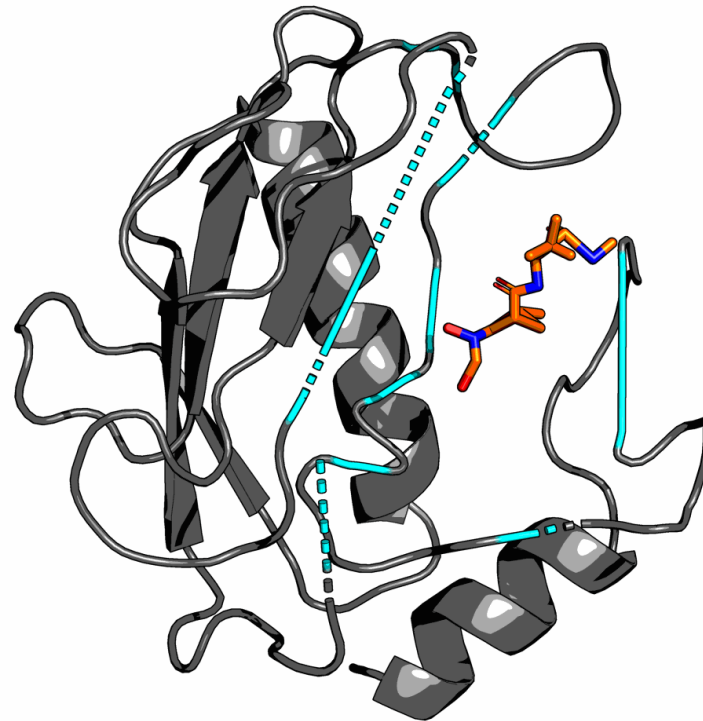
Property	Value
Solubility	-4.07 log mol/L
Lipophilicity	2.62 (log-ratio)
(Absorption) Caco-2	-5.05 cm/s
(Absorption) HIA	86.09 %
(Absorption) Pgp	20.73 %
(Absorption) Bioavailability F20	75.41 %
(Distribution) BBB	41.67 %
(Distribution) PPBR	50.20 %
(Metabolism) CYP2C19	74.68 %
(Metabolism) CYP2D6	44.95 %
(Metabolism) CYP3A4	86.54 %
(Metabolism) CYP1A2	11.20 %

Modern data management
Human-AI collaboration

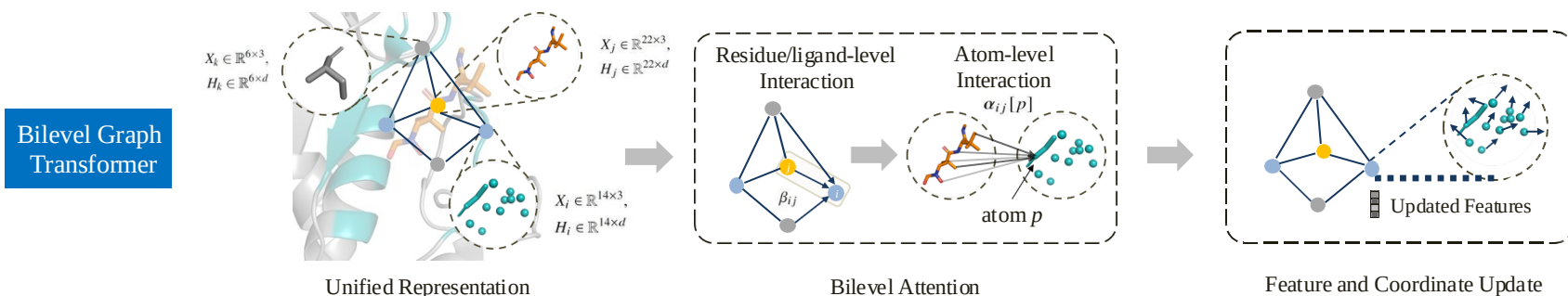
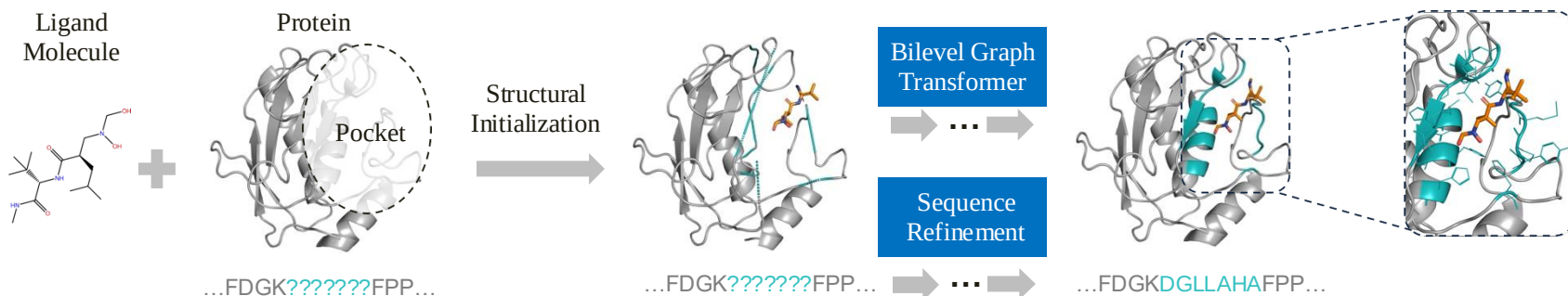
Protein universe

$\sim 20^{300}$

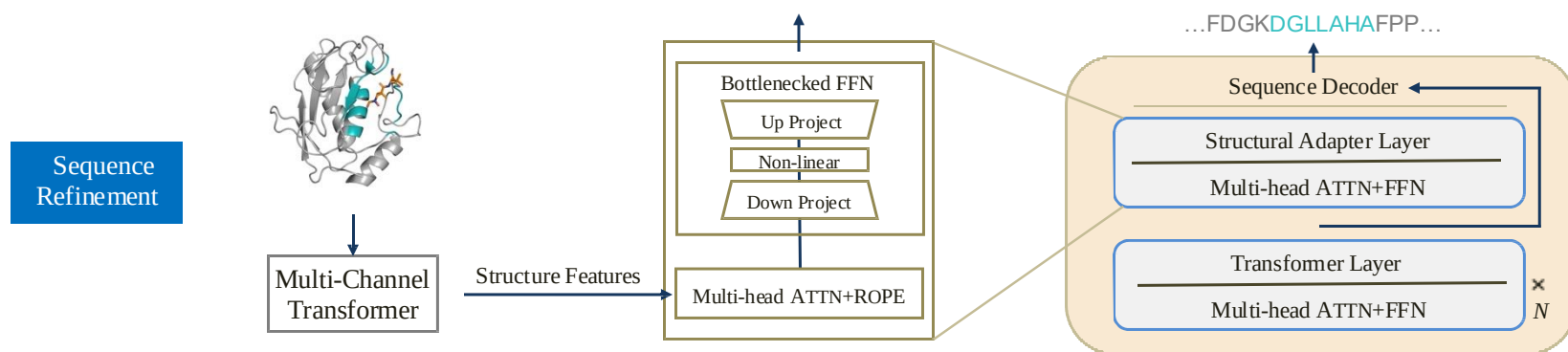
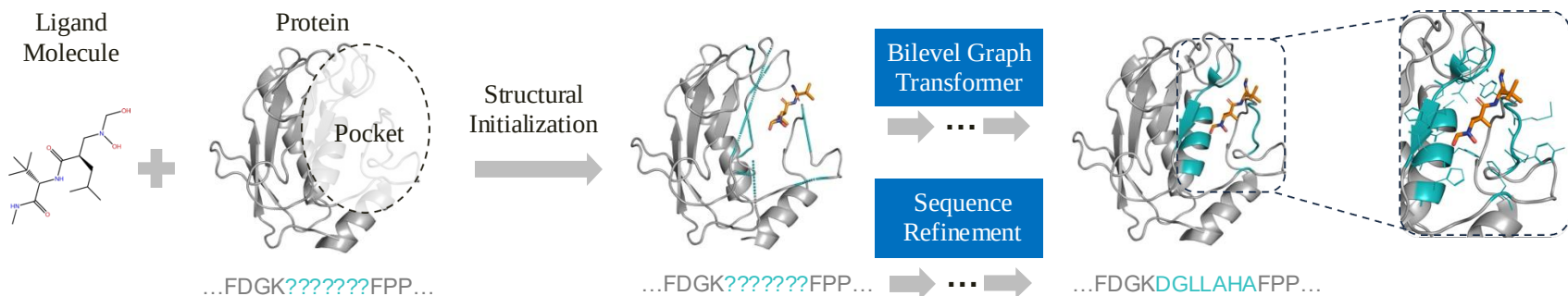
Sequence-structure molecular generation



Iterative refinement of sequence and structure in the protein pocket to maximize binding affinity with small molecule ligands



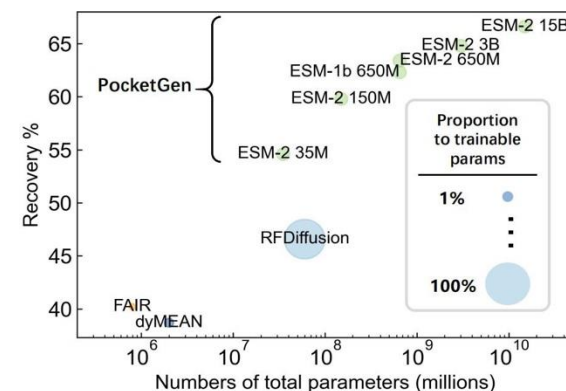
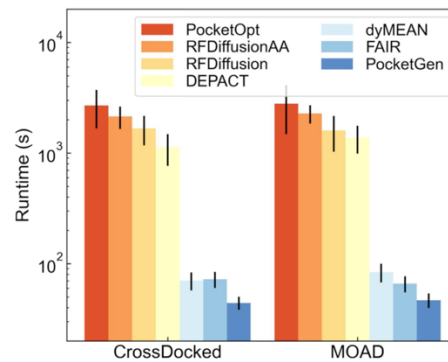
Iterative refinement of sequence and structure in the protein pocket to maximize binding affinity with small molecule ligands



PocketGen generates protein pockets with high binding affinity and structural validity

	PocketOpt	DEPACT	dyMEAN	FAIR	RFDiffusion	RFDiffusionAA	PocketGen
Top-1 generated protein pocket							
Vina score (↓)	-9.216	-8.527	-8.540	-8.792	-9.037	-9.216	-9.655
Success Rate (↑)	0.92	0.75	0.76	0.80	0.89	0.93	0.97
pLDDT (↑)	-	82.1	83.3	83.2	84.5	86.3	86.7
scRMSD (↓)	-	0.705	0.703	0.680	0.676	0.654	0.645
scTM (↑)	-	0.901	0.906	0.899	0.924	0.931	0.937
Top-3 generated protein pockets							
Vina score (↓)	-8.878	-8.131	-8.196	-8.321	-8.876	-8.980	-9.353
pLDDT (↑)	-	81.9	82.8	83.1	84.6	86.2	86.2
scRMSD (↓)	-	0.706	0.724	0.685	0.679	0.653	0.657
scTM (↑)	-	0.896	0.892	0.897	0.929	0.930	0.934
Top-5 generated protein pockets							
Vina score (↓)	-8.702	-7.786	-7.974	-7.943	-8.510	-8.689	-9.239
pLDDT (↑)	-	82.2	82.9	83.3	84.3	85.7	86.1
scRMSD (↓)	-	0.717	0.725	0.690	0.680	0.656	0.652
scTM (↑)	-	0.892	0.903	0.886	0.926	0.929	0.935
Top-10 generated protein pockets							
Vina score (↓)	-8.556	-7.681	-7.690	-7.785	-8.352	-8.524	-9.065
pLDDT (↑)	-	81.5	82.7	83.0	84.2	85.3	85.9
scRMSD (↓)	-	0.710	0.734	0.705	0.684	0.672	0.678
scTM (↑)	-	0.895	0.896	0.884	0.924	0.929	0.931

Model	CrossDocked			Binding MOAD		
	AAR (↑)	Designability (↑)	Vina (↓)	AAR (↑)	Designability (↑)	Vina (↓)
Test set	-	0.77	-7.016	-	0.79	-8.076
DEPACT	31.52±3.26%	0.68±0.04	-6.632±0.18	35.30±2.19%	0.67±0.06	-7.571±0.15
dyMEAN	38.71±2.16%	0.71±0.03	-6.855±0.06	41.22±1.40%	0.70±0.03	-7.675±0.09
FAIR	40.16±1.17%	0.73±0.02	-7.015±0.12	43.68±0.92%	0.72±0.05	-7.930±0.15
RFDiffusion	46.57±2.07%	0.74±0.01	-6.936±0.07	45.31±2.73%	0.75±0.05	-7.942±0.14
RFDiffusionAA	50.85±1.85%	0.75±0.03	-7.012±0.09	49.09±2.49%	0.78±0.03	-8.020±0.11
PocketGen	63.40±1.64%	0.77±0.02	-7.135±0.08	64.43±2.35%	0.80±0.04	-8.112±0.14



Improved structural validity, amino acid sequence recovery, and binding affinity with target ligands

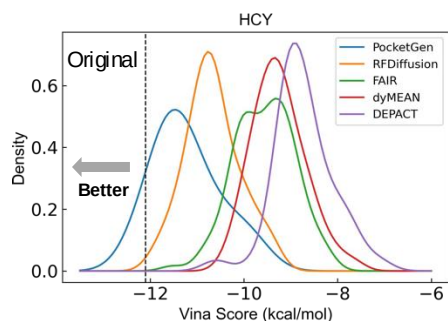
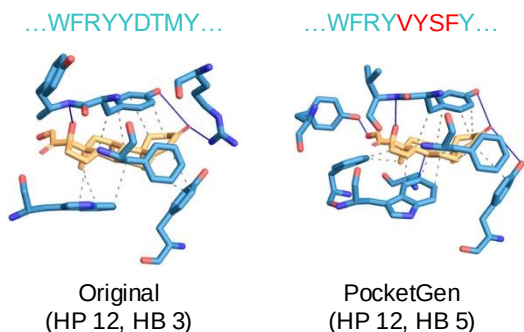
Better generation efficiency

Performance wrt protein LM size

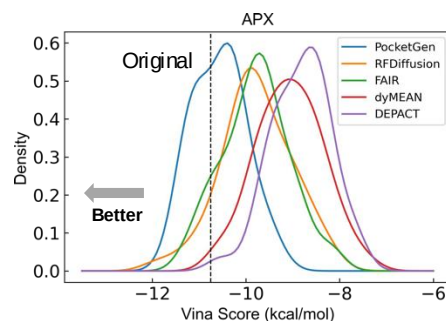
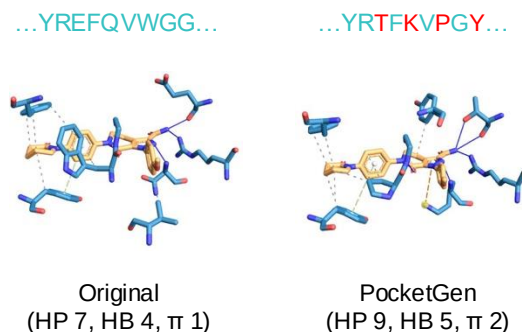
PocketGen can design protein pockets of antibodies, enzymes, and biosensors for target ligand molecules

- Protein
- Ligand
- Aromatic Ring Center
- Hydrophobic Interaction
- Hydrogen Bond
- π -Stacking (parallel)
- π -Stacking (perpendicular)
- π -Cation Interaction

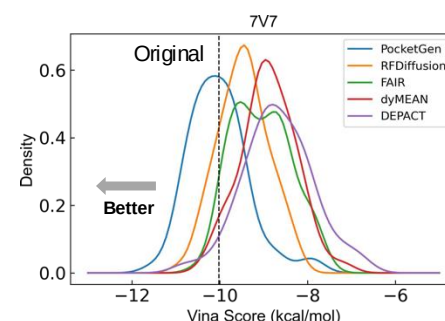
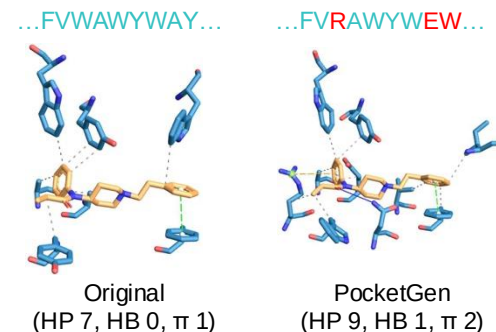
Cortisol (HCY)



Apixaban (APX)



Fentanyl 7V7





Words and genes share a correspondence:
their **meanings** arise from their **context**.

Gene perturbation measurements across diverse cell contexts
induce **semantics for genes**

(under the right approach)

“apple” is a **polysemic** word...



🔍 grow an apple

🔍 buy an apple|

... whose **particular meaning** is resolved via **sentence context**.



🔍 grow an apple

🔍 grow an apple **tree**

🔍 grow an apple **tree from seed**

🔍 grow an apple **tree in a pot**

🔍 grow an apple **tree indoors**



🔍 buy an apple|

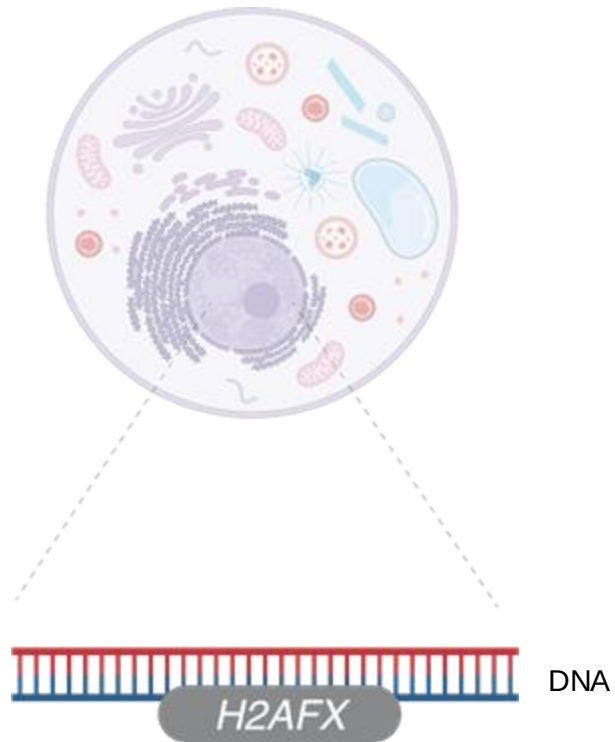
🔍 buy an apple **watch**

🔍 buy an apple **gift card**

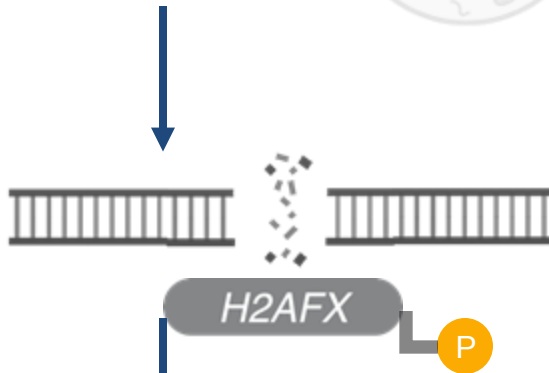
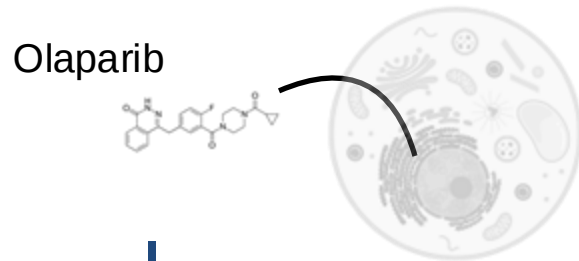
🔍 buy an apple **tv**



H2AFX is a **pleiotropic** gene...



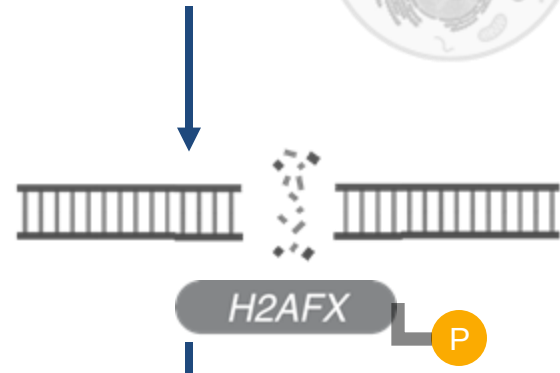
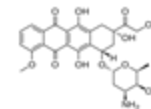
... whose **particular function** is resolved via **cell context**.



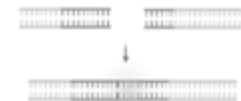
Homologous
Recombination



Doxorubicin



End Joining



While unsupervised learning of word polysemy is **common**...

Data: corpus
of sentence contexts

Approach: word embeddings
w/ linear semantics

king - man + woman $\approx$ queen

unsupervised learning of gene pleiotropy is **unsolved**

Data: ?

Approach: ?

geneA - func1 + func2 $\approx$ geneB

Our goal for today

Unsupervised learning of gene pleiotropy with applications to therapeutic science

Data:

?

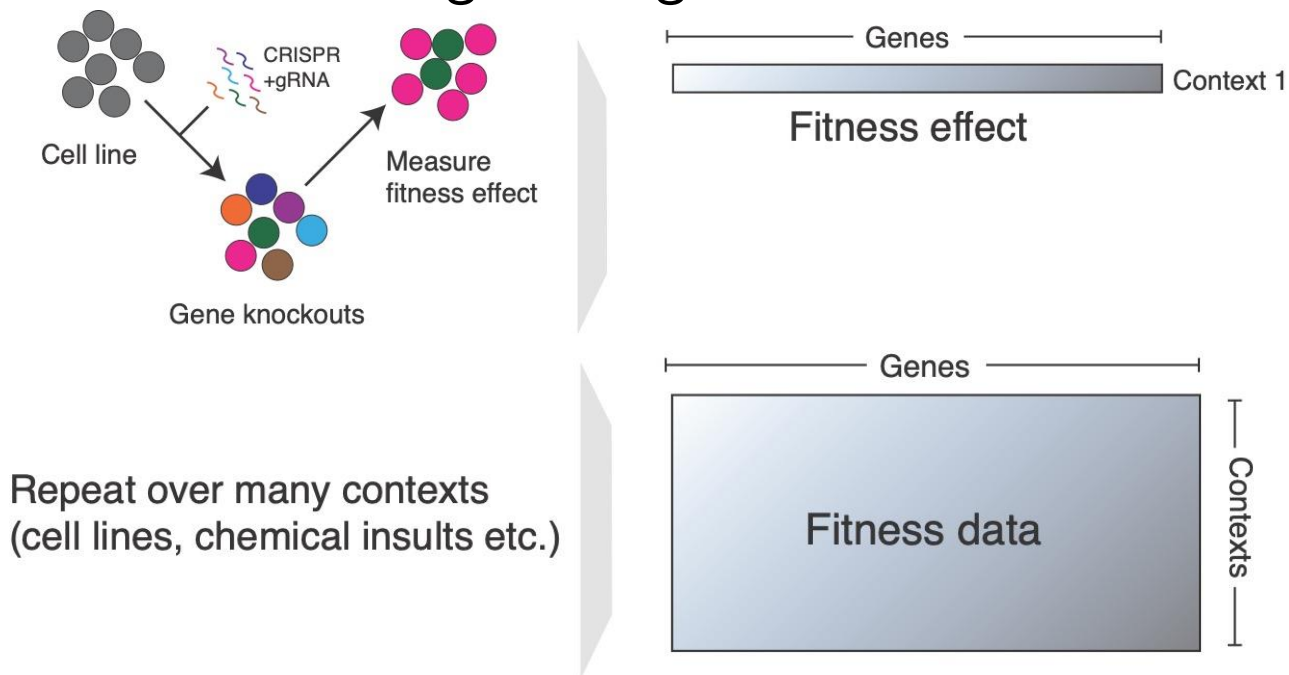
Approach:

?

$$\textit{geneA} - \textit{func1} + \textit{func2} \approx \textit{geneB}$$

Data

Use **gene perturbation effect measurements** for inferring biological functions

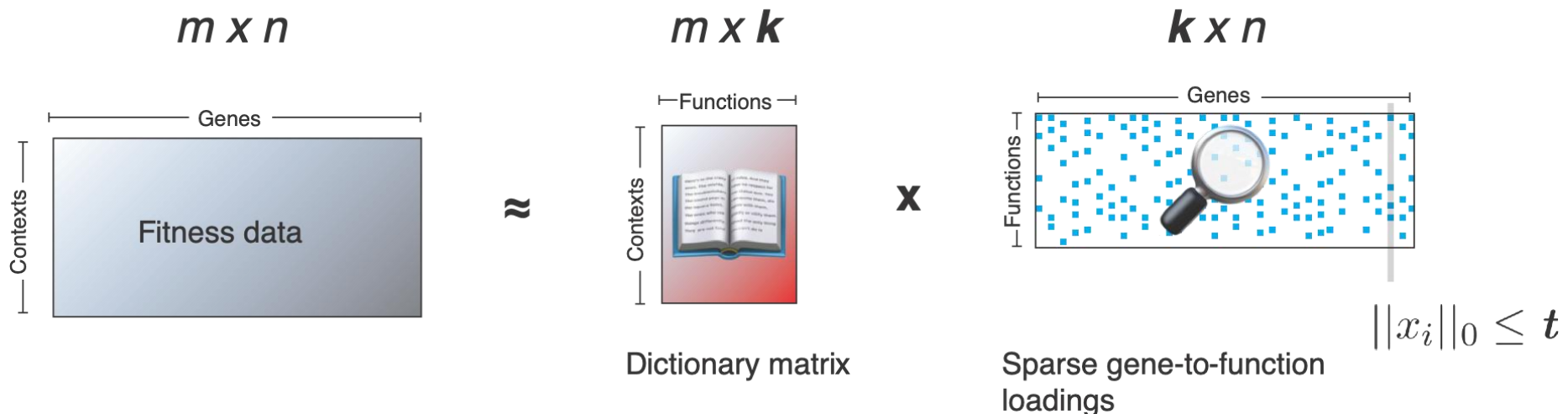


Why perturbation datasets? Alternative data types:

- **Transcriptomics:** gene co-expression is necessary but not sufficient for co-function
- **Protein-protein interactions:** direct interactions are not necessary for co-function

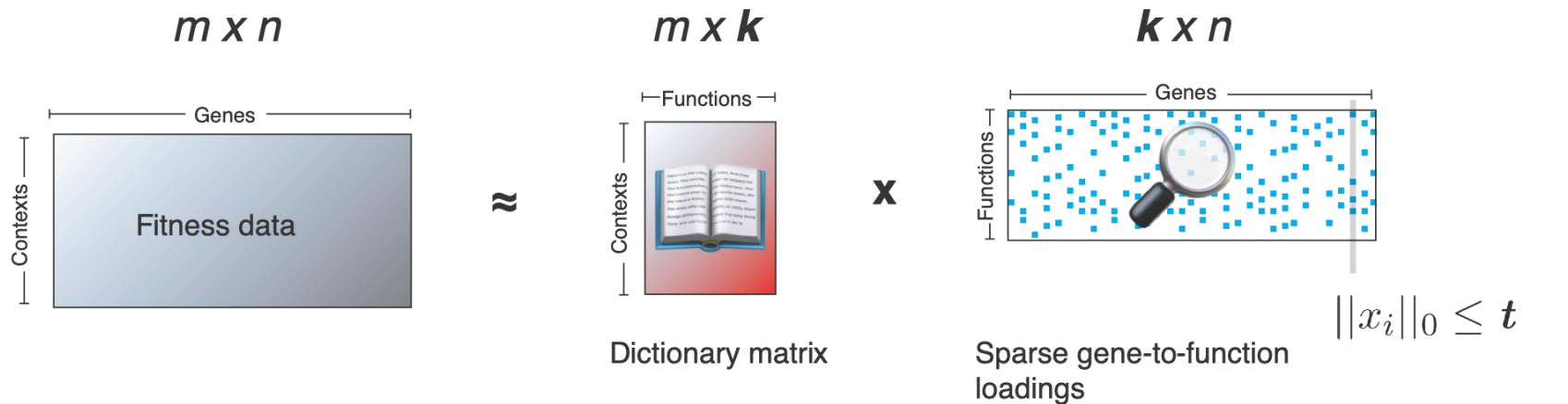
Approach: Webster

- Low-dimensional vector embeddings that satisfy three criteria:
 - Sparse
 - Latents are biologically meaningful
 - Account for redundancy between cell contexts



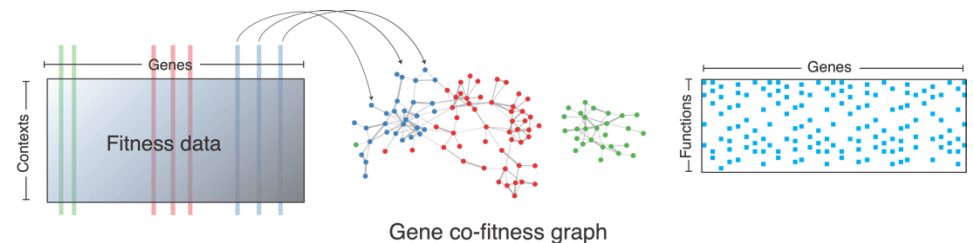
Approach: Webster

Webster learns a dictionary matrix that **sparsely** approximates gene effects...



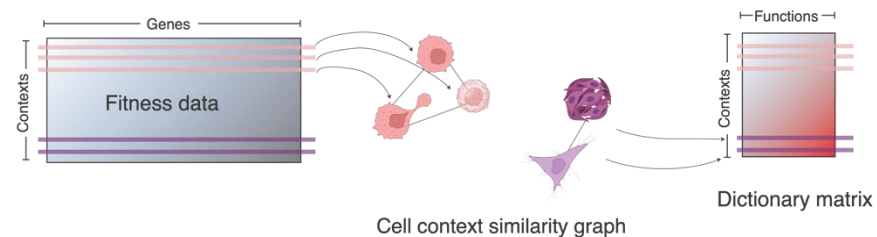
1

... while preserving interpretable relationships between genes



2

... and accounting for redundancies between cell contexts

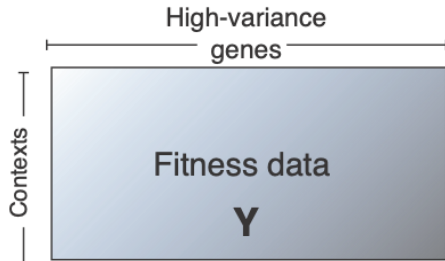


Overview of Webster

Preprocessing

Raw fitness data

Standardize cell lines
Center gene effects
Filter genes by variance



Graph-regularized dictionary learning *Objectives*

Reduce dimensionality

Diagram illustrating the dimensionality reduction objective. A blue box labeled Y is approximately equal to a red box labeled D multiplied by a blue box labeled X . The equation is $Y \approx DX$. Below the equation is the formula $\|Y - DX\|_F^2$.

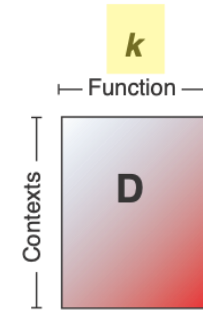
Preserve gene similarity

Diagram illustrating the gene similarity objective. A graph with nodes and edges is shown, with a cluster of nodes highlighted in green. The graph is labeled L_c . Below the graph is the formula $Tr(XL_cX^T)$.

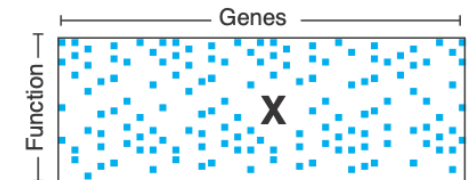
Preserve cell context similarity

Diagram illustrating the cell context similarity objective. A graph with nodes and edges is shown, with a cluster of nodes highlighted in red. The graph is labeled L . Below the graph is the formula $Tr(D^T L D)$.

Output



Dictionary matrix



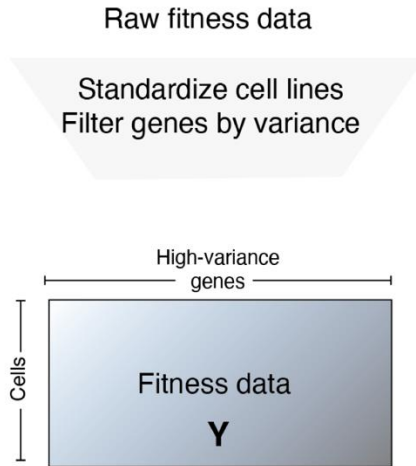
Gene-to-function loadings

$$\|x_i\|_0 \leq t$$

 = key hyperparameters

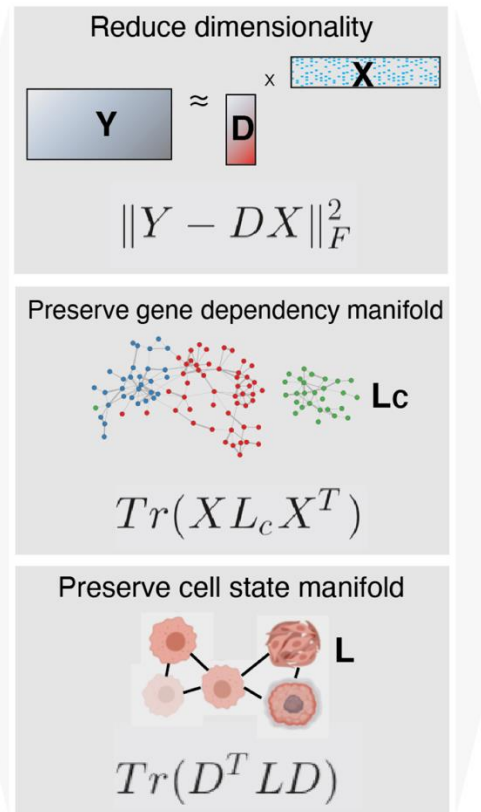
Model optimization

Preprocessing

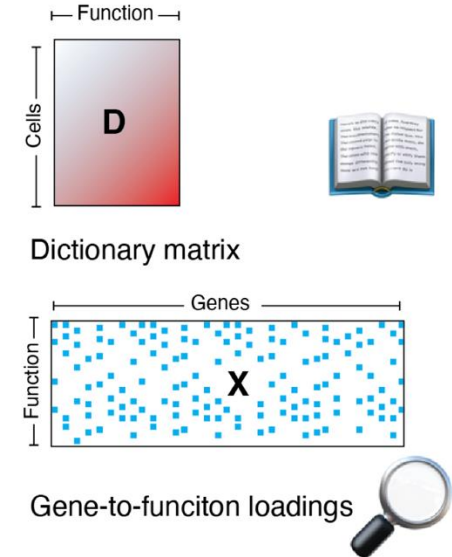


Graph-regularized dictionary learning

Objectives



Output



$$\text{s.t. } \|x_i\|_0 \leq T$$

Parameters

$$\arg \min_{D, X} \|Y - DX\|_F^2 + \alpha Tr(D^TLD)$$

$$+ \beta Tr(XL_cX^T) \quad \text{s.t.} \quad \|x_i\|_0 \leq T \quad \forall i.$$

k = latent dimension size

L = cell Laplacian (num neighbors, metric)

L_c = gene Laplacian (num neighbors, metric)

α = weight of cell Laplacian

β = weight of gene Laplacian

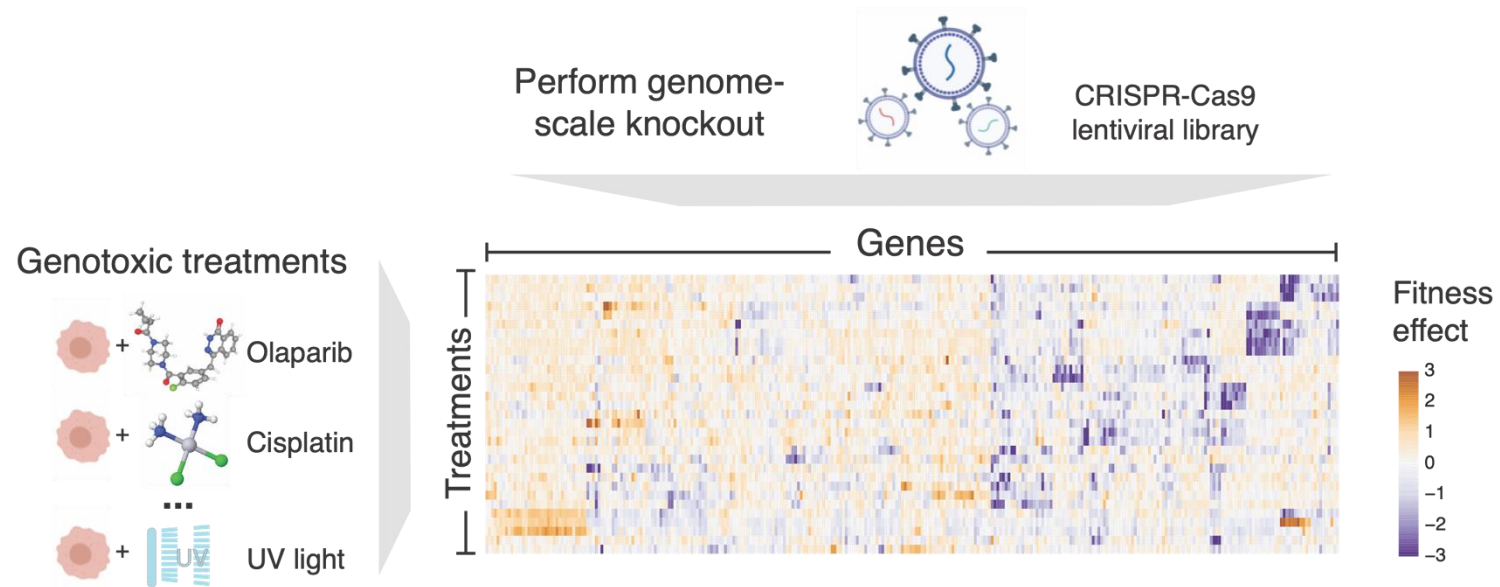
T = sparsity

Applications to three screens of gene perturbation effects

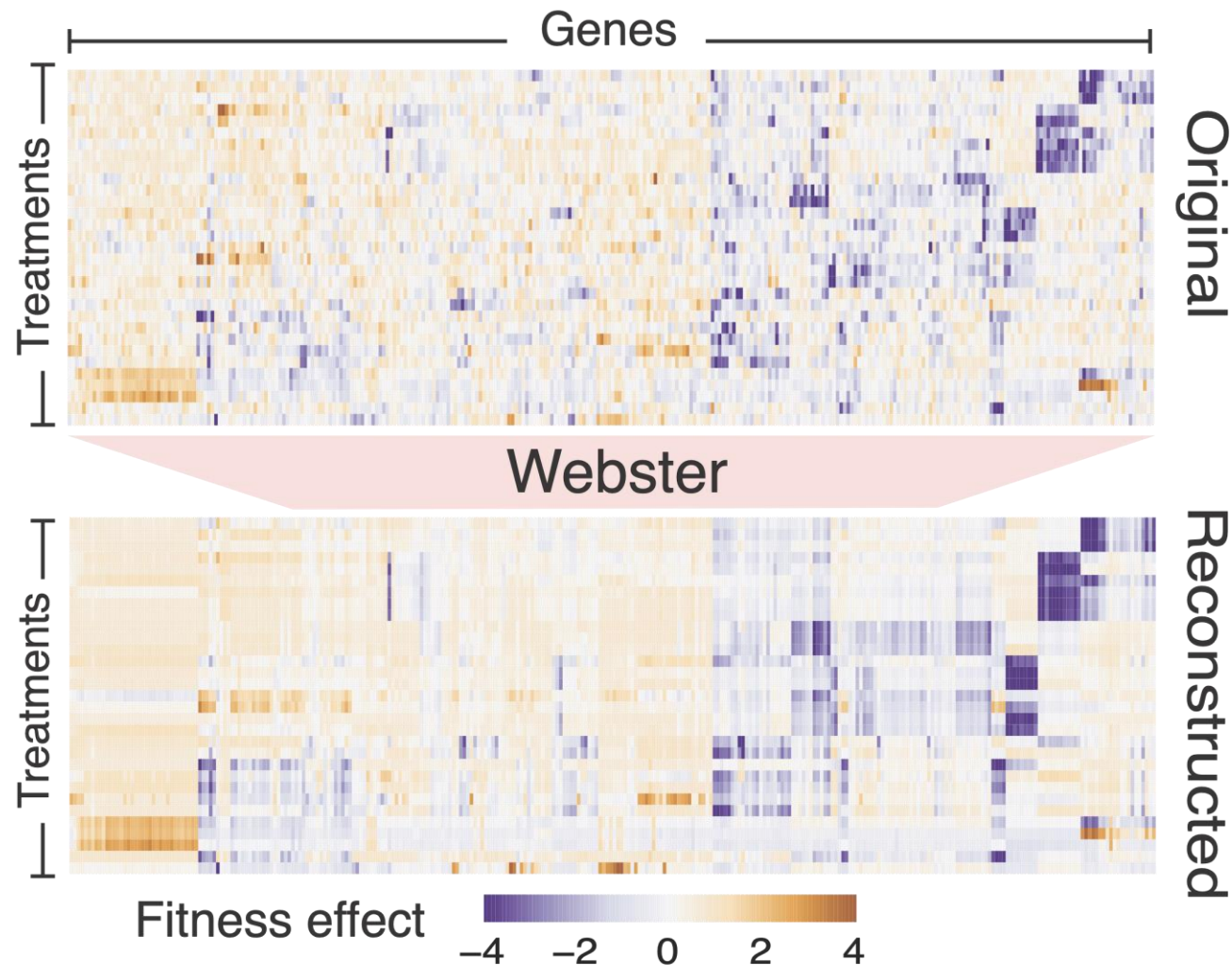
- 1) Genotoxic screens
- 2) Cancer fitness screens
- 3) Compound sensitivity screens

Part 1: Genotoxic screens

Olivieri et al. 2020: fitness effect of gene knockout in presence of genotoxins



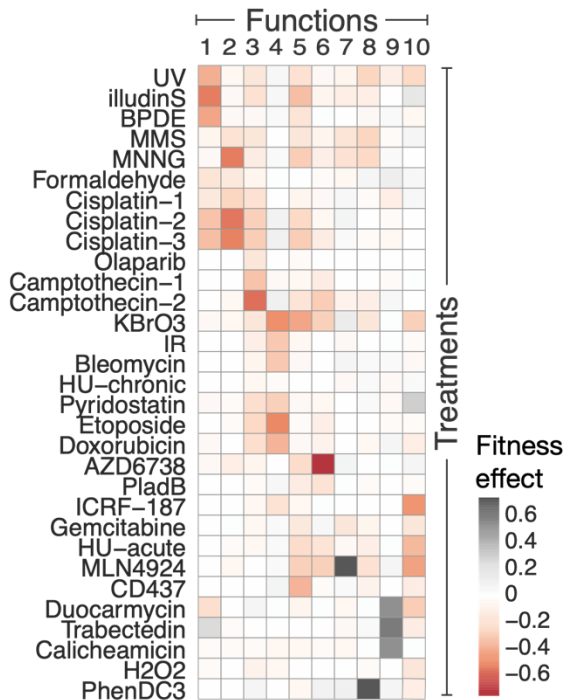
Webster approximates the input data matrix...



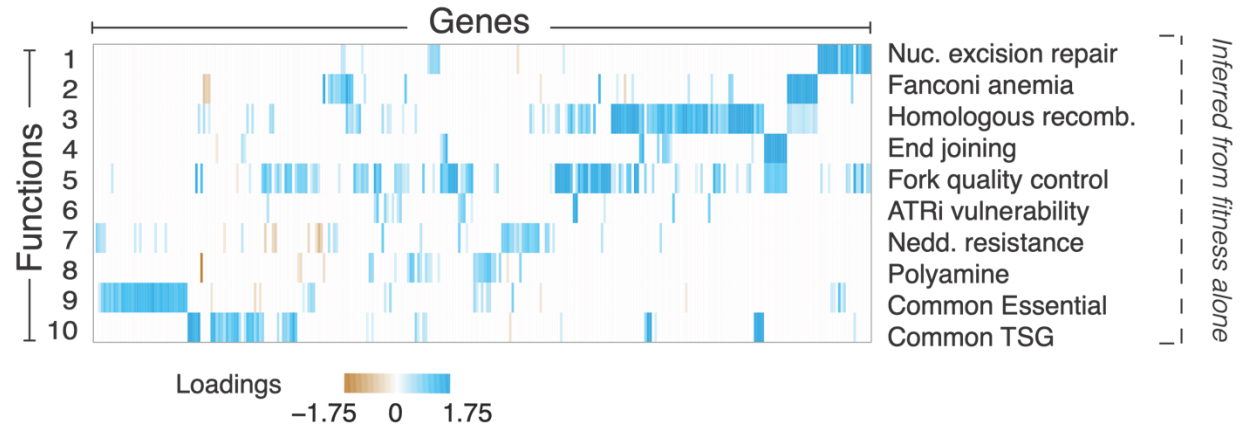
$k=10$
 $t=2$

... as a product between a
dictionary matrix and a loadings
matrix

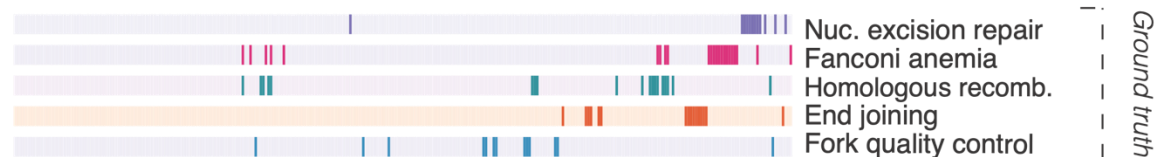
Dictionary matrix



Gene-to-function loadings

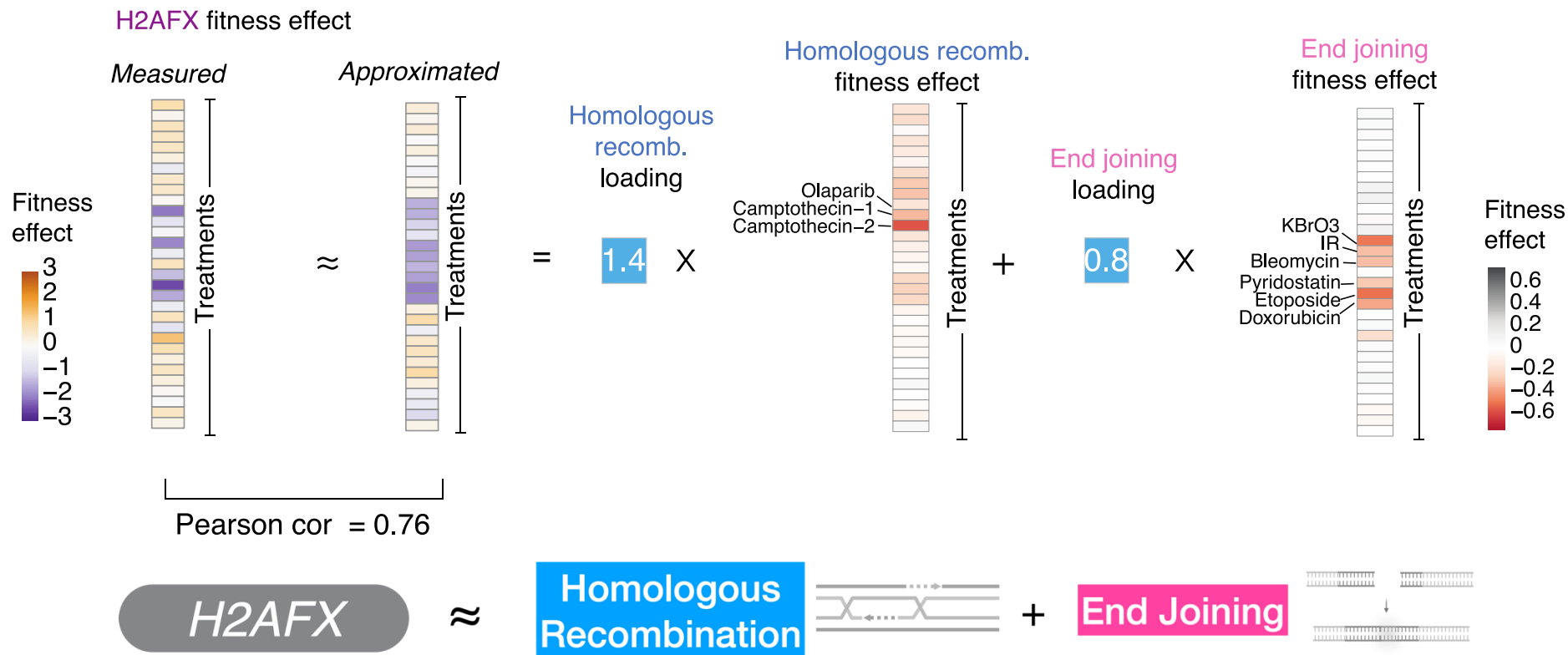


Literature annotations



Learned gene-to-function loadings recover
biological genesets hidden during model training

Latents inferred by the model recapitulate pleiotropy *without prior knowledge*

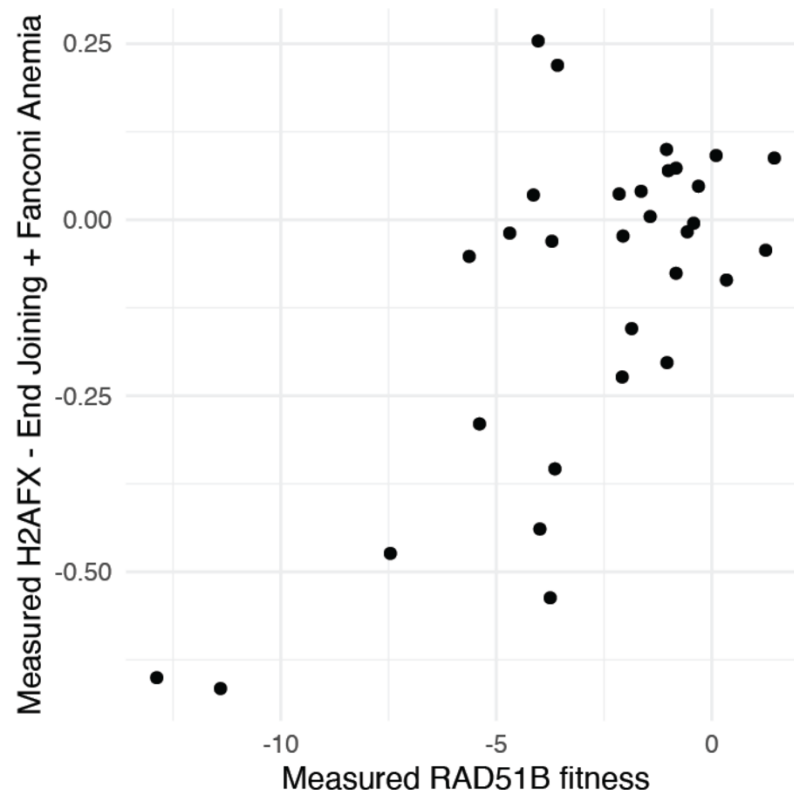


(hidden during model training!)

Latents are biologically meaningful

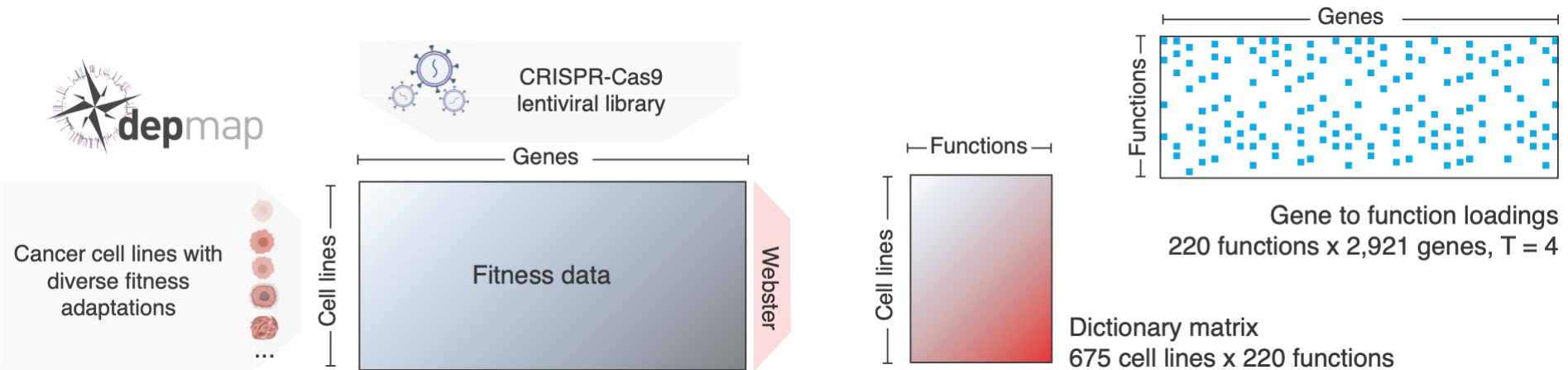
$$\text{geneA} - \text{func1} + \text{func2} \approx \text{geneB}$$

H2AFX - End Joining + Fanconi Anemia $\approx$ RAD51B



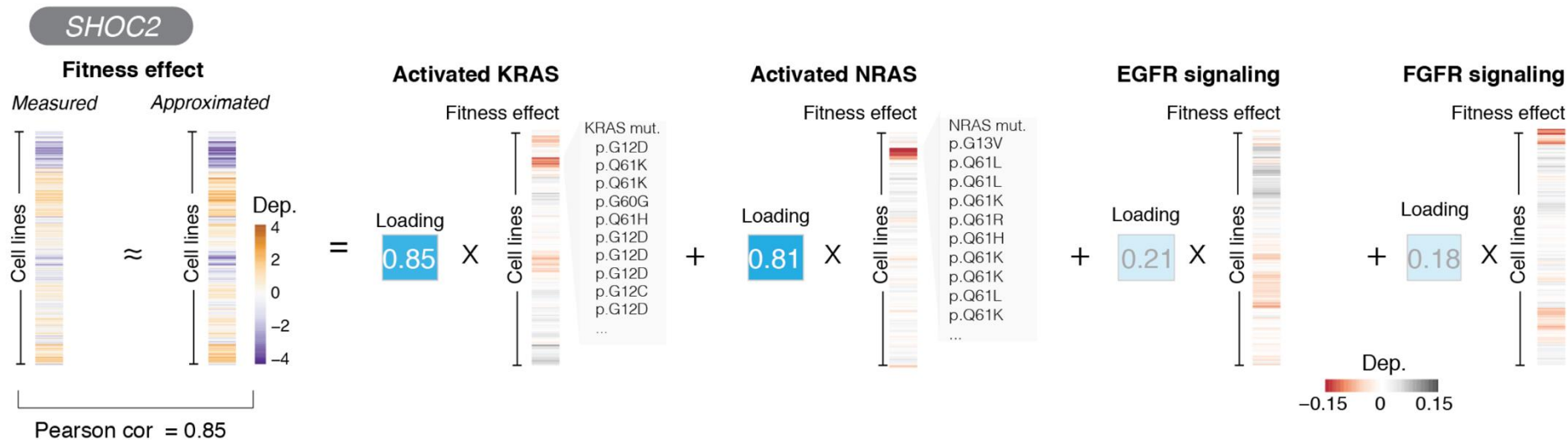
= cell context (treatment)

Part 2: Cancer fitness screens

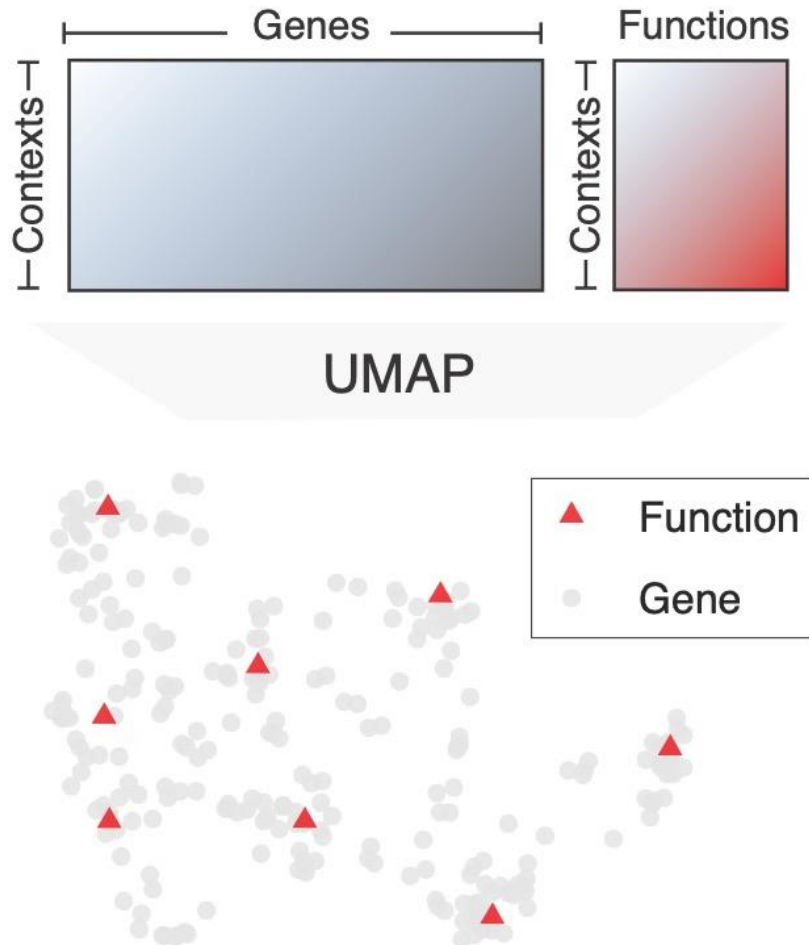


Pleiotropic genes obey linear semantics in the latent space

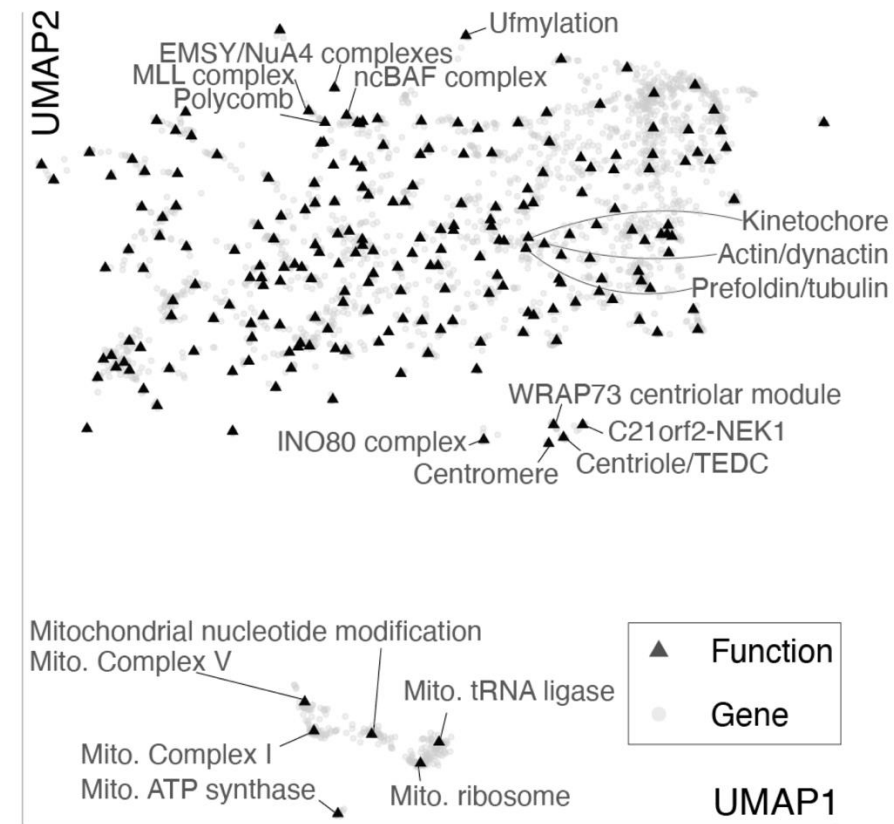
SHOC2 $\approx$ Activated KRAS + Activated NRAS + EGFR Signaling + FGFR Signaling



Joint embedding space of genes and functions



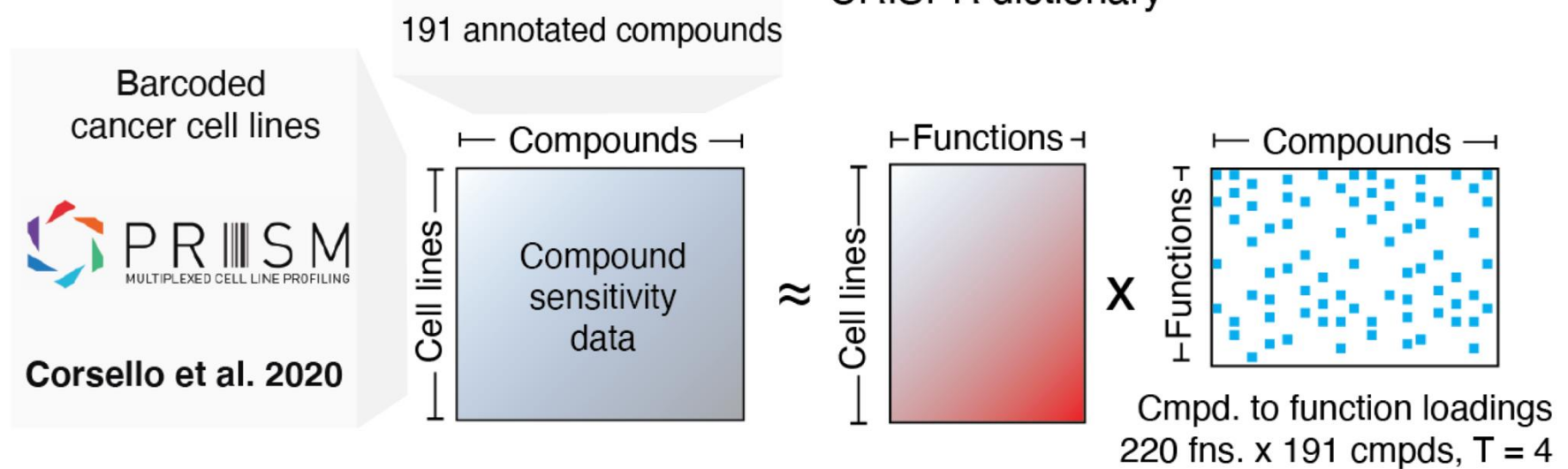
It captures interpretable processes in cancer



Part 3: Compound sensitivity screens

Query: Drug Repurposing dataset

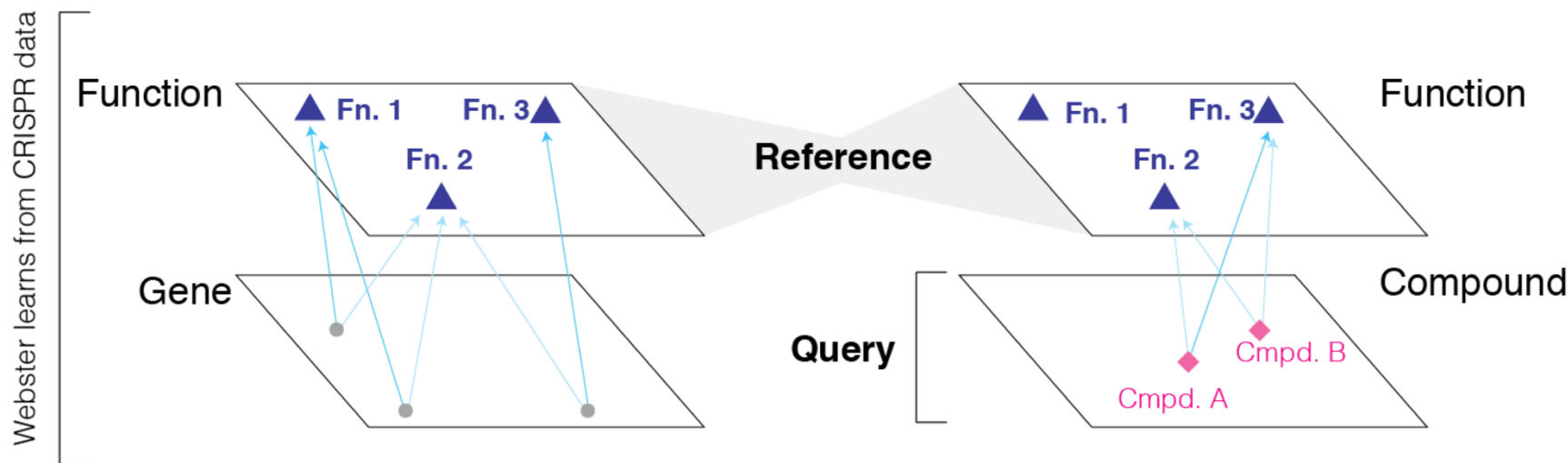
Reference:
CRISPR dictionary



Modeling compound sensitivity profiles as mixtures of functions learned from CRISPR

Modeling compounds as mixtures of latent functions

Reference-query projection

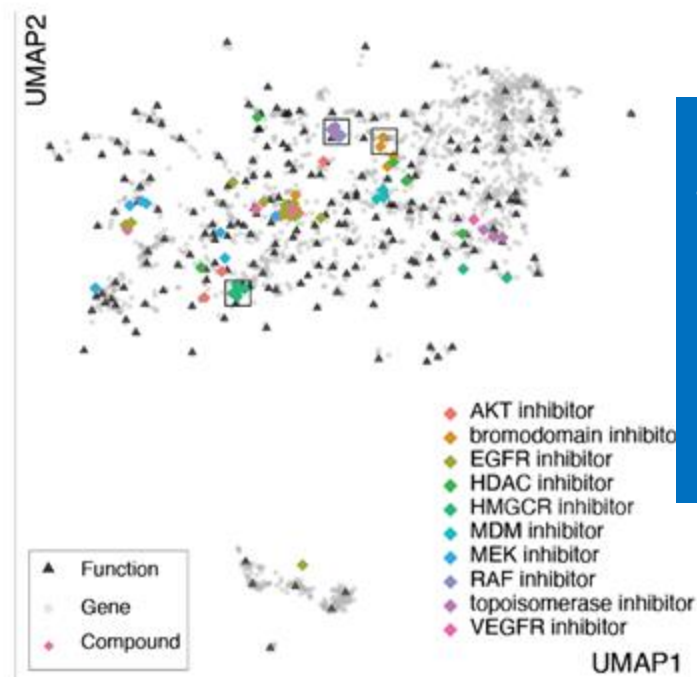


- Modeling compounds as mixtures of functions learned from CRISPR signatures with high similarity represent useful and previously unrecognized connections
 - between two proteins operating in the same pathway
 - between a small-molecule and its protein target
 - between two small-molecules of similar function but structural dissimilarity
- Such a catalog of connections can serve as a functional look-up table of compounds to predict sensitivity and genotoxic profiles and to inform therapeutic use

Compounds' mechanisms of action

Compounds are embedded nearby gene functions, reflecting their mechanism of action

Projecting compound sensitivity into gene fn. map



BRAF signaling

Loadings

BRAF
SOX10
SOX9

H2A.Z maintenance

Loadings

KDM2A
H2AFZ
KANSL3

Mevalonate synthesis

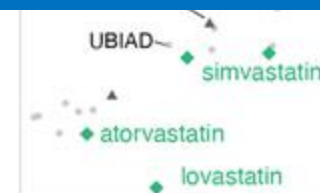
Loadings

UBIAD1
HMGCR
MVK

Refere

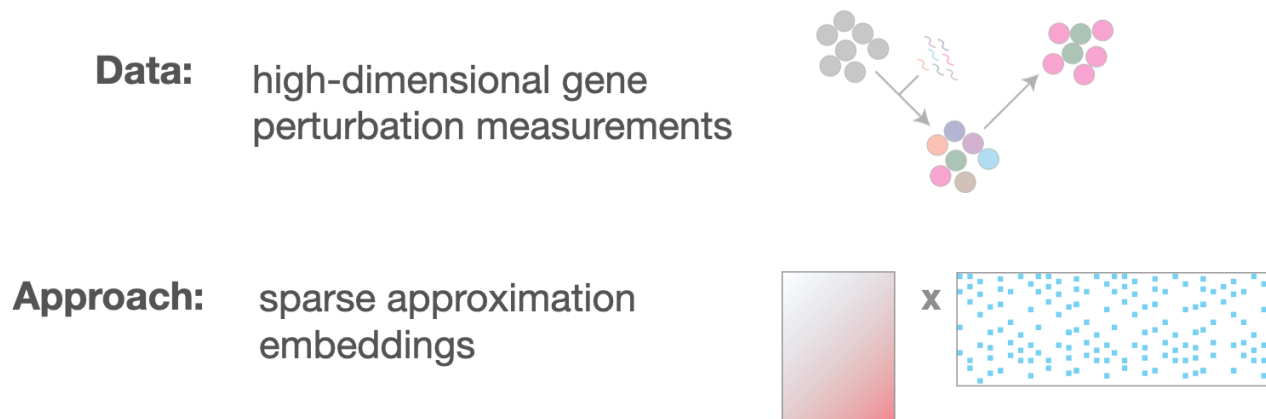
Modeling compounds as mixtures of functions learned from CRISPR signatures with high similarity represent useful and previously unrecognized connections

- between two proteins operating in the same pathway
- between a compound and its protein target
- between two compounds of similar function but structural dissimilarity



Key takeaways

- Analogously to word semantics, genes can be modeled as **distributions over latent bio functions**
 - **Sparse learning** is an effective strategy for learning bio functions from high-dimensional chemical and genetic perturbations
 - New perturbations can be **projected** into learned space



$$geneA - func1 + func2 \approx geneB$$

https://depmap.org/webster

